FHIR-STU3 — Clinical Reasoning (sections 14 – 14.6.10)

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14 Clinical Reasoning

URL: https://hl7.org/fhir/STU3/clinicalreasoning-module.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Work Group Clinical Decision Support & Clinical Quality Information | Ballot Status: Informative

14.0 Clinical Reasoning

14.0.1 Introduction

The Clinical Reasoning module provides resources and operations to enable the representation, distribution, and evaluation of clinical knowledge artifacts such as clinical decision support rules, quality measures, order sets, and protocols. In addition, the module describes how expression languages can be used throughout the specification to provide dynamic capabilities.

Clinical Reasoning involves the ability to represent and encode clinical knowledge in a very broad sense so that it can be integrated into clinical systems. This encoding may be as simple as controlling whether or not a particular section of an order set appears based on the conditions that a patient has, or it may be as complex as representing the care pathway for a patient with multiple conditions.

14.0.1.1 Scope

The Clinical Reasoning module focuses on enabling two primary use cases:

* Sharing - The ability to represent clinical knowledge artifacts such as decision support rules, order sets, protocols, and quality measures, and to do so in a way that enables those artifacts to be shared across organizations.

* Evaluation - The ability to evaluate clinical knowledge artifacts in the context of a specific patient or population, including the ability to request decision support guidance, impact clinical workflow, and retrospectively assess quality metrics.

Note to implementers: The user-facing external clinical decision support use case supported by the ServiceDefinition/$evaluate operation of the Clinical Reasoning module has significant overlap with the functionality provided by the CDS Hooks specification. As part of FHIR Release 4, it is the intention to unify the CDS Hooks specification with the Clinical Reasoning module, ensuring that implementers have a single consistent mechanism to support this use case that meets the requirements of both the Clinical Quality Framework and CDS Hooks communities. Although the functionality in both specifications is conceptually aligned, this unification will likely result in changes to both specifications. The CQF and CDS Hooks project teams are committed to this unification and will work to ensure that the resulting changes have as little impact as possible on current and ongoing implementation efforts, while meeting the needs of both communities. The project team is planning on a workable implementation tested at the connectathon in May of 2017, with a trial ballot following in September 2017, targeting publication as part of the FHIR Release 4 specification.

Specifically, the CDS Hooks/Clinical Reasoning work will impact the ServiceDefinition and GuidanceResponse resources. The ActivityDefinition, PlanDefinition, Library, RequestGroup, Measure, and MeasureReport resources are not expected to be impacted by this work.

To enable these use cases, the module defines several components that can each be used independently, or combined to enable more complex functionality. These components are:

* Expression Logic - the representation of logic using languages such as FHIRPath and Clinical Quality Language (CQL).

* Definitional Resources - resources that are not defined on any specific patient, but are used to define the actions to be performed as part of a clinical knowledge artifact such as an order set or decision support rule. These resources can be used directly, or with profiles to provide intended structure for specific types of resources.

* Knowledge Artifacts - representation of clinical knowledge such as decision support rules and clinical quality measures.

These basic components can then be used to enable a broad variety of clinical decision support and quality measurement use cases, including knowledge sharing, decision support services, and clinical quality assessment and reporting. The topics below provide more detailed discussion on each of these components and uses:

Topic | Description

Overview and Background | If you are interested in the background and development of the FHIR Clinical Reasoning module, this topic covers where it came from and why it exists. See also the general FHIR introductions for clinicians, developers or architects

Using Expressions | If you want to see how to add dynamic capabilities to FHIR resources using expressions, start here.

Definitional Resources | If you want to see how to describe definitional resources using the ActivityDefinition resource, start here.

Representing Knowledge Artifacts | If you want to represent knowledge artifacts such as Event-Condition-Action rules, Order Sets, or Protocols, start here.

Sharing Knowledge Artifacts | If you want to share and distribute knowledge artifacts, start here.

Clinical Decision Support Service | If you want to use the Clinical Reasoning module to provide or use Clinical Decision Support services, start here.

Quality Reporting | If you want to define or report clinical quality measures, start here.

14.0.1.2 Audience

From the perspective of a Knowledge Author, this module describes an approach to representing knowledge artifacts within FHIR.

From the perspective of a Knowledge Content Provider, this module defines search functionality for using a FHIR server as a knowledge artifact repository.

From the perspective of a Knowledge Evaluation Service Provider, this module defines operations and profiles in support of evaluating quality measures, and defining a service for guidance request and response, consistent with the approach taken by the current Decision Support Service specification.

And finally, from the perspective of a Knowledge Evaluation Service Consumer, this module defines the expected available operations and behavior of a knowledge evaluation service.

14.0.2 Index

14.0.2.1 Resources

Resource | Description
ActivityDefinition | A resource to represent definitional resources.
DataRequirement | A data type that represents a general data requirement for a knowledge asset such as a decision support rule or quality measure.
GuidanceResponse | Represents the response to a specific guidance request returned by the $evaluate operation.
Library | Provides a container for knowledge artifacts that includes logic libraries, model definitions, and asset collections.
Measure | Represents a clinical quality measure and provides evaluation through the $evaluate-measure operation.
MeasureReport | Represents the response to a specific measure evaluation request returned by the $evaluate-measure operation.
PlanDefinition | Represents the description of a plan for accomplishing a particular goal. This resource is used to represent a broad variety of clinical knowledge artifacts including decision support rules, order sets, and protocols.
RequestGroup | Represents a group of options for a particular subject that can be used to accomplish a particular goal. This resource is often, but not always, the result of applying a PlanDefinition to a particular patient.
ServiceDefinition | Describes a specific clinical decision support service and supports evaluation using the $evaluate operation.

14.0.2.2 Extensions

Extension | Description
cqlExpression | A general purpose extension that supports the use of Clinical Quality Language within FHIR
fhirPathExpression | A general purpose extension that supports the use of FHIRPath within FHIR
library | A general purpose extension that supports the declaration of dependencies that can be accessed by expression logic
qualityOfEvidence | An extension that can be applied to Attachments to indicate the quality of evidence in support of a particular artifact or recommendation.
strengthOfRecommendation | An extension that can be applied to Attachments to indicate the strength of a recommendation.
optionCode | An extension that can be applied to questionnaire response options to provide meaning to the given option.
sourceValueSet | An extension that can be applied to coding elements to indicate the intended or actual source value set for the coding.
measureInfo | An extension that can be applied to resources to indicate the measure criteria they satisfy. Used in evaluated resource bundles as part of reporting measure results for a patient to identify resources that contributed to the patient's membership in a particular population criteria.

14.0.2.3 Profiles

Profile | Description
CQIF-Questionnaire | Defines extensions to the base Questionnaire that allow it to be used as a DocumentationTemplate with behavior specified via logic in CQL libraries.

14.0.2.4 Services

Service | Description
Knowledge Repository | Defines minimum service capabilities for a knowledge repository
Measure Processor | Defines minimum service capabilities for a measure processor

14.0.3 Security and Privacy

Because Knowledge Artifacts are typically patient-independent, many of the resources in the clinical reasoning module have no patient security and privacy concerns beyond the normal sensitivity that should be paid in any electronic healthcare system environment. However, the evaluation use case, including decision support guidance request/response, as well as quality measure evaluation have significant patient security and privacy concerns.

For the clinical decision support evaluation use case, as with any patient-specific information, care should be taken to ensure that the request and response are properly secured both at rest and in-motion, and that all access to the patient's information is done via a properly authenticated and authorized mechanism. This is particularly true of decision support artifacts where the logic is ingested as part of the definition of the artifact. In this scenario, the evaluation engine must ensure that data access within the ingested logic is subject to the same authentication and authorization requirements as any other access.

For guidance services that receive patient information, ensure that logging and auditing trails do not inadvertently compromise patient privacy and security by logging potentially sensitive information in an unencrypted way. In addition, guidance and recommendations returned from the service must ensure that content that contains patient information is clearly indicated so that consuming clients can take the appropriate care in integrating and displaying the resulting guidance.

For quality measure evaluation, individual and patient-list reports have the potential to contain large amounts of patient information. As with the decision support guidance responses, care must be taken to ensure the patient information is only accessible to properly authenticated and authorized agents, and that inadvertent breaches are minimized by following appropriate logging and auditing protocols.

In particular, because expression languages, depending on their power and scope, can provide the ability to access large amounts of data, as well as the potential for infinite recursion or looping, care should be taken to ensure that implementations adequately safeguard against Denial-of-Service-style attacks that leverage these capabilities to compromise systems by overloading capacity.

For more general considerations, see the Security and Privacy module.

14.0.4 Common use Cases

Use Case | Description
Providing a dynamic value for a resource element | Using expressions to define the value for an element of a FHIR resource.
Defining a CQL library | Using the Library resource to incorporate a Clinical Quality Language library for use in FHIR resources.
Defining a Model Definition artifact | Using the Library resource to incorporate the definition of an information model for use with expressions in FHIR.
Defining an Event Condition Action rule | Using the PlanDefinition resource to represent an event-condition-action rule in FHIR.
Defining a Referral Request activity | Using the ActivityDefinition resource to define a referral request activity that can be used as part of a knowledge artifact.
Defining a Medication Request activity | Using the ActivityDefinition resource to define a medication request activity that can be used as part of a knowledge artifact.
Defining an Order Set | Using the PlanDefinition resource to represent an order set.
Defining a Protocol | Using the PlanDefinition resource to represent a protocol.
Defining a Questionnaire with dynamic content | Using the Questionnaire resource and expression extensions to add dynamic functionality to a FHIR Questionnaire.
Defining a Service Module | Using the ServiceDefinition resource to define a decision support service module.
Obtaining guidance from a Decision Support Service | Using the $evaluate operation to request and process guidance from a decision support service.
Defining a Measure | Using the Measure resource to represent a clinical quality measure.
Evaluating a Measure | Using the $evaluate-measure operation to request calculation of a clinical quality measure.
Applying an ActivityDefinition | Using the $apply operation to realize the intent resource defined by an ActivityDefinition.
Applying a PlanDefinition | Using the $apply operation to realize a plan definition for a specific context.
Representing Quality of Evidence/Strength of Recommendation | Using the qualityOfEvidence and strengthOfRecommendation extensions to indicate ratings associated with evidence for a particular artifact or recommendation.

14.0.5 Developmental Roadmap

The resources defined for the Clinical Reasoning module are the result of the combined efforts of multiple communities working on the shared goal of harmonized standards and specifications for clinical decision support and quality measurement artifacts. The current state of the module reflects changes incorporated both from previous ballots on the FHIR-specific material, as well as content derived from several other balloted specifications in the CDS and CQM domains. The content at this point is capable of supporting the two primary use cases of sharing and evaluation in both domains and for a broad variety of artifacts.

In particular, the use of Clinical Quality Language (CQL) as a foundational mechanism for representing clinical quality logic enables decision support and quality measurement artifacts to share common definitions. For example, a Chlamydia Screening measure and related decision support artifacts focused on improving the measure can share a common library that describes the criteria for detecting when Chlamydia Screening is required in a patient. The decision support rule applies these criteria to determine when and how to impact a workflow, while the quality measure uses these same criteria to determine whether the screening goal has been met for a patient or population. In addition, the resources defined in this module use common patterns for describing the structure of artifacts and their associated metadata, enabling a consistent approach to the sharing and distribution of clinical knowledge artifacts.

The goals of the module over the next year are to provide a stable basis for implementation of the sharing use case, as well as unification with the CDS Hooks specification in support of the evaluation use case. The Clinical Quality Framework Initiative will use these resources as the basis for implementation projects, targeting an FMM level of 3 or 4 for all module resources within a year.

STU Note: We are actively seeking comments on all areas of the module, with particular focus on supporting the following scenarios:

* The use of FHIRPath and CQL to provide dynamic functionality throughout FHIR

* FHIR-Based Knowledge Artifact Repository

* FHIR-Based Clinical Decision Support Service

* FHIR-Based Quality Measure Evaluation and Reporting

* FHIR-Based Order Set Catalog/Ordering Services

* The use of FHIR operations as the protocol for defining artifact evaluation requests and responses. In particular, the ability to bundle multiple requests into a single operation, as well as the use of a generic FHIR operation for any service module evaluation, versus defining an evaluation specific to a given knowledge artifact.

* The use of FHIR interactions in general as a mechanism for enabling knowledge artifact repository and distribution functionality.

Feedback is welcome here .

14.0.6 Background

The FHIR Clinical Reasoning module is sponsored by the Clinical Decision Support (CDS) and Clinical Quality Information (CQI) HL7 Work Groups, with input and coordination from the FHIR Infrastructure and Service Oriented Architecture HL7 Work Groups.

The guidance in this module is prepared as a Universal Realm Specification with support from the Clinical Quality Framework (CQF) initiative , which is a public-private partnership sponsored by the Centers for Medicare & Medicaid Services (CMS) and the U.S. Office of the National Coordinator for Health Information Technology (ONC) to identify, develop, and harmonize standards for clinical decision support and electronic clinical quality measurement.

The Clinical Quality Framework initiative has focused on harmonizing the historically disjointed specifications used by the Clinical Quality Measurement and Clinical Decision Support communities. Specifically, the initiative has focused on the specifications used to represent knowledge artifacts within the two communities. The strategy employed has been to break the conceptual content of knowledge artifacts into three core components, to define common standards for these core components, and to re-use these common standards for both clinical decision support and clinical quality measurement:

* Metadata - Descriptive information about the artifact and its content

* Clinical Information - Information about a patient or population of concern within a given artifact

* Logic - The clinical reasoning involved in an artifact

The first component has resulted in the Clinical Quality Common Metadata Conceptual Model , an informative document harmonizing metadata requirements between Quality Measurement and Decision Support artifacts.

The second component has resulted in the QUICK Conceptual and Logical Models, a harmonization of the Virtual Medical Record (vMR) used in Decision Support and the Quality Data Model (QDM) used in Quality Measurement, and with its core requirements realized in FHIR as the Quality Improvement Core (QICore) profiles . Ongoing work in this area is focusing on coordination with the Clinical Information Modeling Initiative (CIMI) and a methodology for producing FHIR profiles from CIMI models. Currently, the QICore FHIR profiles (which are in turn derived from the US-Core profiles) can be used to model clinical quality data, and to present a consistent model for use in authoring and evaluating clinical quality artifacts.

Finally, the third component has resulted in the Clinical Quality Language specification , a harmonization of the expressive capabilities of the Clinical Decision Support Knowledge Artifact Specification (CDS KAS) (produced by the Health eDecisions (HeD) Standards and Interoperability (S&I) initiative), and the Health Quality Measures Format (HQMF) .

As part of the ongoing CQF initiative pilot efforts, these developing specifications are being used to support knowledge artifact sharing, as well as evaluation of knowledge artifacts as part of decision support request/response and measure evaluation.

This module continues the harmonization of quality domain specifications by defining an approach to using a FHIR server as a component of a knowledge system in both the Knowledge Repository and Knowledge Evaluation Service roles.

14.0.7 Related Specifications

The approach and representations within this guide are derived from and intended to be consistent with the following specifications:

* HL7 Version 3 Standard: Clinical Decision Support Knowledge Artifact Specification, Release 1.3

* HL7 Version 3 Standard: Representation of the Health Quality Measure Format (eMeasure) DSTU, Release 2.1

* HL7 Version 3 Standard: Decision Support Service, Release 2

* HL7 Version 3 Implementation Guide: Decision Support Service, Release 1

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14.0.9 Acknowledgements

The guidance in this module is the work of a joint project between the HL7 Clinical Quality Information and Clinical Decision Support Work Groups with the co-sponsorship of the FHIR Infrastructure, Implementable Technology Specifications, and Service Oriented Architecture Work Groups.

14.1 Resource Library - Content

URL: https://hl7.org/fhir/STU3/library.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

The Library resource is a general-purpose container for knowledge asset definitions. It can be used to describe and expose existing knowledge assets such as logic libraries and information model descriptions, as well as to describe a collection of knowledge assets.

14.1.1 Scope and Usage

The Library resource is a general purpose container for clinical knowledge assets. These assets may be defined using a non-FHIR representation, such as a shareable library of clinical logic, written in Clinical Quality Language (CQL), or the XML Schema for an information model, or they may be defined using FHIR resources such as the description of a protocol represented by a PlanDefinition resource. In addition to representing the metadata of the library, the resource has elements for tracking dependencies, as well as for representing the parameters and data requirements for any expression functionality provided by the library.

For non-FHIR assets, the actual content of the library is represented using the Attachment data type, and may either be referenced with a URL, or the content may be embedded as a base-64 encoded string. Either way, the contentType element of the attachment is used to indicate the representation of the library content.

Note that because the library content may be embedded as well as be retrievable from an external repository via the attachment URL, the possibility exists for the embedded content to be different from the content on the repository. With proper versioning and governance, this should never occur, but to minimize the potential impact of this possibility, implementers SHALL give precedence to the embedded content of a library when it is present.

14.1.2 Boundaries and Relationships

The Library resource is definitional in nature, meaning it is intended to represent shareable knowledge independent of any particular patient. This is in contrast to the DocumentReference resource, which captures non-FHIR content related to a specific patient, and the Media and Binary resources which capture multimedia content and raw binary content, respectively.

This resource is referenced by ActivityDefinition, Measure and PlanDefinition

14.1.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Library | | | DomainResource | Represents a library of quality improvement components Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 0..1 | uri | Logical URI to reference this library (globally unique)
identifier | ? | 0..* | Identifier | Additional identifier for the library
version | ? | 0..1 | string | Business version of the library
name | ? | 0..1 | string | Name for this library (computer friendly)
title | ? | 0..1 | string | Name for this library (human friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
type | ? | 1..1 | CodeableConcept | logic-library | model-definition | asset-collection | module-definition LibraryType (Extensible)
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
description | ? | 0..1 | markdown | Natural language description of the library
purpose | | 0..1 | markdown | Why this library is defined
usage | | 0..1 | string | Describes the clinical usage of the library
approvalDate | | 0..1 | date | When the library was approved by publisher
lastReviewDate | | 0..1 | date | When the library was last reviewed
effectivePeriod | ? | 0..1 | Period | When the library is expected to be used
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for library (if applicable) Jurisdiction ValueSet (Extensible)
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc DefinitionTopic (Example)
contributor | | 0..* | Contributor | A content contributor
contact | ? | 0..* | ContactDetail | Contact details for the publisher
copyright | | 0..1 | markdown | Use and/or publishing restrictions
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.
parameter | | 0..* | ParameterDefinition | Parameters defined by the library

dataRequirement | | 0..* | DataRequirement | What data is referenced by this library
content | | 0..* | Attachment | Contents of the library, either embedded or referenced
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Library xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this library (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the library --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the library -->
 <name value="[string]"/><!-- 0..1 Name for this library (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this library (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <type><!-- 1..1 CodeableConcept logic-library | model-definition | asset-collection | module-definition --></type>
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <description value="[markdown]"/><!-- 0..1 Natural language description of the library -->
 <purpose value="[markdown]"/><!-- 0..1 Why this library is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the library -->
 <approvalDate value="[date]"/><!-- 0..1 When the library was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the library was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the library is expected to be used --></effectivePeriod>
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for library (if applicable) --></jurisdiction>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc --></topic>
 <contributor><!-- 0..* Contributor A content contributor --></contributor>
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <parameter><!-- 0..* ParameterDefinition Parameters defined by the library --></parameter>
 <dataRequirement><!-- 0..* DataRequirement What data is referenced by this library --></dataRequirement>
 <content><!-- 0..* Attachment Contents of the library, either embedded or referenced --></content>
</Library>
```

JSON Template

```
{
  "resourceType" : "Library",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this library (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the library
  "version" : "<string>", // Business version of the library
  "name" : "<string>", // Name for this library (computer friendly)
  "title" : "<string>", // Name for this library (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "type" : { CodeableConcept }, // R!  logic-library | model-definition | asset-collection | module-definition
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "description" : "<markdown>", // Natural language description of the library
  "purpose" : "<markdown>", // Why this library is defined
  "usage" : "<string>", // Describes the clinical usage of the library
  "approvalDate" : "<date>", // When the library was approved by publisher
  "lastReviewDate" : "<date>", // When the library was last reviewed
  "effectivePeriod" : { Period }, // When the library is expected to be used
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for library (if applicable)
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc
  "contributor" : [{ Contributor }], // A content contributor
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "parameter" : [{ ParameterDefinition }], // Parameters defined by the library
  "dataRequirement" : [{ DataRequirement }], // What data is referenced by this library
  "content" : [{ Attachment }] // Contents of the library, either embedded or referenced
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Library;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Library.url [ uri ]; # 0..1 Logical URI to reference this library (globally unique)
  fhir:Library.identifier [ Identifier ], ... ; # 0..* Additional identifier for the library
  fhir:Library.version [ string ]; # 0..1 Business version of the library
  fhir:Library.name [ string ]; # 0..1 Name for this library (computer friendly)
  fhir:Library.title [ string ]; # 0..1 Name for this library (human friendly)
  fhir:Library.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:Library.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:Library.type [ CodeableConcept ]; # 1..1 logic-library | model-definition | asset-collection | module-definition
```

```
  fhir:Library.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:Library.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:Library.description [ markdown ]; # 0..1 Natural language description of the library
  fhir:Library.purpose [ markdown ]; # 0..1 Why this library is defined
  fhir:Library.usage [ string ]; # 0..1 Describes the clinical usage of the library
  fhir:Library.approvalDate [ date ]; # 0..1 When the library was approved by publisher
  fhir:Library.lastReviewDate [ date ]; # 0..1 When the library was last reviewed
  fhir:Library.effectivePeriod [ Period ]; # 0..1 When the library is expected to be used
  fhir:Library.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:Library.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for library (if applicable)
  fhir:Library.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc
  fhir:Library.contributor [ Contributor ], ... ; # 0..* A content contributor
  fhir:Library.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:Library.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:Library.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:Library.parameter [ ParameterDefinition ], ... ; # 0..* Parameters defined by the library
  fhir:Library.dataRequirement [ DataRequirement ], ... ; # 0..* What data is referenced by this library
  fhir:Library.content [ Attachment ], ... ; # 0..* Contents of the library, either embedded or referenced
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

Library | | | DomainResource | Represents a library of quality improvement components Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | ? | 0..1 | uri | Logical URI to reference this library (globally unique)

identifier | ? | 0..* | Identifier | Additional identifier for the library

version | ? | 0..1 | string | Business version of the library

name | ? | 0..1 | string | Name for this library (computer friendly)

title | ? | 0..1 | string | Name for this library (human friendly)

status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage

type | ? | 1..1 | CodeableConcept | logic-library | model-definition | asset-collection | module-definition LibraryType (Extensible)

date | ? | 0..1 | dateTime | Date this was last changed

publisher | ? | 0..1 | string | Name of the publisher (organization or individual)

description | ? | 0..1 | markdown | Natural language description of the library

purpose | | 0..1 | markdown | Why this library is defined

usage | | 0..1 | string | Describes the clinical usage of the library

approvalDate | | 0..1 | date | When the library was approved by publisher

lastReviewDate | | 0..1 | date | When the library was last reviewed

effectivePeriod | ? | 0..1 | Period | When the library is expected to be used

useContext | ? | 0..* | UsageContext | Context the content is intended to support

jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for library (if applicable) Jurisdiction ValueSet (Extensible)

topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc DefinitionTopic (Example)

contributor | | 0..* | Contributor | A content contributor

contact | ? | 0..* | ContactDetail | Contact details for the publisher

copyright | | 0..1 | markdown | Use and/or publishing restrictions

relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc.

parameter | | 0..* | ParameterDefinition | Parameters defined by the library

dataRequirement | | 0..* | DataRequirement | What data is referenced by this library

content | | 0..* | Attachment | Contents of the library, either embedded or referenced

Documentation for this format

UML Diagram (Legend)

XML Template

```
<Library xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this library (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the library --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the library -->
 <name value="[string]"/><!-- 0..1 Name for this library (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this library (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <type><!-- 1..1 CodeableConcept logic-library | model-definition | asset-collection | module-definition --></type>
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <description value="[markdown]"/><!-- 0..1 Natural language description of the library -->
 <purpose value="[markdown]"/><!-- 0..1 Why this library is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the library -->
 <approvalDate value="[date]"/><!-- 0..1 When the library was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the library was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the library is expected to be used --></effectivePeriod>
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for library (if applicable) --></jurisdiction>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc --></topic>
```

```
 <contributor><!-- 0..* Contributor A content contributor --></contributor>
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc. --></relatedArtifact>
 <parameter><!-- 0..* ParameterDefinition Parameters defined by the library --></parameter>
 <dataRequirement><!-- 0..* DataRequirement What data is referenced by this library --></dataRequirement>
 <content><!-- 0..* Attachment Contents of the library, either embedded or referenced --></content>
</Library>
```

JSON Template

```
{
  "resourceType" : "Library",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this library (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the library
  "version" : "<string>", // Business version of the library
  "name" : "<string>", // Name for this library (computer friendly)
  "title" : "<string>", // Name for this library (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "type" : { CodeableConcept }, // R!  logic-library | model-definition | asset-collection | module-definition
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "description" : "<markdown>", // Natural language description of the library
  "purpose" : "<markdown>", // Why this library is defined
  "usage" : "<string>", // Describes the clinical usage of the library
  "approvalDate" : "<date>", // When the library was approved by publisher
  "lastReviewDate" : "<date>", // When the library was last reviewed
  "effectivePeriod" : { Period }, // When the library is expected to be used
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for library (if applicable)
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc
  "contributor" : [{ Contributor }], // A content contributor
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc.
  "parameter" : [{ ParameterDefinition }], // Parameters defined by the library
  "dataRequirement" : [{ DataRequirement }], // What data is referenced by this library
  "content" : [{ Attachment }] // Contents of the library, either embedded or referenced
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Library;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Library.url [ uri ]; # 0..1 Logical URI to reference this library (globally unique)
  fhir:Library.identifier [ Identifier ], ... ; # 0..* Additional identifier for the library
  fhir:Library.version [ string ]; # 0..1 Business version of the library
  fhir:Library.name [ string ]; # 0..1 Name for this library (computer friendly)
  fhir:Library.title [ string ]; # 0..1 Name for this library (human friendly)
  fhir:Library.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:Library.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:Library.type [ CodeableConcept ]; # 1..1 logic-library | model-definition | asset-collection | module-definition
  fhir:Library.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:Library.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:Library.description [ markdown ]; # 0..1 Natural language description of the library
  fhir:Library.purpose [ markdown ]; # 0..1 Why this library is defined
  fhir:Library.usage [ string ]; # 0..1 Describes the clinical usage of the library
  fhir:Library.approvalDate [ date ]; # 0..1 When the library was approved by publisher
  fhir:Library.lastReviewDate [ date ]; # 0..1 When the library was last reviewed
  fhir:Library.effectivePeriod [ Period ]; # 0..1 When the library is expected to be used
  fhir:Library.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:Library.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for library (if applicable)
  fhir:Library.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc
  fhir:Library.contributor [ Contributor ], ... ; # 0..* A content contributor
  fhir:Library.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:Library.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:Library.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc.
  fhir:Library.parameter [ ParameterDefinition ], ... ; # 0..* Parameters defined by the library
  fhir:Library.dataRequirement [ DataRequirement ], ... ; # 0..* What data is referenced by this library
  fhir:Library.content [ Attachment ], ... ; # 0..* Contents of the library, either embedded or referenced
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

14.1.3.1 Terminology Bindings

Path | Definition | Type | Reference

Library.status | The lifecycle status of a Value Set or Concept Map. | Required | PublicationStatus
Library.type | The type of knowledge asset this library contains | Extensible | LibraryType
Library.jurisdiction | Countries and regions within which this artifact is targeted for use | Extensible | Jurisdiction ValueSet
Library.topic | High-level categorization of the definition, used for searching, sorting, and filtering | Example | DefinitionTopic

14.1.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
composed-of | reference | What resource is being referenced | Library.relatedArtifact.where(type='composed-of').resource (Any)
date | date | The library publication date | Library.date
depends-on | reference | What resource is being referenced | Library.relatedArtifact.where(type='depends-on').resource (Any)
derived-from | reference | What resource is being referenced | Library.relatedArtifact.where(type='derived-from').resource (Any)
description | string | The description of the library | Library.description
effective | date | The time during which the library is intended to be in use | Library.effectivePeriod
identifier | token | External identifier for the library | Library.identifier
jurisdiction | token | Intended jurisdiction for the library | Library.jurisdiction
name | string | Computationally friendly name of the library | Library.name
predecessor | reference | What resource is being referenced | Library.relatedArtifact.where(type='predecessor').resource (Any)
publisher | string | Name of the publisher of the library | Library.publisher
status | token | The current status of the library | Library.status
successor | reference | What resource is being referenced | Library.relatedArtifact.where(type='successor').resource (Any)
title | string | The human-friendly name of the library | Library.title
topic | token | Topics associated with the module | Library.topic
url | uri | The uri that identifies the library | Library.url
version | token | The business version of the library | Library.version

14.1.5 Resource Library - Examples

URL: https://hl7.org/fhir/STU3/library-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.1.5 Resource Library - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

These examples illustrate the use of the Library resource to share quality measurement and improvement logic as a FHIR resource. Note that the actual content of each library (the Clinical Quality Language (CQL) source, or the XML for the information model definition, etc.) should be contained in the content element as an Attachment. For illustration purposes, the content of each library is included in its raw form, and linked here.

Example | Description | Content
Chlamydia Screening Common Library | Common Chlamydia Screening library developed as the running example in the Author's Guide of the Clinical Quality Language specification. | CQL Source
Exclusive Breastfeeding Intervention Logic | Clinical Decision Support rules derived from CMS9v4 - Exclusive Breastfeeding measure | CQL Source
Exclusive Breastfeeding Measure Logic | Quality Measurement logic used to express the CMS9v4 measure. | CQL Source
CMS 146 Measure Logic | Measure logic for CMS 146 used as the running example for Quality Reporting guidance in the Clinical Reasoning module. | CQL Source
QUICK Model Definition | The QUICK model information file used by the CQL-to-ELM translator to validate CQL written against the QUICK data model. | XML Information Model
FHIR Model Definition | The FHIR model information file used by the CQL-to-ELM translator to validate CQL written against the FHIR data model. | XML Information Model

Example List:

Chlamydia Screening Common Library | example | XML | JSON | Turtle
Exclusive Breastfeeding Intervention Logic | library-exclusive-breastfeeding-cds-logic | XML | JSON | Turtle
Exclusive Breastfeeding Measurement Logic | library-exclusive-breastfeeding-cqm-logic | XML | JSON | Turtle
CMS146 Measure Logic | library-cms146-example | XML | JSON | Turtle
QUICK Model Definition | library-quick-model-definition | XML | JSON | Turtle
FHIR Model Definition | library-fhir-model-definition | XML | JSON | Turtle
Suicide Risk Order Set Logic | suiciderisk-orderset-logic | XML | JSON | Turtle
FHIR Helpers | library-fhir-helpers | XML | JSON | Turtle
Zika Virus Management Intervention Logic | zika-virus-intervention-logic | XML | JSON | Turtle
Zika Virus Management Library | composition-example | XML | JSON | Turtle
Predecessor library example | library-fhir-helpers-predecessor | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.1.6 Resource Library - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/library-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.1.6 Resource Library - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Library resource.

Library
Definition | The Library resource is a general-purpose container for knowledge asset definitions. It can be used to describe and expose existing knowledge assets such as logic libraries and information model descriptions, as well as to describe a collection of knowledge assets.
Control | 1..1
Library.url
Definition | An absolute URI that is used to identify this library when it is referenced in a specification, model, design or an instance. This SHALL be a URL, SHOULD be globally unique, and SHOULD be an address at which this library is (or will be) published. The URL SHOULD include the major version of the library. For more information see Technical and Business Versions.
Control | 0..1
Type | uri
Requirements | Allows the library to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid:, but real http: addresses are preferred. Multiple instances may share the same url if they have a distinct version.
Library.identifier
Definition | A formal identifier that is used to identify this library when it is represented in other formats, or referenced in a specification, model, design or an instance. e.g. CMS or NQF identifiers for a measure artifact. Note that at least one identifier is required for non-experimental active artifacts.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, e.g., to identify this library outside of FHIR, where it is not possible to use the logical URI.
Library.version
Definition | The identifier that is used to identify this version of the library when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the library author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a version consistent with the Decision Support Service specification, use the format Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification. Note that a version is required for non-experimental active artifacts.
Note | This is a business versionId, not a resource version id (see discussion)
Control | 0..1
Type | string
Summary | true
Comments | There may be different library instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a refrence to a particular business version of the library with the format [url]|[version].
Library.name
Definition | A natural language name identifying the library. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Control | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alpha-numeric type name to ensure that it is computable friendly.
Library.title
Definition | A short, descriptive, user-friendly title for the library.
Control | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
Library.status
Definition | The status of this library. Enables tracking the life-cycle of the content.
Control | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | Allows filtering of librarys that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use a retired library without due consideration.
Library.experimental
Definition | A boolean value to indicate that this library is authored for testing purposes (or education/evaluation/marketing), and is not intended to be used for genuine usage.
Control | 0..1
Type | boolean
Is Modifier | true

Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level library.
Summary | true
Comments | Allows filtering of library that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use an experimental library in production.
Library.type
Definition | Identifies the type of library such as a Logic Library, Model Definition, Asset Collection, or Module Definition.
Control | 1..1
Terminology Binding | LibraryType (Extensible)
Type | CodeableConcept
Summary | true
Library.date
Definition | The date (and optionally time) when the library was published. The date must change if and when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the library changes.
Control | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the library. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
Library.publisher
Definition | The name of the individual or organization that published the library.
Control | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the library. May also allow for contact.
Summary | true
Comments | Usually an organization, but may be an individual. The publisher (or steward) of the library is the organization or individual primarily responsible for the maintenance and upkeep of the library. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the library. This item SHOULD be populated unless the information is available from context.
Library.description
Definition | A free text natural language description of the library from a consumer's perspective.
Control | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the library was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the library as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the profile is presumed to be the predominant language in the place the profile was created).
Library.purpose
Definition | Explaination of why this library is needed and why it has been designed as it has.
Control | 0..1
Type | markdown
Comments | This element does not describe the usage of the library Instead it provides traceability of "why" the resource is either needed or "why" it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this library.
Library.usage
Definition | A detailed description of how the library is used from a clinical perspective.
Control | 0..1
Type | string
Library.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Control | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes / editorial corrections.
Library.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval, but doesn't change the original approval date.
Control | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.
Comments | If specified, this is usually after the approval date.
Library.effectivePeriod
Definition | The period during which the library content was or is planned to be in active use.
Control | 0..1
Type | Period
Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of a the library are or are expected to be used instead.
Summary | true
Comments | The effective period for a library determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
Library.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These terms may be used to assist with indexing and searching for appropriate library instances.

Control | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation whether all or any of the contexts apply.
Library.jurisdiction
Definition | A legal or geographic region in which the library is intended to be used.
Control | 0..*
Terminology Binding | Jurisdiction ValueSet (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the library to be used in jurisdictions other than those for which it was originally designed or intended.
Library.topic
Definition | Descriptive topics related to the content of the library. Topics provide a high-level categorization of the library that can be useful for filtering and searching.
Control | 0..*
Terminology Binding | DefinitionTopic (Example)
Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the library so that it can be found by topical searches.
Library.contributor
Definition | A contributor to the content of the library, including authors, editors, reviewers, and endorsers.
Control | 0..*
Type | Contributor
Requirements | Consumers of the content must be able to quickly determine who contributed to the content of the library.
Library.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Control | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
Library.copyright
Definition | A copyright statement relating to the library and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the library.
Control | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the library and/or its content.
Alternate Names | License; Restrictions
Library.relatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Control | 0..*
Type | RelatedArtifact
Requirements | Libraries must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Comments | Each related artifact is either an attachment, or a reference to another resource, but not both.
Library.parameter
Definition | The parameter element defines parameters used by the library.
Control | 0..*
Type | ParameterDefinition
Library.dataRequirement
Definition | Describes a set of data that must be provided in order to be able to successfully perform the computations defined by the library.
Control | 0..*
Type | DataRequirement
Library.content
Definition | The content of the library as an Attachment. The content may be a reference to a url, or may be directly embedded as a base-64 string. Either way, the contentType of the attachment determines how to interpret the content.
Control | 0..*
Type | Attachment

14.1.7 Resource Library - Mappings

URL: https://hl7.org/fhir/STU3/library-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.1.7 Resource Library - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

RIM Mapping

Mappings for the library resource.

14.1.7.1 Mappings for RIM Mapping (http://hl7.org/v3)

Library | Act[classCode=GROUPER;moodCode=EVN]
url | .identifier[scope=BUSN;reliability=ISS]
identifier | .identifier
version | N/A (to add?)
name | N/A
title | .title
status | .status
experimental | N/A (to add?)
type | .code
date | .participation[typeCode=AUT].time
publisher | .participation[typeCode=AUT].role
description | .text
purpose | .reasonCode.text
usage | N/A (to add?)
approvalDate | .outboundRelationship[typeCode="SUBJ"].act[classCode=CACT;moodCode=EVN;code="approval"].effectiveTime
lastReviewDate | .outboundRelationship[typeCode="SUBJ";
subsetCode="RECENT"].act[classCode=CACT;moodCode=EVN;code="review"].effectiveTime
effectivePeriod | N/A (to add?)
useContext | N/A (to add?)
jurisdiction | N/A (to add?)
topic | N/A (to add?)
contributor | .participation[typeCode=AUT,VRF] {No support yet for editor nor clear whether VRF best corresponds to reviewer or endorser}
contact | .participation[typeCode=CALLBCK].role
copyright | N/A (to add?)
relatedArtifact | .outboundRelationship[typeCode=DOC,RSON,PREV, DRIV, USE, COMP] {successor would be PREV w/ inversionInd=true; No support for citation}
parameter | N/A (to add?)
dataRequirement | N/A (to add?)
content | .text

14.1.8 Resource Library - Design Notes

URL: https://hl7.org/fhir/STU3/library-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.1.8 Resource Library - Design Notes

to do

14.1.9 Resource Library - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/library-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.1.9 Resource Library - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Library resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

14.1.9.1 Profiles

No Profiles defined for this resource

14.1.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

cqif-library | library | for CQIF Extensions

14.1.9.3 Search Extensions

No Search Extensions defined for this resource

14.1.10 Resource Library - Operations

URL: https://hl7.org/fhir/STU3/library-operations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.1.10 Resource Library - Operations

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

This resource has 1 operation associated with it:

$data-requirements | Data Requirements

For more information about operations, including how they are invoked, see Operations.

14.1.10.1 Data Requirements

The data-requirements operation aggregates and returns the parameters and data requirements for a resource and all its dependencies as a single module definition

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Library-data-requirements
```

Formal Definition (as a OperationDefinition).

URL: [base]/$data-requirements

URL: [base]/Library/[id]/$data-requirements

This is not an idempotent operation

In Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
target	0..1	Reference(Any)			The target of the data requirements operation

Out Parameters:

Name	Cardinality	Type	Binding	Profile	Documentation
return	1..1	Library			The result of the requirements gathering Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The effect of invoking this operation is to determine the aggregate set of data requirements and dependencies for a given target resource. The result is a Library resource with a type of module-definition that contains all the parameter definitions and data requirements of the target resource and any libraries referenced by it. Implementations SHOULD aggregate data requirements intelligently (i.e. by collapsing overlapping data requirements)

14.1.10.1.1 Examples

Request data requirements for the CMS146 library (Request):

```
GET [base]/Library/library-cms146-example/$data-requirements
```

Returned on successful evaluation (Response):

```
HTTP/1.1 200 OK

{
  "resourceType": "Library",
  "type": {
    "coding": [
      {
        "code": "module-definition"
      }
    ]
  },
  "relatedArtifact": [
    {
      "type": "depends-on",
      "resource": {
        "reference": "Library/library-quick-model-definition"
      }
    }
  ],
  "dataRequirement": [
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "code",
          "valueSetString": "Other Female Reproductive Conditions"
        }
      ]
    },
    {
      "type": "Patient"
    },
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "category",
          "valueCode": [
            "diagnosis"
          ]
```

```
    },
    {
      "path": "clinicalStatus",
      "valueCode": [
        "confirmed"
      ]
    },
    {
      "path": "code",
      "valueSetString": "2.16.840.1.113883.3.464.1003.102.12.1011"
    }
  ]
},
{
  "type": "Condition",
  "codeFilter": [
    {
      "path": "category",
      "valueCode": [
        "diagnosis"
      ]
    },
    {
      "path": "clinicalStatus",
      "valueCode": [
        "confirmed"
      ]
    },
    {
      "path": "code",
      "valueSetString": "2.16.840.1.113883.3.464.1003.102.12.1012"
    }
  ]
},
{
  "type": "Encounter",
  "codeFilter": [
    {
      "path": "status",
      "valueCode": [
        "finished"
      ]
    },
    {
      "path": "class",
      "valueCode": [
        "ambulatory"
      ]
    },
    {
      "path": "type",
      "valueSetString": "2.16.840.1.113883.3.464.1003.101.12.1061"
    }
  ]
},
{
  "type": "DiagnosticReport",
  "codeFilter": [
    {
      "path": "diagnosis",
      "valueSetString": "2.16.840.1.113883.3.464.1003.198.12.1012"
    }
  ]
},
{
  "type": "Medication",
  "codeFilter": [
    {
      "path": "code",
      "valueSetString": "2.16.840.1.113883.3.464.1003.196.12.1001"
    }
  ]
},
{
  "type": "MedicationRequest",
  "codeFilter": [
    {
      "path": "status",
      "valueCode": [
        "active"
      ]
    },
    {
      "path": "medication.code",
      "valueSetString": "2.16.840.1.113883.3.464.1003.196.12.1001"
    }
  ]
},
{
```

```
      "type": "MedicationStatement",
      "codeFilter": [
        {
          "path": "status",
          "valueCode": [
            "completed"
          ]
        },
        {
          "path": "medication.code",
          "valueSetString": "2.16.840.1.113883.3.464.1003.196.12.1001"
        }
      ]
    }
  ]
}
```

14.2 Resource ServiceDefinition - Content

URL: https://hl7.org/fhir/STU3/servicedefinition.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Clinical Decision Support Work Group | Maturity Level: 0 | Draft | Compartments: Not linked to any defined compartments

The ServiceDefinition describes a unit of decision support functionality that is made available as a service, such as immunization modules or drug-drug interaction checking.

14.2.1 Scope and Usage

This resource is a definition resource from a FHIR workflow perspective - see Workflow, specifically Definition.

The ServiceDefinition defines a module of clinical decision support functionality made available by a decision support service. For example, a service may provide immunization modules, drug-drug interaction checking, or appropriate use assessment.

Each module defines three main features related to its functionality:

* Input and output parameters

* Data requirements

* Triggers

Input and output parameters are used to specify any named parameters used by the module. These are typically patient-independent configuration parameters such as an A1C threshold for a diabetes control module, but they may also be used to return calculations performed by the module.

Data requirements are used to specify the set of data that must be provided (or available) to the module in order to achieve a successful evaluation. For example, if the module requires A1C lab results within the last 6 months, or information on bilateral or both left and right amputation at or below the knee.

Triggers are used to advertise when the module should be invoked. On encountering a specific trigger, a clinical application can invoke the modules associated with the trigger using the $evaluate operation. Any data required by the module can be sent as part of the request, and any suggested actions and other output data are returned via the GuidanceResponse resource.

For a more detailed discussion of the evaluation process, refer to the Clinical Decision Support topic in the Clinical Reasoning module.

Note to implementers: The user-facing external clinical decision support use case supported by the ServiceDefinition/$evaluate operation of the Clinical Reasoning module has significant overlap with the functionality provided by the CDS Hooks specification. As part of FHIR Release 4, it is the intention to unify the CDS Hooks specification with the Clinical Reasoning module, ensuring that implementers have a single consistent mechanism to support this use case that meets the requirements of both the Clinical Quality Framework and CDS Hooks communities. Although the functionality in both specifications is conceptually aligned, this unification will likely result in changes to both specifications. The CQF and CDS Hooks project teams are committed to this unification and will work to ensure that the resulting changes have as little impact as possible on current and ongoing implementation efforts, while meeting the needs of both communities. The project team is planning on a workable implementation tested at the connectathon in May of 2017, with a trial ballot following in September 2017, targeting publication as part of the FHIR Release 4 specification.

14.2.2 Boundaries and Relationships

The ServiceDefinition is similar in structure and purpose to OperationDefinition. However, ServiceDefinition also represents more information about what service is being described than an operation definition supports. This information is critical to the effective use of the resource as part of a decision support service repository, as it enables searching and selection of functionality based on this additional information.

In addition, the way the $evaluate operation of the ServiceDefinition resource is defined enables all decision support services to share a common set of parameters without having to redeclare them for every module, while still supporting service-specific paramters on each module.

This resource is referenced by guidanceresponse

14.2.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
ServiceDefinition | | | DomainResource | A description of decision support service functionality Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 0..1 | uri | Logical URI to reference this service definition (globally unique)
identifier | ? | 0..* | Identifier | Additional identifier for the service definition
version | ? | 0..1 | string | Business version of the service definition
name | ? | 0..1 | string | Name for this service definition (computer friendly)
title | ? | 0..1 | string | Name for this service definition (human friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
description | | 0..1 | markdown | Natural language description of the service definition
purpose | | 0..1 | markdown | Why this service definition is defined
usage | | 0..1 | string | Describes the clinical usage of the module
approvalDate | | 0..1 | date | When the service definition was approved by publisher
lastReviewDate | | 0..1 | date | When the service definition was last reviewed
effectivePeriod | ? | 0..1 | Period | When the service definition is expected to be used
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for service definition (if applicable) Jurisdiction ValueSet (Extensible)
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc DefinitionTopic (Example)
contributor | | 0..* | Contributor | A content contributor
contact | ? | 0..* | ContactDetail | Contact details for the publisher
copyright | | 0..1 | markdown | Use and/or publishing restrictions
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc
trigger | | 0..* | TriggerDefinition | "when" the module should be invoked
dataRequirement | | 0..* | DataRequirement | What data is used by the module
operationDefinition | | 0..1 | Reference(OperationDefinition) | Operation to invoke
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ServiceDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this service definition (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the service definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the service definition -->
 <name value="[string]"/><!-- 0..1 Name for this service definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this service definition (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <description value="[markdown]"/><!-- 0..1 Natural language description of the service definition -->
 <purpose value="[markdown]"/><!-- 0..1 Why this service definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the module -->
 <approvalDate value="[date]"/><!-- 0..1 When the service definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the service definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the service definition is expected to be used --></effectivePeriod>
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for service definition (if applicable) --></jurisdiction>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc --></topic>
 <contributor><!-- 0..* Contributor A content contributor --></contributor>
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc --></relatedArtifact>
 <trigger><!-- 0..* TriggerDefinition "when" the module should be invoked --></trigger>
 <dataRequirement><!-- 0..* DataRequirement What data is used by the module --></dataRequirement>
 <operationDefinition><!-- 0..1 Reference(OperationDefinition) Operation to invoke --></operationDefinition>
</ServiceDefinition>
```

JSON Template

```
{
  "resourceType" : "ServiceDefinition",
  // from Resource: id, meta, implicitRules, and language
```

```
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this service definition (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the service definition
  "version" : "<string>", // Business version of the service definition
  "name" : "<string>", // Name for this service definition (computer friendly)
  "title" : "<string>", // Name for this service definition (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "description" : "<markdown>", // Natural language description of the service definition
  "purpose" : "<markdown>", // Why this service definition is defined
  "usage" : "<string>", // Describes the clinical usage of the module
  "approvalDate" : "<date>", // When the service definition was approved by publisher
  "lastReviewDate" : "<date>", // When the service definition was last reviewed
  "effectivePeriod" : { Period }, // When the service definition is expected to be used
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for service definition (if applicable)
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc
  "contributor" : [{ Contributor }], // A content contributor
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc
  "trigger" : [{ TriggerDefinition }], // "when" the module should be invoked
  "dataRequirement" : [{ DataRequirement }], // What data is used by the module
  "operationDefinition" : { Reference(OperationDefinition) } // Operation to invoke
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:ServiceDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ServiceDefinition.url [ uri ]; # 0..1 Logical URI to reference this service definition (globally unique)
  fhir:ServiceDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the service definition
  fhir:ServiceDefinition.version [ string ]; # 0..1 Business version of the service definition
  fhir:ServiceDefinition.name [ string ]; # 0..1 Name for this service definition (computer friendly)
  fhir:ServiceDefinition.title [ string ]; # 0..1 Name for this service definition (human friendly)
  fhir:ServiceDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ServiceDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:ServiceDefinition.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:ServiceDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ServiceDefinition.description [ markdown ]; # 0..1 Natural language description of the service definition
  fhir:ServiceDefinition.purpose [ markdown ]; # 0..1 Why this service definition is defined
  fhir:ServiceDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the module
  fhir:ServiceDefinition.approvalDate [ date ]; # 0..1 When the service definition was approved by publisher
  fhir:ServiceDefinition.lastReviewDate [ date ]; # 0..1 When the service definition was last reviewed
  fhir:ServiceDefinition.effectivePeriod [ Period ]; # 0..1 When the service definition is expected to be used
  fhir:ServiceDefinition.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:ServiceDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for service definition (if
     applicable)
  fhir:ServiceDefinition.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc
  fhir:ServiceDefinition.contributor [ Contributor ], ... ; # 0..* A content contributor
  fhir:ServiceDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ServiceDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ServiceDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc
  fhir:ServiceDefinition.trigger [ TriggerDefinition ], ... ; # 0..* "when" the module should be invoked
  fhir:ServiceDefinition.dataRequirement [ DataRequirement ], ... ; # 0..* What data is used by the module
  fhir:ServiceDefinition.operationDefinition [ Reference(OperationDefinition) ]; # 0..1 Operation to invoke
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

ServiceDefinition | | | DomainResource | A description of decision support service functionality Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

url | ? | 0..1 | uri | Logical URI to reference this service definition (globally unique)

identifier | ? | 0..* | Identifier | Additional identifier for the service definition

version | ? | 0..1 | string | Business version of the service definition

name | ? | 0..1 | string | Name for this service definition (computer friendly)

title | ? | 0..1 | string | Name for this service definition (human friendly)

status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)

experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage

date | ? | 0..1 | dateTime | Date this was last changed

publisher | ? | 0..1 | string | Name of the publisher (organization or individual)

description | | 0..1 | markdown | Natural language description of the service definition

purpose | | 0..1 | markdown | Why this service definition is defined

usage | | 0..1 | string | Describes the clinical usage of the module
approvalDate | | 0..1 | date | When the service definition was approved by publisher
lastReviewDate | | 0..1 | date | When the service definition was last reviewed
effectivePeriod | ? | 0..1 | Period | When the service definition is expected to be used
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for service definition (if applicable) Jurisdiction ValueSet (Extensible)
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc DefinitionTopic (Example)
contributor | | 0..* | Contributor | A content contributor
contact | ? | 0..* | ContactDetail | Contact details for the publisher
copyright | | 0..1 | markdown | Use and/or publishing restrictions
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc
trigger | | 0..* | TriggerDefinition | "when" the module should be invoked
dataRequirement | | 0..* | DataRequirement | What data is used by the module
operationDefinition | | 0..1 | Reference(OperationDefinition) | Operation to invoke
Documentation for this format

UML Diagram (Legend)

XML Template

```
<ServiceDefinition xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this service definition (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the service definition --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the service definition -->
 <name value="[string]"/><!-- 0..1 Name for this service definition (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this service definition (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <description value="[markdown]"/><!-- 0..1 Natural language description of the service definition -->
 <purpose value="[markdown]"/><!-- 0..1 Why this service definition is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the module -->
 <approvalDate value="[date]"/><!-- 0..1 When the service definition was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the service definition was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the service definition is expected to be used --></effectivePeriod>
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for service definition (if applicable) --></jurisdiction>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc --></topic>
 <contributor><!-- 0..* Contributor A content contributor --></contributor>
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc --></relatedArtifact>
 <trigger><!-- 0..* TriggerDefinition "when" the module should be invoked --></trigger>
 <dataRequirement><!-- 0..* DataRequirement What data is used by the module --></dataRequirement>
 <operationDefinition><!-- 0..1 Reference(OperationDefinition) Operation to invoke --></operationDefinition>
</ServiceDefinition>
```

JSON Template

```
{
  "resourceType" : "ServiceDefinition",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this service definition (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the service definition
  "version" : "<string>", // Business version of the service definition
  "name" : "<string>", // Name for this service definition (computer friendly)
  "title" : "<string>", // Name for this service definition (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "description" : "<markdown>", // Natural language description of the service definition
  "purpose" : "<markdown>", // Why this service definition is defined
  "usage" : "<string>", // Describes the clinical usage of the module
  "approvalDate" : "<date>", // When the service definition was approved by publisher
  "lastReviewDate" : "<date>", // When the service definition was last reviewed
  "effectivePeriod" : { Period }, // When the service definition is expected to be used
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for service definition (if applicable)
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc
  "contributor" : [{ Contributor }], // A content contributor
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc
  "trigger" : [{ TriggerDefinition }], // "when" the module should be invoked
  "dataRequirement" : [{ DataRequirement }], // What data is used by the module
  "operationDefinition" : { Reference(OperationDefinition) } // Operation to invoke
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[ a fhir:ServiceDefinition;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root
```

```
  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:ServiceDefinition.url [ uri ]; # 0..1 Logical URI to reference this service definition (globally unique)
  fhir:ServiceDefinition.identifier [ Identifier ], ... ; # 0..* Additional identifier for the service definition
  fhir:ServiceDefinition.version [ string ]; # 0..1 Business version of the service definition
  fhir:ServiceDefinition.name [ string ]; # 0..1 Name for this service definition (computer friendly)
  fhir:ServiceDefinition.title [ string ]; # 0..1 Name for this service definition (human friendly)
  fhir:ServiceDefinition.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:ServiceDefinition.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:ServiceDefinition.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:ServiceDefinition.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:ServiceDefinition.description [ markdown ]; # 0..1 Natural language description of the service definition
  fhir:ServiceDefinition.purpose [ markdown ]; # 0..1 Why this service definition is defined
  fhir:ServiceDefinition.usage [ string ]; # 0..1 Describes the clinical usage of the module
  fhir:ServiceDefinition.approvalDate [ date ]; # 0..1 When the service definition was approved by publisher
  fhir:ServiceDefinition.lastReviewDate [ date ]; # 0..1 When the service definition was last reviewed
  fhir:ServiceDefinition.effectivePeriod [ Period ]; # 0..1 When the service definition is expected to be used
  fhir:ServiceDefinition.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:ServiceDefinition.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for service definition (if
     applicable)
  fhir:ServiceDefinition.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc
  fhir:ServiceDefinition.contributor [ Contributor ], ... ; # 0..* A content contributor
  fhir:ServiceDefinition.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:ServiceDefinition.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:ServiceDefinition.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc
  fhir:ServiceDefinition.trigger [ TriggerDefinition ], ... ; # 0..* "when" the module should be invoked
  fhir:ServiceDefinition.dataRequirement [ DataRequirement ], ... ; # 0..* What data is used by the module
  fhir:ServiceDefinition.operationDefinition [ Reference(OperationDefinition) ]; # 0..1 Operation to invoke
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

14.2.3.1 Terminology Bindings

Path	Definition	Type	Reference
ServiceDefinition.status	The lifecycle status of a Value Set or Concept Map.	Required	PublicationStatus
ServiceDefinition.jurisdiction	Countries and regions within which this artifact is targeted for use	Extensible	Jurisdiction ValueSet
ServiceDefinition.topic	High-level categorization of the definition, used for searching, sorting, and filtering	Example	DefinitionTopic

14.2.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name	Type	Description	Expression	In Common
composed-of	reference	What resource is being referenced	ServiceDefinition.relatedArtifact.where(type='composed-of').resource (Any)	
date	date	The service definition publication date	ServiceDefinition.date	
depends-on	reference	What resource is being referenced	ServiceDefinition.relatedArtifact.where(type='depends-on').resource (Any)	
derived-from	reference	What resource is being referenced	ServiceDefinition.relatedArtifact.where(type='derived-from').resource (Any)	
description	string	The description of the service definition	ServiceDefinition.description	
effective	date	The time during which the service definition is intended to be in use	ServiceDefinition.effectivePeriod	
identifier	token	External identifier for the service definition	ServiceDefinition.identifier	
jurisdiction	token	Intended jurisdiction for the service definition	ServiceDefinition.jurisdiction	
name	string	Computationally friendly name of the service definition	ServiceDefinition.name	
predecessor	reference	What resource is being referenced	ServiceDefinition.relatedArtifact.where(type='predecessor').resource (Any)	
publisher	string	Name of the publisher of the service definition	ServiceDefinition.publisher	
status	token	The current status of the service definition	ServiceDefinition.status	
successor	reference	What resource is being referenced	ServiceDefinition.relatedArtifact.where(type='successor').resource (Any)	
title	string	The human-friendly name of the service definition	ServiceDefinition.title	
topic	token	Topics associated with the module	ServiceDefinition.topic	
url	uri	The uri that identifies the service definition	ServiceDefinition.url	
version	token	The business version of the service definition	ServiceDefinition.version	

14.2.5 Resource ServiceDefinition - Examples

URL: https://hl7.org/fhir/STU3/servicedefinition-examples.html

This example illustrates the use of the DecisionSupportServiceModule resource to describe a Guideline Appropriate Ordering assessment module. This module is also the focus of the $evaluate operation example.

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.2.5 Resource ServiceDefinition - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

Assessment Example | example | XML | JSON | Turtle
InfoButton Module Example | infobutton | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.2.6 Resource ServiceDefinition - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/servicedefinition-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.2.6 Resource ServiceDefinition - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 0 | Draft | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the ServiceDefinition resource.

ServiceDefinition
Definition | The ServiceDefinition describes a unit of decision support functionality that is made available as a service, such as immunization modules or drug-drug interaction checking.
Control | 1..1
ServiceDefinition.url
Definition | An absolute URI that is used to identify this service definition when it is referenced in a specification, model, design or an instance. This SHALL be a URL, SHOULD be globally unique, and SHOULD be an address at which this service definition is (or will be) published. The URL SHOULD include the major version of the service definition. For more information see Technical and Business Versions.
Control | 0..1
Type | uri
Requirements | Allows the service definition to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid:, but real http: addresses are preferred. Multiple instances may share the same url if they have a distinct version.
ServiceDefinition.identifier
Definition | A formal identifier that is used to identify this service definition when it is represented in other formats, or referenced in a specification, model, design or an instance. This is used for CMS or NQF identifiers for a measure artifact. Note that at least one identifier is required for non-experimental active artifacts.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, e.g., to identify this service definition outside of FHIR, where it is not possible to use the logical URI.
ServiceDefinition.version
Definition | The identifier that is used to identify this version of the service definition when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the service definition author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence.
Note | This is a business versionId, not a resource version id (see discussion)
Control | 0..1
Type | string
Summary | true
Comments | There may be different service definition instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a refrence to a particular business version of the service definition with the format [url]|[version].
ServiceDefinition.name
Definition | A natural language name identifying the service definition. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Control | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alpha-numeric type name to ensure that it is computable friendly.
ServiceDefinition.title
Definition | A short, descriptive, user-friendly title for the service definition.
Control | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
ServiceDefinition.status
Definition | The status of this service definition. Enables tracking the life-cycle of the content.
Control | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true
Summary | true

Comments | Allows filtering of service definitions that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use a retired service definition without due consideration.

ServiceDefinition.experimental

Definition | A boolean value to indicate that this service definition is authored for testing purposes (or education/evaluation/marketing), and is not intended to be used for genuine usage.

Control | 0..1

Type | boolean

Is Modifier | true

Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level service definition.

Summary | true

Comments | Allows filtering of service definition that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use an experimental service definition in production.

ServiceDefinition.date

Definition | The date (and optionally time) when the service definition was published. The date must change if and when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the service definition changes.

Control | 0..1

Type | dateTime

Alternate Names | Revision Date

Summary | true

Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the service definition. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.

ServiceDefinition.publisher

Definition | The name of the individual or organization that published the service definition.

Control | 0..1

Type | string

Requirements | Helps establish the "authority/credibility" of the service definition. May also allow for contact.

Summary | true

Comments | Usually an organization, but may be an individual. The publisher (or steward) of the service definition is the organization or individual primarily responsible for the maintenance and upkeep of the service definition. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the service definition. This item SHOULD be populated unless the information is available from context.

ServiceDefinition.description

Definition | A free text natural language description of the service definition from a consumer's perspective.

Control | 0..1

Type | markdown

Comments | This description can be used to capture details such as why the service definition was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the service definition as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the profile is presumed to be the predominant language in the place the profile was created).

ServiceDefinition.purpose

Definition | Explaination of why this service definition is needed and why it has been designed as it has.

Control | 0..1

Type | markdown

Comments | This element does not describe the usage of the service definition Instead it provides traceability of "why" the resource is either needed or "why" it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this service definition.

ServiceDefinition.usage

Definition | A detailed description of how the module is used from a clinical perspective.

Control | 0..1

Type | string

ServiceDefinition.approvalDate

Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.

Control | 0..1

Type | date

Comments | The 'date' element may be more recent than the approval date because of minor changes / editorial corrections.

ServiceDefinition.lastReviewDate

Definition | The date on which the resource content was last reviewed. Review happens periodically after approval, but doesn't change the original approval date.

Control | 0..1

Type | date

Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.

Comments | If specified, this is usually after the approval date.

ServiceDefinition.effectivePeriod

Definition | The period during which the service definition content was or is planned to be in active use.

Control | 0..1

Type | Period

Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of a the service definition are or are expected to be used instead.

Summary | true

Comments | The effective period for a service definition determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.

ServiceDefinition.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These terms may be used to assist with indexing and searching for appropriate service definition instances.
Control | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation whether all or any of the contexts apply.
ServiceDefinition.jurisdiction
Definition | A legal or geographic region in which the service definition is intended to be used.
Control | 0..*
Terminology Binding | Jurisdiction ValueSet (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the service definition to be used in jurisdictions other than those for which it was originally designed or intended.
ServiceDefinition.topic
Definition | Descriptive topics related to the module. Topics provide a high-level categorization of the module that can be useful for filtering and searching.
Control | 0..*
Terminology Binding | DefinitionTopic (Example)
Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the module so that it can be found by topical searches.
ServiceDefinition.contributor
Definition | A contributor to the content of the module, including authors, editors, reviewers, and endorsers.
Control | 0..*
Type | Contributor
Requirements | Consumers of the content must be able to quickly determine who contributed to the content of the knowledge module.
ServiceDefinition.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Control | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
ServiceDefinition.copyright
Definition | A copyright statement relating to the service definition and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the service definition.
Control | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the service definition and/or its content.
Alternate Names | License; Restrictions
ServiceDefinition.relatedArtifact
Definition | Related resources such as additional documentation, justification, or bibliographic references.
Control | 0..*
Type | RelatedArtifact
Requirements | Modules must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Comments | Each related resource is either an attachment, or a reference to another resource, but not both.
ServiceDefinition.trigger
Definition | The trigger element defines when the rule should be invoked. This information is used by consumers of the rule to determine how to integrate the rule into a specific workflow.
Control | 0..*
Type | TriggerDefinition
ServiceDefinition.dataRequirement
Definition | Data requirements are a machine processable description of the data required by the module in order to perform a successful evaluation.
Control | 0..*
Type | DataRequirement
ServiceDefinition.operationDefinition
Definition | A reference to the operation that is used to invoke this service.
Control | 0..1
Type | Reference(OperationDefinition)

14.2.7 Resource ServiceDefinition - Mappings

URL: https://hl7.org/fhir/STU3/servicedefinition-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.2.7 Resource ServiceDefinition - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

Workflow Mapping

Mappings for the servicedefinition resource.

14.2.7.1 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

ServiceDefinition | Definition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
name
title | Definition.title
status | Definition.status
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
description | Definition.description
purpose | Definition.purpose
usage
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
topic
contributor
contact | Definition.contact
copyright | Definition.copyright
relatedArtifact
trigger
dataRequirement
operationDefinition

14.2.8 Resource ServiceDefinition - Design Notes

URL: https://hl7.org/fhir/STU3/servicedefinition-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.2.8 Resource ServiceDefinition - Design Notes

to do

14.2.9 Resource ServiceDefinition - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/servicedefinition-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.2.9 Resource ServiceDefinition - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the ServiceDefinition resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

14.2.9.1 Profiles

No Profiles defined for this resource

14.2.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

14.2.9.3 Search Extensions

No Search Extensions defined for this resource

14.2.10 Resource ServiceDefinition - Operations

URL: https://hl7.org/fhir/STU3/servicedefinition-operations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.2.10 Resource ServiceDefinition - Operations

Clinical Decision Support Work Group | Maturity Level: 0 | Draft | Compartments: Not linked to any defined compartments

This resource has 2 operations associated with it:

$evaluate | Evaluate
$data-requirements | Data Requirements

For more information about operations, including how they are invoked, see Operations.

14.2.10.1 Evaluate

The evaluate operation requests clinical decision support guidance based on a specific decision support module

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/ServiceDefinition-evaluate
```

Formal Definition (as a OperationDefinition).

URL: [base]/ServiceDefinition/[id]/$evaluate

This is not an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation
requestId | 0..1 | id | | | An optional client-provided identifier to track the request.
evaluateAtDateTime | 0..1 | dateTime | | | An optional date and time specifying that the evaluation should be performed as though it was the given date and time. The most direct implication of this is that references to "Now" within the evaluation logic of the module should result in this value. In addition, wherever possible, the data accessed by the module should appear as though it was accessed at this time. The evaluateAtDateTime value may be any time in the past or future, enabling both retrospective and prospective scenarios. If no value is provided, the date and time of the request is assumed.
inputParameters | 0..1 | Parameters | | | The input parameters for a request, if any. These parameters are defined by the module that is the target of the evaluation, and typically supply patient-independent information to the module.
inputData | 0..* | Any | | | The input data for the request. These data are defined by the data requirements of the module and typically provide patient-dependent information.
patient | 0..1 | Reference(Patient) | | | The patient in context, if any.
encounter | 0..1 | Reference(Encounter) | | | The encounter in context, if any.
initiatingOrganization | 0..1 | Reference(Organization) | | | The organization initiating the request.
initiatingPerson | 0..1 | Reference(Patient|Practitioner|RelatedPerson) | | | The person initiating the request.
userType | 0..1 | CodeableConcept | | | The type of user initiating the request, e.g. patient, healthcare provider, or specific type of healthcare provider (physician, nurse, etc.).
userLanguage | 0..1 | CodeableConcept | | | Preferred language of the person using the system.
userTaskContext | 0..1 | CodeableConcept | | | The task the system user is performing, e.g. laboratory results review, medication list review, etc. This information can be used to tailor decision support outputs, such as recommended information resources.
receivingOrganization | 0..1 | Reference(Organization) | | | The organization that will receive the response.
receivingPerson | 0..1 | Reference(Patient|Practitioner|RelatedPerson) | | | The person in the receiving organization that will receive the response.
recipientType | 0..1 | CodeableConcept | | | The type of individual that will consume the response content. This may be different from the requesting user type (e.g. if a clinician is getting disease management guidance for provision to a patient). E.g. patient, healthcare provider or specific type of healthcare provider (physician, nurse, etc.).
recipientLanguage | 0..1 | CodeableConcept | | | Preferred language of the person that will consume the content.
setting | 0..1 | CodeableConcept | | | The current setting of the request (inpatient, outpatient, etc).
settingContext | 0..1 | CodeableConcept | | | Additional detail about the setting of the request, if any.

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation
return | 1..1 | GuidanceResponse | | | The result of the request as a GuidanceResponse resource. Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

14.2.10.1.1 Examples

Request guideline appropriate ordering assessment (Request):

```
POST [base]/ServiceDefinition/example/$evaluate

<?xml version="1.0" encoding="UTF-8"?>
<Parameters xmlns="http://hl7.org/fhir">
  <id value="evaluate-operation-request-example"/>
  <!--  This example demonstrates the $evaluate operation against a DecisionSupportServiceModule resource. -->
  <parameter>
    <name value="requestId"/>
    <valueString value="guidanceRequest1"/>
  </parameter>
  <parameter>
    <name value="setting"/>
    <valueCoding>
      <system value="http://hl7.org/fhir/encounter-class"/>
      <code value="inpatient"/>
    </valueCoding>
  </parameter>
  <parameter>
    <name value="patient"/>
    <resource>
      <Patient>
        <id value="example"/>
        <meta>
```

```
      <profile value="http://hl7.org/fhir/StructureDefinition/qicore-patient"/>
      <profile value="http://hl7.org/fhir/StructureDefinition/gao-patient"/>
    </meta>
    <text>
      <status value="generated"/>
      <div xmlns="http://www.w3.org/1999/xhtml">
    </div>
    </text>
    <!--  MRN assigned by ACME healthcare on 6-May 2001  -->
    <identifier>
      <use value="usual"/>
      <type>
        <coding>
          <system value="http://hl7.org/fhir/v2/0203"/>
          <code value="MRN"/>
        </coding>
      </type>
      <system value="urn:oid:1.2.36.146.595.217.0.1"/>
      <value value="12345"/>
      <period>
        <start value="2001-05-06"/>
      </period>
      <assigner>
        <display value="Acme Healthcare"/>
      </assigner>
    </identifier>
    <active value="true"/>
    <!--  Peter James Chalmers, but called "Jim"  -->
    <name>
      <use value="official"/>
      <family value="Chalmers"/>
      <given value="Peter"/>
      <given value="James"/>
    </name>
    <name>
      <use value="usual"/>
      <given value="Jim"/>
    </name>
    <telecom>
      <use value="home"/>
      <!--  home communication details aren't known  -->
    </telecom>
    <telecom>
      <system value="phone"/>
      <value value="(03) 5555 6473"/>
      <use value="work"/>
    </telecom>
    <!--  use FHIR code system for male / female  -->
    <gender value="male"/>
    <birthDate value="1974-12-25">
      <extension url="http://hl7.org/fhir/StructureDefinition/patient-birthTime">
        <valueDateTime value="1974-12-25T14:35:45-05:00"/>
      </extension>
    </birthDate>
    <deceasedBoolean value="false"/>
    <address>
      <use value="home"/>
      <line value="534 Erewhon St"/>
      <city value="PleasantVille"/>
      <state value="Vic"/>
      <postalCode value="3999"/>
    </address>
    <contact>
      <relationship>
        <coding>
          <system value="http://hl7.org/fhir/patient-contact-relationship"/>
          <code value="partner"/>
        </coding>
      </relationship>
      <name>
        <family value="du">
          <!--  the "du" part is a family name prefix (VV in iso 21090)  -->
          <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-EN-qualifier">
            <valueCode value="VV"/>
          </extension>
        </family>
        <family value="Marché"/>
        <given value="Bénédicte"/>
      </name>
      <telecom>
        <system value="phone"/>
        <value value="+33 (237) 998327"/>
      </telecom>
    </contact>
    <managingOrganization>
      <reference value="Organization/1"/>
    </managingOrganization>
  </Patient>
</resource>
```

```
  </parameter>
  <parameter>
    <name value="order"/>
    <resource>
      <Order>
        <id value="order1"/>
        <meta>
          <profile value="http://hl7.org/fhir/StructureDefinition/gao-order"/>
        </meta>
        <text>
          <status value="generated"/>
          <div xmlns="http://www.w3.org/1999/xhtml">

        </div>
        </text>
        <contained>
          <ProcedureRequest>
            <id value="ProcedureRequest1"/>
            <meta>
              <profile value="http://hl7.org/fhir/StructureDefinition/gao-procedurerequest"/>
            </meta>
            <subject>
              <reference value="Patient/example"/>
            </subject>
            <item>
              <code>
                <coding>
                  <!--  TODO: This will need to use the ACR Commons code  -->
                  <system value="http://www.ama-assn.org/go/cpt"/>
                  <code value="70450"/>
                  <display value="CT, head, wo iv contrast"/>
                </coding>
              </code>
            </item>
          </ProcedureRequest>
        </contained>
        <date value="2012-12-28T09:03:04+11:00"/>
        <subject>
          <reference value="Patient/example"/>
        </subject>
        <source>
          <reference value="Practitioner/example"/>
        </source>
        <reasonCodeableConcept>
          <coding>
            <!--  TODO: This will need to use the ACR Commons code  -->
            <system value="http://hl7.org/fhir/sid/icd-9"/>
            <code value="70450"/>
            <display value="Headache, Chronic, no new features"/>
          </coding>
        </reasonCodeableConcept>
        <detail>
          <reference value="#ProcedureRequest1"/>
        </detail>
      </Order>
    </resource>
  </parameter>
</Parameters>
```

Request an InfoButton evaluate (Request):

```
POST [base]/ServiceDefinition/infobutton/$evaluate

<?xml version="1.0" encoding="UTF-8"?>
<Parameters xmlns="http://hl7.org/fhir">
  <id value="infobutton-evaluate-request"/>
  <!--  This example demonstrates the $evaluate operation to represent InfoButton using an appropriately defined
    ServiceDefinition. -->
  <parameter>
    <name value="requestId"/>
    <valueString value="InfoButtonRequest2"/>
  </parameter>
  <parameter>
    <name value="userTaskContext"/>
    <valueCoding>
      <system value="urn:oid:2.16.840.1.113883.5.4"/>
      <code value="PROBLISTREV"/>
      <display value="Problem List Review"/>
    </valueCoding>
  </parameter>
  <parameter>
    <name value="patient"/>
    <resource>
      <Patient xmlns="http://hl7.org/fhir">
        <id value="example"/>
        <text>
          <status value="generated"/>
          <div xmlns="http://www.w3.org/1999/xhtml">
            <table>
```

```
      <tbody>
        <tr>
          <td>Name</td>
          <td>Peter James
                 <b>Chalmers</b>(&amp;quot;Jim&amp;quot;)</td>
        </tr>
        <tr>
          <td>Address</td>
          <td>534 Erewhon, Pleasantville, Vic, 3999</td>
        </tr>
        <tr>
          <td>Contacts</td>
          <td>Home: unknown. Work: (03) 5555 6473</td>
        </tr>
        <tr>
          <td>Id</td>
          <td>MRN: 12345 (Acme Healthcare)</td>
        </tr>
      </tbody>
    </table>
  </div>
</text>
<!--    MRN assigned by ACME healthcare on 6-May 2001    -->
<identifier>
  <use value="usual"/>
  <type>
    <coding>
      <system value="http://hl7.org/fhir/v2/0203"/>
      <code value="MR"/>
    </coding>
  </type>
  <system value="urn:oid:1.2.36.146.595.217.0.1"/>
  <value value="12345"/>
  <period>
    <start value="2001-05-06"/>
  </period>
  <assigner>
    <display value="Acme Healthcare"/>
  </assigner>
</identifier>
<active value="true"/>
<!--    Peter James Chalmers, but called "Jim"    -->
<name>
  <use value="official"/>
  <family value="Chalmers"/>
  <given value="Peter"/>
  <given value="James"/>
</name>
<name>
  <use value="usual"/>
  <given value="Jim"/>
</name>
<telecom>
  <use value="home"/>
  <!--    home communication details aren't known    -->
</telecom>
<telecom>
  <system value="phone"/>
  <value value="(03) 5555 6473"/>
  <use value="work"/>
</telecom>
<!--    use FHIR code system for male / female    -->
<gender value="male"/>
<birthDate value="1974-12-25">
  <extension url="http://hl7.org/fhir/StructureDefinition/patient-birthTime">
    <valueDateTime value="1974-12-25T14:35:45-05:00"/>
  </extension>
</birthDate>
<deceasedBoolean value="false"/>
<address>
  <use value="home"/>
  <type value="both"/>
  <line value="534 Erewhon St"/>
  <city value="PleasantVille"/>
  <district value="Rainbow"/>
  <state value="Vic"/>
  <postalCode value="3999"/>
  <period>
    <start value="1974-12-25"/>
  </period>
</address>
<contact>
  <relationship>
    <coding>
      <system value="http://hl7.org/fhir/patient-contact-relationship"/>
      <code value="partner"/>
    </coding>
  </relationship>
  <name>
```

```
            <family value="du">
              <!--    the "du" part is a family name prefix (VV in iso 21090)    -->
              <extension url="http://hl7.org/fhir/StructureDefinition/iso21090-EN-qualifier">
                <valueCode value="VV"/>
              </extension>
            </family>
            <family value="Marché"/>
            <given value="Bénédicte"/>
          </name>
          <telecom>
            <system value="phone"/>
            <value value="+33 (237) 998327"/>
          </telecom>
          <gender value="female"/>
          <period>
            <!--    The contact relationship started in 2012    -->
            <start value="2012"/>
          </period>
        </contact>
        <managingOrganization>
          <reference value="Organization/1"/>
        </managingOrganization>
      </Patient>
    </resource>
  </parameter>
  <parameter>
    <name value="mainSearchCriteria"/>
    <valueCodeableConcept>
      <coding>
        <system value="http://hl7.org/fhir/sid/icd-9"/>
        <code value="493"/>
        <display value="Asthma"/>
      </coding>
    </valueCodeableConcept>
  </parameter>
</Parameters>
```

Returned on successful assessment (Response):

```
HTTP/1.1 200 OK

<?xml version="1.0" encoding="UTF-8"?>
<GuidanceResponse xmlns="http://hl7.org/fhir">
  <!-- This example demonstrates the response to an $evaluate operation using the GuidanceResponse resource. -->
  <contained>
    <Parameters >
      <id value="outputParameters1"/>
      <parameter>
        <name value="order"/>
        <valueReference>
          <reference value="#order1"/>
        </valueReference>
      </parameter>
      <parameter>
        <name value="result"/>
        <resource>
          <!-- GAO Result, profile of Basic -->
          <Basic>
            <id value="result1"/> <!-- Could also be the Response Identifier (guidanceResponse1 in this example)? -->
            <meta>
              <!-- TODO: Profile url for GAO Result profile -->
              <!--profile value="http://hl7.org/fhir/StructureDefinition/result-gao-gao-result"/-->
            </meta>
            <extension url="http://hl7.org/fhir/StructureDefinition/result-gao-score">
              <valueDecimal value="7"/>
            </extension>
            <extension url="http://hl7.org/fhir/StructureDefinition/result-gao-item">
              <extension url="http://hl7.org/fhir/StructureDefinition/result-gao-item#code">
                <valueCodeableConcept>
                  <coding>
                    <!-- TODO: This will need to use the ACR Commons code -->
                    <system value="http://www.ama-assn.org/go/cpt"/>
                    <code value="70450"/>
                    <display value="CT, head, wo iv contrast"/>
                  </coding>
                </valueCodeableConcept>
              </extension>
            </extension>
            <!-- Appropriate Use Criteria Identifier -->
            <extension url="http://hl7.org/fhir/StructureDefinition/result-gao-guideline">
              <valueUri value="http://someguidelineprovider.org/radiology-appropriateness-guidelines.html"/>
            </extension>
            <!-- Certified HealthCare Information Technology Identifier -->
            <extension url="http://hl7.org/fhir/StructureDefinition/result-gao-device">
              <valueReference>
                <reference value="#device1"/>
              </valueReference>
            </extension>
            <!-- Provider NPI -->
```

```
          <extension url="http://hl7.org/fhir/StructureDefinition/result-gao-practitioner">
            <valueIdentifier>
              <value value="exampleNPI"/>
            </valueIdentifier>
          </extension>
          <identifier>
            <value value="result1"/>
          </identifier>
          <code>
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              <code value="within"/>
              <display value="Order is within identified guidelines"/>
            </coding>
          </code>
          <subject>
            <reference value="#order1"/>
          </subject>
        </Basic>
      </resource>
    </parameter>
  </Parameters>
</contained>
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<module>
  <reference value="ServiceDefinition/example"/>
</module>
<status value="success"/>
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</outputParameters>
</GuidanceResponse>
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Returned on successful InfoButton request (Response):

```
HTTP/1.1 200 OK

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          <code value="special"/>
        </coding>
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      </author>
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    </class>
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      <reference value="Organization/healthwise"/>
    </author>
    <custodian>
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    <class>
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    </author>
    <custodian>
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        <code value="34895-3"/>
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    <class>
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    <author>
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    <custodian>
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        <code value="34895-3"/>
        <display value="Education note"/>
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        <system value="http://healthwise.org/doctype"/>
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    </author>
    <custodian>
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        <display value="final"/>
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    <content>
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    </author>
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      <reference value="Organization/healthwise"/>
    </author>
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  and dry powder inhalers and nebulizers."/>
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  an action plan. Covers what medicines to take in each zone. Reviews what to do if an attack becomes an emergency."/>
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   metered-dose..."/>
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  <description value="Learn how following an asthma action plan can help you take control of your asthma."/>
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  <description value="Covers control of asthma with asthma action plan. Includes working with doctor on a plan and
  medicines for symptoms and attacks. Includes use of controller medicine and peak flow meter. Reviews avoiding
  triggers and keeping asthma diary."/>
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  <description value="Exercise challenge and inhalation challenge tests are sometimes used to diagnose asthma and
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  <description value="An asthma action plan is based on zones defined by your symptoms or your peak flow, or both. It
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  <description value="An asthma trigger is a factor that can lead to sudden difficulty breathing or other symptoms of
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  <description value="An asthma action plan is based on zones defined by your symptoms, your peak flow, or both. It
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  juices into the esophagus, the tube that leads from the throat to the stomach."/>
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  <description value="Occupational asthma is the most common form of work-related lung disease in many countries. When
   a person develops asthma as an adult,&#x9;occupational exposure is a likely cause."/>
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  <title value="Asthma: Symptoms of Difficulty Breathing"/>
  <description value="It is important to know the symptoms of difficulty&#x9;breathing in asthma. If you or your child
   is having trouble&#x9;breathing, follow your asthma action plan."/>
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  <title value="Asthma Treatment Goals"/>
  <description value="The American Academy of Allergy, Asthma, and Immunology (AAAAI) and the American College of
  Allergy, Asthma, and Immunology (ACAAI) have developed guidelines for getting asthma under control. They list the
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  <description value="Asthma is a long-lasting (chronic) disease that may last throughout your life—you must treat it
  long term. Taking medicines and following a management plan can be difficult over a long period of time."/>
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  <title value="Asthma: Controlling Cockroaches"/>
  <description value="Exposure to cockroaches may increase asthma symptoms. Cockroaches leave behind particles from
  their feces, eggs, and shells that can cause an allergic reaction. Cockroaches are a problem in many homes,
  especially in the southern part of the..."/>
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  <description value="Educating yourself and your family about asthma is essential for you and your child to have
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  control of the disease. If you understand asthma, you will have an easier time&#x9;following the different aspects
  of treatment, such as avoiding..."/>
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  <title value="Allergy Shots for Asthma"/>
  <description value="Allergy shots are a type of immunotherapy treatment in which small doses of substances to which
  you are allergic (allergens) are injected under your skin. Over time, your body may become less responsive to the
  allergens, which means you may..."/>
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  warning not to overuse."/>
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  brand names like budesonide and formoterol (Symbicort) and fluticasone and salmeterol (Advair). Covers how well they
   work and side effects."/>
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  <description value="Dust builds up throughout your home. The dust may contain substances that trigger asthma
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  stuffy nose of&#x9;allergic rhinitis. These..."/>
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</action>
<action>
  <title value="Inhaled corticosteroids for asthma"/>
```

```
  <description value="What are the most important things you need to know about your medicines?"/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse27"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Inhaled quick-relief medicines for asthma"/>
  <description value="What are the most important things you need to know about your medicines?"/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse28"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Steroid Medicine for Asthma: Myths and Facts"/>
  <description value="Asthma is more than a day-to-day problem that makes your life difficult. Over time it damages
  your airways and lungs. That damage can lead to serious health problems.  So it's important to keep your asthma
  under control with medicine."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse29"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Nebulizer for Asthma Treatment"/>
  <description value="A nebulizer is a device used to deliver liquid medicine in the form of a fine mist (aerosol). It
   is sometimes used for&#x9;asthma because:"/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse30"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Asthma and Vocal Cord Problems"/>
  <description value="Vocal cord dysfunction is the uncontrolled closing of the vocal cords when you breathe in. The
  symptoms can seem to be the same as those of asthma and may occur alone or along with asthma. If you have asthma and
   vocal cord dysfunction, it may..."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse31"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Asthma Action Plan: Green Zone"/>
  <description value="An asthma action plan is based on zones defined by your symptoms, your peak flow, or both. It
  tells you what to do if you or your child has a sudden increase in asthma symptoms (asthma attack)."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse32"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Asthma: Peak Expiratory Flow and Personal Best"/>
  <description value="Peak expiratory flow (PEF) measures how much air you or your child can breathe out using the
  greatest effort. It is used in the monitoring and treatment of asthma to determine how well your lungs are
  functioning. Your peak flow drops when..."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse33"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Asthma: How to Overcome Treatment Obstacles"/>
  <description value="Asthma is a long-lasting (chronic) disease that may&#xA;&#x9;&#x9;last throughout your life—you
  must treat it long term. But following a&#xA;&#x9;&#x9;management plan can be difficult over a long period of
  time."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse34"/>
    </resource>
  </documentation>
</action>
<action>
```

```
  <title value="Asthma and Wheezing"/>
  <description value="Wheezing is a whistling noise that occurs when the&#xA;&#x9;&#x9;bronchial tubes, which carry
  air to the lungs, narrow&#xA;&#x9;&#x9;because of&#xA;&#x9;&#x9;inflammation or&#xA;&#x9;&#x9;mucus buildup.
  Wheezing is often present in&#xA;&#x9;&#x9;asthma."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse35"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Asthma: Measuring Peak Flow"/>
  <description value="As someone with&#xA;&#x9;&#x9;  asthma, you know how important it is to monitor
  your&#xA;&#x9;&#x9;  condition. Your doctor may want you to know how well your lungs are &quot;working.&quot; Is
  their ability&#xA;&#x9;&#x9;  to move air in and out staying the same, or is it getting better or&#xA;&#x9;&#x9;
  worse?"/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse36"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Asthma: Identifying Your Triggers"/>
  <description value="Asthma is a&#xA;&#x9;&#x9;  long-lasting (chronic) disease of the&#xA;&#x9;&#x9;  respiratory
  system. It causes&#xA;&#x9;&#x9;  inflammation in tubes that carry air to the lungs&#xA;&#x9;&#x9;  (bronchial
  tubes). The inflammation makes your bronchial tubes likely to&#xA;&#x9;&#x9;  overreact to certain..."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse37"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Asthma: Using an Inhaler and Spacer"/>
  <description value="Learn the correct way to use a spacer with your inhaler."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse38"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Stories From People Controlling Their Asthma"/>
  <description value="Hear how others with asthma have learned to control their symptoms."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse39"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Asthma: Myths About Inhaled Steroids"/>
  <description value="Learn common myths about inhaled steroids, which are some of the best medicines for controlling
  asthma."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse40"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Dealing With Asthma Triggers"/>
  <description value="Learn how to avoid common asthma triggers to better control your symptoms."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse41"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Asthma: The Importance of Controller Medicines"/>
  <description value="Learn why your controller medicines are so important."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse42"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Asthma: What Happens in Your Lungs"/>
```

```
  <description value="Learn what's going on in the lungs that can make it hard to breathe when you have asthma."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse43"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Using a Metered-Dose Inhaler Without a Spacer"/>
  <description value="Learn the correct way to use an inhaler that doesn't have a spacer."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse44"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Using a Dry Powder Inhaler"/>
  <description value="Learn the right way to use a dry powder inhaler to help control your asthma."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse45"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Checking Peak Flow"/>
  <description value="Learn how to use a peak flow meter to check how well your lungs are working."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse46"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Asthma (Holistic)"/>
  <description value="Don’t let asthma knock the wind out of you. Asthma is a lung disorder characterized by sudden
  fits of wheezing, coughing, or shortness of breath. According to research or other evidence, the following self-care
   steps may be helpful. Try proven herbal remedies Boswellia extract (900 mg a day), ivy leaf extract (50 drops..."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse47"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Asthma (Homeopathy)"/>
  <description value="Primary RemediesArsenicum album A person needing this remedy can feel exhausted, yet be very
  restless and anxious. Breathing problems tend to be worse while lying down, better when sitting up, and may begin,
  or be the most intense, between midnight and two a.m. Dry wheezing may progress to a cough that brings up
  frothy..."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse48"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Omalizumab for Asthma"/>
  <description value="Omalizumab (Xolair) is a medicine approved by the U.S. Food&#xA;&#x9;&#x9;and Drug
  Administration (FDA) for use in people age 12 and older who have moderate&#xA;&#x9;&#x9;or severe
  persistent&#xA;&#x9;&#x9;asthma. This medicine costs a lot more than any of the
  standard&#xA;&#x9;&#x9;treatments..."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse49"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Asthma: Ways to Take Inhaled Medicines"/>
  <description value="When you use inhaled&#xA;&#x9;&#x9;asthma medicine, you usually use a device
  that&#xA;&#x9;&#x9;delivers the medicine directly to your lungs. Different types of delivery&#xA;&#x9;&#x9;systems
  are available. And one type may be more suitable for certain people,&#xA;&#x9;&#x9;age groups, or..."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse50"/>
    </resource>
  </documentation>
</action>
<action>
```

```
  <title value="Asthma: Overuse of Quick-Relief Medicines"/>
  <description value="Medicines for quick relief of the narrowed&#xA;&#x9;&#x9;bronchial tubes caused
  by&#xA;&#x9;&#x9;asthma include short-acting beta2-agonists. These&#xA;&#x9;&#x9;medicines relieve sudden increases
  of symptoms (asthma attacks)&#xA;&#x9;&#x9;quickly. But overuse may be harmful."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse51"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Asthma Diary"/>
  <description value="An asthma diary helps you keep track of how well you are
  managing&#xA;&#x9;&#x9;your&#xA;&#x9;&#x9;asthma."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse52"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Inhaled Corticosteroids for Long-Term Control of Asthma"/>
  <description value="Covers medicines used in metered-dose or dry powder inhaler to treat asthma. Lists generic and
  brand names such as budesonide (Pulmicort) and fluticasone and salmeterol (Advair). Covers how well they work and
  side effects."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse53"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Asthma Action Plan"/>
  <description value="An&#xA;&#x9;&#x9;  asthma action plan is a written plan that tells you&#xA;&#x9;&#x9;  how to
  treat your asthma on a daily basis. The plan also helps you deal with&#xA;&#x9;&#x9;  sudden increases in your or
  your child&apos;s asthma symptoms (asthma attacks). You need to treat the&#xA;&#x9;&#x9; ..."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse54"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Assessing Your Asthma Knowledge"/>
  <description value="Understanding&#xA;&#x9;&#x9;asthma can help you control your symptoms and
  reduce&#xA;&#x9;&#x9;your risk of&#xA;&#x9;&#x9;asthma attacks."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse55"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Monitoring Asthma Treatment"/>
  <description value="The severity of&#xA;&#x9;&#x9;asthma can vary, and asthma often requires changes
  in&#xA;&#x9;&#x9;your treatment to control it. To ensure that you are getting the proper&#xA;&#x9;&#x9;treatment,
  you have to continuously monitor and evaluate the disease and&#xA;&#x9;&#x9;communicate with your doctor."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse56"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Breathing Problems: Using a Dry Powder Inhaler"/>
  <description value="Covers using an inhaler to get needed medicine into lungs quickly. Describes dry powder
  inhalers, how they work, and why to use them. Includes pictures on how to use a dry powder inhaler."/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse57"/>
    </resource>
  </documentation>
</action>
<action>
  <title value="Complications of Allergic Rhinitis"/>
  <description value="Long-term (chronic)&#xA;&#x9;&#x9;allergic rhinitis may cause a number of
  health&#xA;&#x9;&#x9;problems, including:"/>
  <documentation>
    <type value="documentation"/>
    <resource>
      <reference value="#InfoButtonResponse58"/>
    </resource>
```

```
    </documentation>
  </action>
  </RequestGroup>
  </contained>
  <requestId value="InfoButtonRequest2"/>
  <module>
    <reference value="ServiceDefinition/infobutton"/>
  </module>
  <status value="success"/>
</GuidanceResponse>
```

14.2.10.2 Data Requirements

The data-requirements operation aggregates and returns the parameters and data requirements for the service module and all its dependencies as a single module definition library.

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/ServiceDefinition-data-requirements
```

Formal Definition (as a OperationDefinition).

URL: [base]/ServiceDefinition/[id]/$data-requirements

This is not an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

evaluateAtDateTime | 0..1 | dateTime | | | An optional date and time specifying that the evaluation should be performed as though it was the given date and time. Data requirements returned will use this as the "Now" point for the purposes of determining the data requirements.

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Library | | | The result of the requirements gathering is a module-definition Library that describes the aggregate parameters, data requirements, and dependencies of the service. Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The effect of invoking this operation is to determine the aggregate set of data requirements and dependencies for the service module. The result is a Library resource with a type of module-definition that contains all the parameter definitions and data requirements of the service module and any libraries referenced by it. Implementations SHOULD aggregate data requirements intelligently (i.e. by collapsing overlapping data requirements).

14.3 Resource GuidanceResponse - Content

URL: https://hl7.org/fhir/STU3/guidanceresponse.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

A guidance response is the formal response to a guidance request, including any output parameters returned by the evaluation, as well as the description of any proposed actions to be taken.

14.3.1 Scope and Usage

The GuidanceResponse resource is used to represent the result of invoking a decision support service. It provides a container for the status of the response, any warnings or messages returned by the service, as well as the output data of the module and any suggested actions to be performed.

For a detailed discussion of the evaluation process, refer to the Guidance Request topic in the Clinical Reasoning module.

Note to implementers: The user-facing external clinical decision support use case supported by the ServiceDefinition/$evaluate operation of the Clinical Reasoning module has significant overlap with the functionality provided by the CDS Hooks specification. As part of FHIR Release 4, it is the intention to unify the CDS Hooks specification with the Clinical Reasoning module, ensuring that implementers have a single consistent mechanism to support this use case that meets the requirements of both the Clinical Quality Framework and CDS Hooks communities. Although the functionality in both specifications is conceptually aligned, this unification will likely result in changes to both specifications. The CQF and CDS Hooks project teams are committed to this unification and will work to ensure that the resulting changes have as little impact as possible on current and ongoing implementation efforts, while meeting the needs of both communities. The project team is planning on a workable implementation tested at the connectathon in May of 2017, with a trial ballot following in September 2017, targeting publication as part of the FHIR Release 4 specification.

14.3.2 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
GuidanceResponse | | | DomainResource | The formal response to a guidance request Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
requestId | ? | 0..1 | id | The id of the request associated with this response, if any
identifier | ? | 0..1 | Identifier | Business identifier
module | ? | 1..1 | Reference(ServiceDefinition) | A reference to a knowledge module
status | ?!? | 1..1 | code | success | data-requested | data-required | in-progress | failure | entered-in-error GuidanceResponseStatus (Required)
subject | | 0..1 | Reference(Patient | Group) | Patient the request was performed for
context | | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or Episode during which the response was returned
occurrenceDateTime | | 0..1 | dateTime | When the guidance response was processed
performer | | 0..1 | Reference(Device) | Device returning the guidance
reason[x] | | 0..1 | | Reason for the response
reasonCodeableConcept | | | CodeableConcept
reasonReference | | | Reference(Any)
note | | 0..* | Annotation | Additional notes about the response
evaluationMessage | | 0..* | Reference(OperationOutcome) | Messages resulting from the evaluation of the artifact or artifacts
outputParameters | | 0..1 | Reference(Parameters) | The output parameters of the evaluation, if any
result | | 0..1 | Reference(CarePlan | RequestGroup) | Proposed actions, if any
dataRequirement | | 0..* | DataRequirement | Additional required data
Documentation for this format

UML Diagram (Legend)

XML Template

```
<GuidanceResponse xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <requestId value="[id]"/><!-- 0..1 The id of the request associated with this response, if any -->
 <identifier><!-- 0..1 Identifier Business identifier --></identifier>
 <module><!-- 1..1 Reference(ServiceDefinition) A reference to a knowledge module --></module>
 <status value="[code]"/><!-- 1..1 success | data-requested | data-required | in-progress | failure | entered-in-error
    -->
 <subject><!-- 0..1 Reference(Patient|Group) Patient the request was performed for --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or Episode during which the response was returned
    --></context>
 <occurrenceDateTime value="[dateTime]"/><!-- 0..1 When the guidance response was processed -->
 <performer><!-- 0..1 Reference(Device) Device returning the guidance --></performer>
 <reason[x]><!-- 0..1 CodeableConcept|Reference(Any) Reason for the response --></reason[x]>
 <note><!-- 0..* Annotation Additional notes about the response --></note>
 <evaluationMessage><!-- 0..* Reference(OperationOutcome) Messages resulting from the evaluation of the artifact or
    artifacts --></evaluationMessage>
 <outputParameters><!-- 0..1 Reference(Parameters) The output parameters of the evaluation, if any
    --></outputParameters>
 <result><!-- 0..1 Reference(CarePlan|RequestGroup) Proposed actions, if any --></result>
 <dataRequirement><!-- 0..* DataRequirement Additional required data --></dataRequirement>
</GuidanceResponse>
```

JSON Template

```
{
  "resourceType" : "GuidanceResponse",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "requestId" : "<id>", // The id of the request associated with this response, if any
  "identifier" : { Identifier }, // Business identifier
  "module" : { Reference(ServiceDefinition) }, // R!  A reference to a knowledge module
  "status" : "<code>", // R!  success | data-requested | data-required | in-progress | failure | entered-in-error
  "subject" : { Reference(Patient|Group) }, // Patient the request was performed for
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or Episode during which the response was returned
  "occurrenceDateTime" : "<dateTime>", // When the guidance response was processed
  "performer" : { Reference(Device) }, // Device returning the guidance
  // reason[x]: Reason for the response. One of these 2:
  "reasonCodeableConcept" : { CodeableConcept },
  "reasonReference" : { Reference(Any) },
  "note" : [{ Annotation }], // Additional notes about the response
  "evaluationMessage" : [{ Reference(OperationOutcome) }], // Messages resulting from the evaluation of the artifact or
    artifacts
  "outputParameters" : { Reference(Parameters) }, // The output parameters of the evaluation, if any
  "result" : { Reference(CarePlan|RequestGroup) }, // Proposed actions, if any
  "dataRequirement" : [{ DataRequirement }] // Additional required data
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:GuidanceResponse;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:GuidanceResponse.requestId [ id ]; # 0..1 The id of the request associated with this response, if any
```

```
  fhir:GuidanceResponse.identifier [ Identifier ]; # 0..1 Business identifier
  fhir:GuidanceResponse.module [ Reference(ServiceDefinition) ]; # 1..1 A reference to a knowledge module
  fhir:GuidanceResponse.status [ code ]; # 1..1 success | data-requested | data-required | in-progress | failure |
    entered-in-error
  fhir:GuidanceResponse.subject [ Reference(Patient|Group) ]; # 0..1 Patient the request was performed for
  fhir:GuidanceResponse.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or Episode during which the
    response was returned
  fhir:GuidanceResponse.occurrenceDateTime [ dateTime ]; # 0..1 When the guidance response was processed
  fhir:GuidanceResponse.performer [ Reference(Device) ]; # 0..1 Device returning the guidance
  # GuidanceResponse.reason[x] : 0..1 Reason for the response. One of these 2
    fhir:GuidanceResponse.reasonCodeableConcept [ CodeableConcept ]
    fhir:GuidanceResponse.reasonReference [ Reference(Any) ]
  fhir:GuidanceResponse.note [ Annotation ], ... ; # 0..* Additional notes about the response
  fhir:GuidanceResponse.evaluationMessage [ Reference(OperationOutcome) ], ... ; # 0..* Messages resulting from the
    evaluation of the artifact or artifacts
  fhir:GuidanceResponse.outputParameters [ Reference(Parameters) ]; # 0..1 The output parameters of the evaluation, if
    any
  fhir:GuidanceResponse.result [ Reference(CarePlan|RequestGroup) ]; # 0..1 Proposed actions, if any
  fhir:GuidanceResponse.dataRequirement [ DataRequirement ], ... ; # 0..* Additional required data
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name	Flags	Card.	Type	Description & Constraints
GuidanceResponse			DomainResource	The formal response to a guidance request Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
requestId	?	0..1	id	The id of the request associated with this response, if any
identifier	?	0..1	Identifier	Business identifier
module	?	1..1	Reference(ServiceDefinition)	A reference to a knowledge module
status	?!?	1..1	code	success | data-requested | data-required | in-progress | failure | entered-in-error GuidanceResponseStatus (Required)
subject		0..1	Reference(Patient | Group)	Patient the request was performed for
context		0..1	Reference(Encounter | EpisodeOfCare)	Encounter or Episode during which the response was returned
occurrenceDateTime		0..1	dateTime	When the guidance response was processed
performer		0..1	Reference(Device)	Device returning the guidance
reason[x]		0..1		Reason for the response
reasonCodeableConcept			CodeableConcept	
reasonReference			Reference(Any)	
note		0..*	Annotation	Additional notes about the response
evaluationMessage		0..*	Reference(OperationOutcome)	Messages resulting from the evaluation of the artifact or artifacts
outputParameters		0..1	Reference(Parameters)	The output parameters of the evaluation, if any
result		0..1	Reference(CarePlan | RequestGroup)	Proposed actions, if any
dataRequirement		0..*	DataRequirement	Additional required data

Documentation for this format

UML Diagram (Legend)

XML Template

```
<GuidanceResponse xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <requestId value="[id]"/><!-- 0..1 The id of the request associated with this response, if any -->
 <identifier><!-- 0..1 Identifier Business identifier --></identifier>
 <module><!-- 1..1 Reference(ServiceDefinition) A reference to a knowledge module --></module>
 <status value="[code]"/><!-- 1..1 success | data-requested | data-required | in-progress | failure | entered-in-error
    -->
 <subject><!-- 0..1 Reference(Patient|Group) Patient the request was performed for --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or Episode during which the response was returned
    --></context>
 <occurrenceDateTime value="[dateTime]"/><!-- 0..1 When the guidance response was processed -->
 <performer><!-- 0..1 Reference(Device) Device returning the guidance --></performer>
 <reason[x]><!-- 0..1 CodeableConcept|Reference(Any) Reason for the response --></reason[x]>
 <note><!-- 0..* Annotation Additional notes about the response --></note>
 <evaluationMessage><!-- 0..* Reference(OperationOutcome) Messages resulting from the evaluation of the artifact or
    artifacts --></evaluationMessage>
 <outputParameters><!-- 0..1 Reference(Parameters) The output parameters of the evaluation, if any
    --></outputParameters>
 <result><!-- 0..1 Reference(CarePlan|RequestGroup) Proposed actions, if any --></result>
 <dataRequirement><!-- 0..* DataRequirement Additional required data --></dataRequirement>
</GuidanceResponse>
```

JSON Template

```
{
  "resourceType" : "GuidanceResponse",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "requestId" : "<id>", // The id of the request associated with this response, if any
  "identifier" : { Identifier }, // Business identifier
  "module" : { Reference(ServiceDefinition) }, // R!  A reference to a knowledge module
  "status" : "<code>", // R!  success | data-requested | data-required | in-progress | failure | entered-in-error
  "subject" : { Reference(Patient|Group) }, // Patient the request was performed for
```

```
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or Episode during which the response was returned
  "occurrenceDateTime" : "<dateTime>", // When the guidance response was processed
  "performer" : { Reference(Device) }, // Device returning the guidance
  // reason[x]: Reason for the response. One of these 2:
  "reasonCodeableConcept" : { CodeableConcept },
  "reasonReference" : { Reference(Any) },
  "note" : [{ Annotation }], // Additional notes about the response
  "evaluationMessage" : [{ Reference(OperationOutcome) }], // Messages resulting from the evaluation of the artifact or
    artifacts
  "outputParameters" : { Reference(Parameters) }, // The output parameters of the evaluation, if any
  "result" : { Reference(CarePlan|RequestGroup) }, // Proposed actions, if any
  "dataRequirement" : [{ DataRequirement }] // Additional required data
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:GuidanceResponse;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:GuidanceResponse.requestId [ id ]; # 0..1 The id of the request associated with this response, if any
  fhir:GuidanceResponse.identifier [ Identifier ]; # 0..1 Business identifier
  fhir:GuidanceResponse.module [ Reference(ServiceDefinition) ]; # 1..1 A reference to a knowledge module
  fhir:GuidanceResponse.status [ code ]; # 1..1 success | data-requested | data-required | in-progress | failure |
    entered-in-error
  fhir:GuidanceResponse.subject [ Reference(Patient|Group) ]; # 0..1 Patient the request was performed for
  fhir:GuidanceResponse.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or Episode during which the
    response was returned
  fhir:GuidanceResponse.occurrenceDateTime [ dateTime ]; # 0..1 When the guidance response was processed
  fhir:GuidanceResponse.performer [ Reference(Device) ]; # 0..1 Device returning the guidance
  # GuidanceResponse.reason[x] : 0..1 Reason for the response. One of these 2
    fhir:GuidanceResponse.reasonCodeableConcept [ CodeableConcept ]
    fhir:GuidanceResponse.reasonReference [ Reference(Any) ]
  fhir:GuidanceResponse.note [ Annotation ], ... ; # 0..* Additional notes about the response
  fhir:GuidanceResponse.evaluationMessage [ Reference(OperationOutcome) ], ... ; # 0..* Messages resulting from the
    evaluation of the artifact or artifacts
  fhir:GuidanceResponse.outputParameters [ Reference(Parameters) ]; # 0..1 The output parameters of the evaluation, if
    any
  fhir:GuidanceResponse.result [ Reference(CarePlan|RequestGroup) ]; # 0..1 Proposed actions, if any
  fhir:GuidanceResponse.dataRequirement [ DataRequirement ], ... ; # 0..* Additional required data
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

14.3.2.1 Terminology Bindings

Path	Definition	Type	Reference
GuidanceResponse.status	The status of a guidance response	Required	GuidanceResponseStatus

14.3.3 GuidanceResponse Search

Although the GuidanceResponse resource does define search parameters, it is up to the individual decision support service whether or not historical records of guidance requests are preserved. A service may not support searching on GuidanceResponses at all, or it may support searching only for a pre-defined expiration period. The search parameters are defined to provide consumers with a consistent interface to searching if it is available for a specific service implementation. In any case, services should detail the support they do provide using a CapabilityStatement.

14.3.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name	Type	Description	Expression	In Common
identifier	token	The identifier of the guidance response	GuidanceResponse.identifier	
patient	reference	The identity of a patient to search for guidance response results	GuidanceResponse.subject (Patient)	
request	token	The identifier of the request associated with the response	GuidanceResponse.requestId	
subject	reference	The subject that the guidance response is about	GuidanceResponse.subject (Group, Patient)	

14.3.5 Resource GuidanceResponse - Examples

URL: https://hl7.org/fhir/STU3/guidanceresponse-examples.html

This example illustrates the use of the GuidanceResponse resource to represent the response returned by an $evaluate operation against the Guideline Appropriate Ordering decision support service module.

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.3.5 Resource GuidanceResponse - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

Example of guidanceresponse | example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.3.6 Resource GuidanceResponse - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/guidanceresponse-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.3.6 Resource GuidanceResponse - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the GuidanceResponse resource.

GuidanceResponse
Definition | A guidance response is the formal response to a guidance request, including any output parameters returned by the evaluation, as well as the description of any proposed actions to be taken.
Control | 1..1
GuidanceResponse.requestId
Definition | The id of the request associated with this response. If an id was given as part of the request, it will be reproduced here to enable the requester to more easily identify the response in a multi-request scenario.
Control | 0..1
Type | id
Summary | true
GuidanceResponse.identifier
Definition | Allows a service to provide a unique, business identifier for the response.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Summary | true
GuidanceResponse.module
Definition | A reference to the knowledge module that was invoked.
Control | 1..1
Type | Reference(ServiceDefinition)
Summary | true
GuidanceResponse.status
Definition | The status of the response. If the evaluation is completed successfully, the status will indicate success. However, in order to complete the evaluation, the engine may require more information. In this case, the status will be data-required, and the response will contain a description of the additional required information. If the evaluation completed successfully, but the engine determines that a potentially more accurate response could be provided if more data was available, the status will be data-requested, and the response will contain a description of the additional requested information.
Control | 1..1
Terminology Binding | GuidanceResponseStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
GuidanceResponse.subject
Definition | The patient for which the request was processed.
Control | 0..1
Type | Reference(Patient | Group)
GuidanceResponse.context
Definition | Allows the context of the guidance response to be provided if available. In a service context, this would likely be unavailable.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
GuidanceResponse.occurrenceDateTime
Definition | Indicates when the guidance response was processed.
Control | 0..1
Type | dateTime
GuidanceResponse.performer
Definition | Provides a reference to the device that performed the guidance.
Control | 0..1
Type | Reference(Device)
GuidanceResponse.reason[x]

Definition | Indicates the reason the request was initiated. This is typically provided as a parameter to the evaluation and echoed by the service, although for some use cases, such as subscription- or event-based scenarios, it may provide an indication of the cause for the response.
Control | 0..1
Type | CodeableConcept|Reference(Any)
[x] Note | See Choice of Data Types for further information about how to use [x]
GuidanceResponse.note
Definition | Provides a mechanism to communicate additional information about the response.
Control | 0..*
Type | Annotation
GuidanceResponse.evaluationMessage
Definition | Messages resulting from the evaluation of the artifact or artifacts. As part of evaluating the request, the engine may produce informational or warning messages. These messages will be provided by this element.
Control | 0..*
Type | Reference(OperationOutcome)
GuidanceResponse.outputParameters
Definition | The output parameters of the evaluation, if any. Many modules will result in the return of specific resources such as procedure or communication requests that are returned as part of the operation result. However, modules may define specific outputs that would be returned as the result of the evaluation, and these would be returned in this element.
Control | 0..1
Type | Reference(Parameters)
GuidanceResponse.result
Definition | The actions, if any, produced by the evaluation of the artifact.
Control | 0..1
Type | Reference(CarePlan | RequestGroup)
GuidanceResponse.dataRequirement
Definition | If the evaluation could not be completed due to lack of information, or additional information would potentially result in a more accurate response, this element will a description of the data required in order to proceed with the evaluation. A subsequent request to the service should include this data.
Control | 0..*
Type | DataRequirement

14.3.7 Resource GuidanceResponse - Mappings

URL: https://hl7.org/fhir/STU3/guidanceresponse-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.3.7 Resource GuidanceResponse - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings for the guidanceresponse resource.

14.3.8 Resource GuidanceResponse - Design Notes

URL: https://hl7.org/fhir/STU3/guidanceresponse-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.3.8 Resource GuidanceResponse - Design Notes

to do

14.3.9 Resource GuidanceResponse - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/guidanceresponse-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.3.9 Resource GuidanceResponse - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the GuidanceResponse resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

14.3.9.1 Profiles

No Profiles defined for this resource

14.3.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

14.3.9.3 Search Extensions

No Search Extensions defined for this resource

14.4 Resource RequestGroup - Content

URL: https://hl7.org/fhir/STU3/requestgroup.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Clinical Decision Support Work Group | Maturity Level: 2 | Draft | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

A group of related requests that can be used to capture intended activities that have inter-dependencies such as "give this medication after that one".

14.4.1 Scope and Usage

This resource is a request resource from a FHIR workflow perspective - see Workflow, specifically Request.

The RequestGroup resource is used to represent a group of optional activities that may be performed for a specific patient or context. This resource is often, but not always, the result of applying a specific PlanDefinition to a particular patient. Other than differences that tie the RequestGroup to a particular subject and setting, the actionDefinition element of PlanDefinition is identical to the action element of the RequestGroup, allowing the same features and functionality to be used in both places to describe optionality of and relationships between activities in a workflow.

RequestGroups can contain hierarchical groups of actions, where each specific action references the action to be performed (in terms of a Request resource), and each group describes additional behavior, relationships, and applicable conditions between the actions in the overall group.

This resource is referenced by CarePlan and GuidanceResponse

14.4.2 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
RequestGroup | | | DomainResource | A group of related requests Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..* | Identifier | Business identifier
definition | | 0..* | Reference(Any) | Instantiates protocol or definition
basedOn | | 0..* | Reference(Any) | Fulfills plan, proposal, or order
replaces | | 0..* | Reference(Any) | Request(s) replaced by this request
groupIdentifier | ? | 0..1 | Identifier | Composite request this is part of
status | ?!? | 1..1 | code | draft | active | suspended | cancelled | completed | entered-in-error | unknown RequestStatus (Required)
intent | ?!? | 1..1 | code | proposal | plan | order RequestIntent (Required)
priority | ? | 0..1 | code | routine | urgent | asap | stat RequestPriority (Required)
subject | | 0..1 | Reference(Patient | Group) | Who the request group is about
context | | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or Episode for the request group
authoredOn | | 0..1 | dateTime | When the request group was authored
author | | 0..1 | Reference(Device | Practitioner) | Device or practitioner that authored the request group
reason[x] | | 0..1 | | Reason for the request group
reasonCodeableConcept | | | CodeableConcept
reasonReference | | | Reference(Any)
note | | 0..* | Annotation | Additional notes about the response
action | I | 0..* | BackboneElement | Proposed actions, if any + Must have resource or action but not both
label | | 0..1 | string | User-visible label for the action (e.g. 1. or A.)
title | | 0..1 | string | User-visible title
description | ? | 0..1 | string | Short description of the action
textEquivalent | ? | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system
code | | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions
documentation | | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action
condition | | 0..* | BackboneElement | Whether or not the action is applicable
kind | | 1..1 | code | applicability | start | stop ActionConditionKind (Required)
description | | 0..1 | string | Natural language description of the condition
language | | 0..1 | string | Language of the expression

expression | | 0..1 | string | Boolean-valued expression
relatedAction | | 0..* | BackboneElement | Relationship to another action
actionId | | 1..1 | id | What action this is related to
relationship | | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end ActionRelationshipType (Required)
offset[x] | | 0..1 | | Time offset for the relationship
offsetDuration | | | Duration
offsetRange | | | Range
timing[x] | | 0..1 | | When the action should take place
timingDateTime | | | dateTime
timingPeriod | | | Period
timingDuration | | | Duration
timingRange | | | Range
timingTiming | | | Timing
participant | | 0..* | Reference(Patient | Person | Practitioner | RelatedPerson) | Who should perform the action
type | | 0..1 | Coding | create | update | remove | fire-event ActionType (Extensible)
groupingBehavior | | 0..1 | code | visual-group | logical-group | sentence-group ActionGroupingBehavior (Required)
selectionBehavior | | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more ActionSelectionBehavior (Required)
requiredBehavior | | 0..1 | code | must | could | must-unless-documented ActionRequiredBehavior (Required)
precheckBehavior | | 0..1 | code | yes | no ActionPrecheckBehavior (Required)
cardinalityBehavior | | 0..1 | code | single | multiple ActionCardinalityBehavior (Required)
resource | I | 0..1 | Reference(Any) | The target of the action
action | I | 0..* | see action | Sub action
Documentation for this format

UML Diagram (Legend)

XML Template

```
<RequestGroup xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <definition><!-- 0..* Reference(Any) Instantiates protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(Any) Fulfills plan, proposal, or order --></basedOn>
 <replaces><!-- 0..* Reference(Any) Request(s) replaced by this request --></replaces>
 <groupIdentifier><!-- 0..1 Identifier Composite request this is part of --></groupIdentifier>
 <status value="[code]"/><!-- 1..1 draft | active | suspended | cancelled | completed | entered-in-error | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | order -->
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <subject><!-- 0..1 Reference(Patient|Group) Who the request group is about --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or Episode for the request group --></context>
 <authoredOn value="[dateTime]"/><!-- 0..1 When the request group was authored -->
 <author><!-- 0..1 Reference(Device|Practitioner) Device or practitioner that authored the request group --></author>
 <reason[x]><!-- 0..1 CodeableConcept|Reference(Any) Reason for the request group --></reason[x]>
 <note><!-- 0..* Annotation Additional notes about the response --></note>
 <action>  <!-- 0..* Proposed actions, if any -->
  <label value="[string]"/><!-- 0..1 User-visible label for the action (e.g. 1. or A.) -->
  <title value="[string]"/><!-- 0..1 User-visible title -->
  <description value="[string]"/><!-- 0..1 Short description of the action -->
  <textEquivalent value="[string]"/><!-- 0..1 Static text equivalent of the action, used if the dynamic aspects cannot
    be interpreted by the receiving system -->
  <code><!-- 0..* CodeableConcept Code representing the meaning of the action or sub-actions --></code>
  <documentation><!-- 0..* RelatedArtifact Supporting documentation for the intended performer of the action
    --></documentation>
  <condition>  <!-- 0..* Whether or not the action is applicable -->
   <kind value="[code]"/><!-- 1..1 applicability | start | stop -->
   <description value="[string]"/><!-- 0..1 Natural language description of the condition -->
   <language value="[string]"/><!-- 0..1 Language of the expression -->
   <expression value="[string]"/><!-- 0..1 Boolean-valued expression -->
  </condition>
  <relatedAction>  <!-- 0..* Relationship to another action -->
   <actionId value="[id]"/><!-- 1..1 What action this is related to -->
   <relationship value="[code]"/><!-- 1..1 before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end -->
   <offset[x]><!-- 0..1 Duration|Range Time offset for the relationship --></offset[x]>
  </relatedAction>
  <timing[x]><!-- 0..1 dateTime|Period|Duration|Range|Timing When the action should take place --></timing[x]>
  <participant><!-- 0..* Reference(Patient|Person|Practitioner|RelatedPerson) Who should perform the action
    --></participant>
  <type><!-- 0..1 Coding create | update | remove | fire-event --></type>
  <groupingBehavior value="[code]"/><!-- 0..1 visual-group | logical-group | sentence-group -->
  <selectionBehavior value="[code]"/><!-- 0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more -->
  <requiredBehavior value="[code]"/><!-- 0..1 must | could | must-unless-documented -->
  <precheckBehavior value="[code]"/><!-- 0..1 yes | no -->
  <cardinalityBehavior value="[code]"/><!-- 0..1 single | multiple -->
  <resource><!--  0..1 Reference(Any) The target of the action --></resource>
  <action><!--  0..* Content as for RequestGroup.action Sub action --></action>
 </action>
</RequestGroup>
```

JSON Template

```
{
  "resourceType" : "RequestGroup",
  // from Resource: id, meta, implicitRules, and language
```

```
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "definition" : [{ Reference(Any) }], // Instantiates protocol or definition
  "basedOn" : [{ Reference(Any) }], // Fulfills plan, proposal, or order
  "replaces" : [{ Reference(Any) }], // Request(s) replaced by this request
  "groupIdentifier" : { Identifier }, // Composite request this is part of
  "status" : "<code>", // R!  draft | active | suspended | cancelled | completed | entered-in-error | unknown
  "intent" : "<code>", // R!  proposal | plan | order
  "priority" : "<code>", // routine | urgent | asap | stat
  "subject" : { Reference(Patient|Group) }, // Who the request group is about
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or Episode for the request group
  "authoredOn" : "<dateTime>", // When the request group was authored
  "author" : { Reference(Device|Practitioner) }, // Device or practitioner that authored the request group
  // reason[x]: Reason for the request group. One of these 2:
  "reasonCodeableConcept" : { CodeableConcept },
  "reasonReference" : { Reference(Any) },
  "note" : [{ Annotation }], // Additional notes about the response
  "action" : [{ // Proposed actions, if any
    "label" : "<string>", // User-visible label for the action (e.g. 1. or A.)
    "title" : "<string>", // User-visible title
    "description" : "<string>", // Short description of the action
    "textEquivalent" : "<string>", // Static text equivalent of the action, used if the dynamic aspects cannot be
    interpreted by the receiving system
    "code" : [{ CodeableConcept }], // Code representing the meaning of the action or sub-actions
    "documentation" : [{ RelatedArtifact }], // Supporting documentation for the intended performer of the action
    "condition" : [{ // Whether or not the action is applicable
      "kind" : "<code>", // R!  applicability | start | stop
      "description" : "<string>", // Natural language description of the condition
      "language" : "<string>", // Language of the expression
      "expression" : "<string>" // Boolean-valued expression
    }],
    "relatedAction" : [{ // Relationship to another action
      "actionId" : "<id>", // R!  What action this is related to
      "relationship" : "<code>", // R!  before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end
      // offset[x]: Time offset for the relationship. One of these 2:
      "offsetDuration" : { Duration }
      "offsetRange" : { Range }
    }],
    // timing[x]: When the action should take place. One of these 5:
    "timingDateTime" : "<dateTime>",
    "timingPeriod" : { Period },
    "timingDuration" : { Duration },
    "timingRange" : { Range },
    "timingTiming" : { Timing },
    "participant" : [{ Reference(Patient|Person|Practitioner|RelatedPerson) }], // Who should perform the action
    "type" : { Coding }, // create | update | remove | fire-event
    "groupingBehavior" : "<code>", // visual-group | logical-group | sentence-group
    "selectionBehavior" : "<code>", // any | all | all-or-none | exactly-one | at-most-one | one-or-more
    "requiredBehavior" : "<code>", // must | could | must-unless-documented
    "precheckBehavior" : "<code>", // yes | no
    "cardinalityBehavior" : "<code>", // single | multiple
    "resource" : { Reference(Any) }, // C? The target of the action
    "action" : [{ Content as for RequestGroup.action }] // C? Sub action
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:RequestGroup;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:RequestGroup.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:RequestGroup.definition [ Reference(Any) ], ... ; # 0..* Instantiates protocol or definition
  fhir:RequestGroup.basedOn [ Reference(Any) ], ... ; # 0..* Fulfills plan, proposal, or order
  fhir:RequestGroup.replaces [ Reference(Any) ], ... ; # 0..* Request(s) replaced by this request
  fhir:RequestGroup.groupIdentifier [ Identifier ]; # 0..1 Composite request this is part of
  fhir:RequestGroup.status [ code ]; # 1..1 draft | active | suspended | cancelled | completed | entered-in-error |
    unknown
  fhir:RequestGroup.intent [ code ]; # 1..1 proposal | plan | order
  fhir:RequestGroup.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:RequestGroup.subject [ Reference(Patient|Group) ]; # 0..1 Who the request group is about
  fhir:RequestGroup.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or Episode for the request group
  fhir:RequestGroup.authoredOn [ dateTime ]; # 0..1 When the request group was authored
  fhir:RequestGroup.author [ Reference(Device|Practitioner) ]; # 0..1 Device or practitioner that authored the request
    group
  # RequestGroup.reason[x] : 0..1 Reason for the request group. One of these 2
    fhir:RequestGroup.reasonCodeableConcept [ CodeableConcept ]
    fhir:RequestGroup.reasonReference [ Reference(Any) ]
  fhir:RequestGroup.note [ Annotation ], ... ; # 0..* Additional notes about the response
  fhir:RequestGroup.action [ # 0..* Proposed actions, if any
    fhir:RequestGroup.action.label [ string ]; # 0..1 User-visible label for the action (e.g. 1. or A.)
    fhir:RequestGroup.action.title [ string ]; # 0..1 User-visible title
    fhir:RequestGroup.action.description [ string ]; # 0..1 Short description of the action
    fhir:RequestGroup.action.textEquivalent [ string ]; # 0..1 Static text equivalent of the action, used if the dynamic
```

```
     aspects cannot be interpreted by the receiving system
    fhir:RequestGroup.action.code [ CodeableConcept ], ... ; # 0..* Code representing the meaning of the action or sub-
    actions
    fhir:RequestGroup.action.documentation [ RelatedArtifact ], ... ; # 0..* Supporting documentation for the intended
    performer of the action
    fhir:RequestGroup.action.condition [ # 0..* Whether or not the action is applicable
      fhir:RequestGroup.action.condition.kind [ code ]; # 1..1 applicability | start | stop
      fhir:RequestGroup.action.condition.description [ string ]; # 0..1 Natural language description of the condition
      fhir:RequestGroup.action.condition.language [ string ]; # 0..1 Language of the expression
      fhir:RequestGroup.action.condition.expression [ string ]; # 0..1 Boolean-valued expression
    ], ...;
    fhir:RequestGroup.action.relatedAction [ # 0..* Relationship to another action
      fhir:RequestGroup.action.relatedAction.actionId [ id ]; # 1..1 What action this is related to
      fhir:RequestGroup.action.relatedAction.relationship [ code ]; # 1..1 before-start | before | before-end |
    concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end
      # RequestGroup.action.relatedAction.offset[x] : 0..1 Time offset for the relationship. One of these 2
        fhir:RequestGroup.action.relatedAction.offsetDuration [ Duration ]
        fhir:RequestGroup.action.relatedAction.offsetRange [ Range ]
    ], ...;
    # RequestGroup.action.timing[x] : 0..1 When the action should take place. One of these 5
      fhir:RequestGroup.action.timingDateTime [ dateTime ]
      fhir:RequestGroup.action.timingPeriod [ Period ]
      fhir:RequestGroup.action.timingDuration [ Duration ]
      fhir:RequestGroup.action.timingRange [ Range ]
      fhir:RequestGroup.action.timingTiming [ Timing ]
    fhir:RequestGroup.action.participant [ Reference(Patient|Person|Practitioner|RelatedPerson) ], ... ; # 0..* Who
    should perform the action
    fhir:RequestGroup.action.type [ Coding ]; # 0..1 create | update | remove | fire-event
    fhir:RequestGroup.action.groupingBehavior [ code ]; # 0..1 visual-group | logical-group | sentence-group
    fhir:RequestGroup.action.selectionBehavior [ code ]; # 0..1 any | all | all-or-none | exactly-one | at-most-one |
    one-or-more
    fhir:RequestGroup.action.requiredBehavior [ code ]; # 0..1 must | could | must-unless-documented
    fhir:RequestGroup.action.precheckBehavior [ code ]; # 0..1 yes | no
    fhir:RequestGroup.action.cardinalityBehavior [ code ]; # 0..1 single | multiple
    fhir:RequestGroup.action.resource [ Reference(Any) ]; # 0..1 The target of the action
    fhir:RequestGroup.action.action [ See RequestGroup.action ], ... ; # 0..* Sub action
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints

RequestGroup | | | DomainResource | A group of related requests Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension

identifier | ? | 0..* | Identifier | Business identifier

definition | | 0..* | Reference(Any) | Instantiates protocol or definition

basedOn | | 0..* | Reference(Any) | Fulfills plan, proposal, or order

replaces | | 0..* | Reference(Any) | Request(s) replaced by this request

groupIdentifier | ? | 0..1 | Identifier | Composite request this is part of

status | ?!? | 1..1 | code | draft | active | suspended | cancelled | completed | entered-in-error | unknown RequestStatus (Required)

intent | ?!? | 1..1 | code | proposal | plan | order RequestIntent (Required)

priority | ? | 0..1 | code | routine | urgent | asap | stat RequestPriority (Required)

subject | | 0..1 | Reference(Patient | Group) | Who the request group is about

context | | 0..1 | Reference(Encounter | EpisodeOfCare) | Encounter or Episode for the request group

authoredOn | | 0..1 | dateTime | When the request group was authored

author | | 0..1 | Reference(Device | Practitioner) | Device or practitioner that authored the request group

reason[x] | | 0..1 | | Reason for the request group

reasonCodeableConcept | | | CodeableConcept

reasonReference | | | Reference(Any)

note | | 0..* | Annotation | Additional notes about the response

action | I | 0..* | BackboneElement | Proposed actions, if any + Must have resource or action but not both

label | | 0..1 | string | User-visible label for the action (e.g. 1. or A.)

title | | 0..1 | string | User-visible title

description | ? | 0..1 | string | Short description of the action

textEquivalent | ? | 0..1 | string | Static text equivalent of the action, used if the dynamic aspects cannot be interpreted by the receiving system

code | | 0..* | CodeableConcept | Code representing the meaning of the action or sub-actions

documentation | | 0..* | RelatedArtifact | Supporting documentation for the intended performer of the action

condition | | 0..* | BackboneElement | Whether or not the action is applicable

kind | | 1..1 | code | applicability | start | stop ActionConditionKind (Required)

description | | 0..1 | string | Natural language description of the condition

language | | 0..1 | string | Language of the expression

expression | | 0..1 | string | Boolean-valued expression

relatedAction | | 0..* | BackboneElement | Relationship to another action

actionId | | 1..1 | id | What action this is related to

relationship | | 1..1 | code | before-start | before | before-end | concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end ActionRelationshipType (Required)

offset[x] | | 0..1 | | Time offset for the relationship

offsetDuration | | | Duration
offsetRange | | | Range
timing[x] | | 0..1 | | When the action should take place
timingDateTime | | | dateTime
timingPeriod | | | Period
timingDuration | | | Duration
timingRange | | | Range
timingTiming | | | Timing
participant | | 0..* | Reference(Patient | Person | Practitioner | RelatedPerson) | Who should perform the action
type | | 0..1 | Coding | create | update | remove | fire-event ActionType (Extensible)
groupingBehavior | | 0..1 | code | visual-group | logical-group | sentence-group ActionGroupingBehavior (Required)
selectionBehavior | | 0..1 | code | any | all | all-or-none | exactly-one | at-most-one | one-or-more ActionSelectionBehavior (Required)
requiredBehavior | | 0..1 | code | must | could | must-unless-documented ActionRequiredBehavior (Required)
precheckBehavior | | 0..1 | code | yes | no ActionPrecheckBehavior (Required)
cardinalityBehavior | | 0..1 | code | single | multiple ActionCardinalityBehavior (Required)
resource | I | 0..1 | Reference(Any) | The target of the action
action | I | 0..* | see action | Sub action
Documentation for this format

UML Diagram (Legend)

XML Template

```
<RequestGroup xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..* Identifier Business identifier --></identifier>
 <definition><!-- 0..* Reference(Any) Instantiates protocol or definition --></definition>
 <basedOn><!-- 0..* Reference(Any) Fulfills plan, proposal, or order --></basedOn>
 <replaces><!-- 0..* Reference(Any) Request(s) replaced by this request --></replaces>
 <groupIdentifier><!-- 0..1 Identifier Composite request this is part of --></groupIdentifier>
 <status value="[code]"/><!-- 1..1 draft | active | suspended | cancelled | completed | entered-in-error | unknown -->
 <intent value="[code]"/><!-- 1..1 proposal | plan | order -->
 <priority value="[code]"/><!-- 0..1 routine | urgent | asap | stat -->
 <subject><!-- 0..1 Reference(Patient|Group) Who the request group is about --></subject>
 <context><!-- 0..1 Reference(Encounter|EpisodeOfCare) Encounter or Episode for the request group --></context>
 <authoredOn value="[dateTime]"/><!-- 0..1 When the request group was authored -->
 <author><!-- 0..1 Reference(Device|Practitioner) Device or practitioner that authored the request group --></author>
 <reason[x]><!-- 0..1 CodeableConcept|Reference(Any) Reason for the request group --></reason[x]>
 <note><!-- 0..* Annotation Additional notes about the response --></note>
 <action>  <!-- 0..* Proposed actions, if any -->
  <label value="[string]"/><!-- 0..1 User-visible label for the action (e.g. 1. or A.) -->
  <title value="[string]"/><!-- 0..1 User-visible title -->
  <description value="[string]"/><!-- 0..1 Short description of the action -->
  <textEquivalent value="[string]"/><!-- 0..1 Static text equivalent of the action, used if the dynamic aspects cannot
    be interpreted by the receiving system -->
  <code><!-- 0..* CodeableConcept Code representing the meaning of the action or sub-actions --></code>
  <documentation><!-- 0..* RelatedArtifact Supporting documentation for the intended performer of the action
    --></documentation>
  <condition>  <!-- 0..* Whether or not the action is applicable -->
   <kind value="[code]"/><!-- 1..1 applicability | start | stop -->
   <description value="[string]"/><!-- 0..1 Natural language description of the condition -->
   <language value="[string]"/><!-- 0..1 Language of the expression -->
   <expression value="[string]"/><!-- 0..1 Boolean-valued expression -->
  </condition>
  <relatedAction>  <!-- 0..* Relationship to another action -->
   <actionId value="[id]"/><!-- 1..1 What action this is related to -->
   <relationship value="[code]"/><!-- 1..1 before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end -->
   <offset[x]><!-- 0..1 Duration|Range Time offset for the relationship --></offset[x]>
  </relatedAction>
  <timing[x]><!-- 0..1 dateTime|Period|Duration|Range|Timing When the action should take place --></timing[x]>
  <participant><!-- 0..* Reference(Patient|Person|Practitioner|RelatedPerson) Who should perform the action
    --></participant>
  <type><!-- 0..1 Coding create | update | remove | fire-event --></type>
  <groupingBehavior value="[code]"/><!-- 0..1 visual-group | logical-group | sentence-group -->
  <selectionBehavior value="[code]"/><!-- 0..1 any | all | all-or-none | exactly-one | at-most-one | one-or-more -->
  <requiredBehavior value="[code]"/><!-- 0..1 must | could | must-unless-documented -->
  <precheckBehavior value="[code]"/><!-- 0..1 yes | no -->
  <cardinalityBehavior value="[code]"/><!-- 0..1 single | multiple -->
  <resource><!--  0..1 Reference(Any) The target of the action --></resource>
  <action><!--  0..* Content as for RequestGroup.action Sub action --></action>
 </action>
</RequestGroup>
```

JSON Template

```
{
  "resourceType" : "RequestGroup",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : [{ Identifier }], // Business identifier
  "definition" : [{ Reference(Any) }], // Instantiates protocol or definition
  "basedOn" : [{ Reference(Any) }], // Fulfills plan, proposal, or order
  "replaces" : [{ Reference(Any) }], // Request(s) replaced by this request
  "groupIdentifier" : { Identifier }, // Composite request this is part of
  "status" : "<code>", // R!  draft | active | suspended | cancelled | completed | entered-in-error | unknown
```

```
  "intent" : "<code>", // R!  proposal | plan | order
  "priority" : "<code>", // routine | urgent | asap | stat
  "subject" : { Reference(Patient|Group) }, // Who the request group is about
  "context" : { Reference(Encounter|EpisodeOfCare) }, // Encounter or Episode for the request group
  "authoredOn" : "<dateTime>", // When the request group was authored
  "author" : { Reference(Device|Practitioner) }, // Device or practitioner that authored the request group
  // reason[x]: Reason for the request group. One of these 2:
  "reasonCodeableConcept" : { CodeableConcept },
  "reasonReference" : { Reference(Any) },
  "note" : [{ Annotation }], // Additional notes about the response
  "action" : [{ // Proposed actions, if any
    "label" : "<string>", // User-visible label for the action (e.g. 1. or A.)
    "title" : "<string>", // User-visible title
    "description" : "<string>", // Short description of the action
    "textEquivalent" : "<string>", // Static text equivalent of the action, used if the dynamic aspects cannot be
    interpreted by the receiving system
    "code" : [{ CodeableConcept }], // Code representing the meaning of the action or sub-actions
    "documentation" : [{ RelatedArtifact }], // Supporting documentation for the intended performer of the action
    "condition" : [{ // Whether or not the action is applicable
      "kind" : "<code>", // R!  applicability | start | stop
      "description" : "<string>", // Natural language description of the condition
      "language" : "<string>", // Language of the expression
      "expression" : "<string>" // Boolean-valued expression
    }],
    "relatedAction" : [{ // Relationship to another action
      "actionId" : "<id>", // R!  What action this is related to
      "relationship" : "<code>", // R!  before-start | before | before-end | concurrent-with-start | concurrent |
    concurrent-with-end | after-start | after | after-end
      // offset[x]: Time offset for the relationship. One of these 2:
      "offsetDuration" : { Duration }
      "offsetRange" : { Range }
    }],
    // timing[x]: When the action should take place. One of these 5:
    "timingDateTime" : "<dateTime>",
    "timingPeriod" : { Period },
    "timingDuration" : { Duration },
    "timingRange" : { Range },
    "timingTiming" : { Timing },
    "participant" : [{ Reference(Patient|Person|Practitioner|RelatedPerson) }], // Who should perform the action
    "type" : { Coding }, // create | update | remove | fire-event
    "groupingBehavior" : "<code>", // visual-group | logical-group | sentence-group
    "selectionBehavior" : "<code>", // any | all | all-or-none | exactly-one | at-most-one | one-or-more
    "requiredBehavior" : "<code>", // must | could | must-unless-documented
    "precheckBehavior" : "<code>", // yes | no
    "cardinalityBehavior" : "<code>", // single | multiple
    "resource" : { Reference(Any) }, // C? The target of the action
    "action" : [{ Content as for RequestGroup.action }] // C? Sub action
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:RequestGroup;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:RequestGroup.identifier [ Identifier ], ... ; # 0..* Business identifier
  fhir:RequestGroup.definition [ Reference(Any) ], ... ; # 0..* Instantiates protocol or definition
  fhir:RequestGroup.basedOn [ Reference(Any) ], ... ; # 0..* Fulfills plan, proposal, or order
  fhir:RequestGroup.replaces [ Reference(Any) ], ... ; # 0..* Request(s) replaced by this request
  fhir:RequestGroup.groupIdentifier [ Identifier ]; # 0..1 Composite request this is part of
  fhir:RequestGroup.status [ code ]; # 1..1 draft | active | suspended | cancelled | completed | entered-in-error |
    unknown
  fhir:RequestGroup.intent [ code ]; # 1..1 proposal | plan | order
  fhir:RequestGroup.priority [ code ]; # 0..1 routine | urgent | asap | stat
  fhir:RequestGroup.subject [ Reference(Patient|Group) ]; # 0..1 Who the request group is about
  fhir:RequestGroup.context [ Reference(Encounter|EpisodeOfCare) ]; # 0..1 Encounter or Episode for the request group
  fhir:RequestGroup.authoredOn [ dateTime ]; # 0..1 When the request group was authored
  fhir:RequestGroup.author [ Reference(Device|Practitioner) ]; # 0..1 Device or practitioner that authored the request
    group
  # RequestGroup.reason[x] : 0..1 Reason for the request group. One of these 2
    fhir:RequestGroup.reasonCodeableConcept [ CodeableConcept ]
    fhir:RequestGroup.reasonReference [ Reference(Any) ]
  fhir:RequestGroup.note [ Annotation ], ... ; # 0..* Additional notes about the response
  fhir:RequestGroup.action [ # 0..* Proposed actions, if any
    fhir:RequestGroup.action.label [ string ]; # 0..1 User-visible label for the action (e.g. 1. or A.)
    fhir:RequestGroup.action.title [ string ]; # 0..1 User-visible title
    fhir:RequestGroup.action.description [ string ]; # 0..1 Short description of the action
    fhir:RequestGroup.action.textEquivalent [ string ]; # 0..1 Static text equivalent of the action, used if the dynamic
     aspects cannot be interpreted by the receiving system
    fhir:RequestGroup.action.code [ CodeableConcept ], ... ; # 0..* Code representing the meaning of the action or sub-
    actions
    fhir:RequestGroup.action.documentation [ RelatedArtifact ], ... ; # 0..* Supporting documentation for the intended
    performer of the action
    fhir:RequestGroup.action.condition [ # 0..* Whether or not the action is applicable
      fhir:RequestGroup.action.condition.kind [ code ]; # 1..1 applicability | start | stop
```

```
      fhir:RequestGroup.action.condition.description [ string ]; # 0..1 Natural language description of the condition
      fhir:RequestGroup.action.condition.language [ string ]; # 0..1 Language of the expression
      fhir:RequestGroup.action.condition.expression [ string ]; # 0..1 Boolean-valued expression
    ], ...;
    fhir:RequestGroup.action.relatedAction [ # 0..* Relationship to another action
      fhir:RequestGroup.action.relatedAction.actionId [ id ]; # 1..1 What action this is related to
      fhir:RequestGroup.action.relatedAction.relationship [ code ]; # 1..1 before-start | before | before-end |
    concurrent-with-start | concurrent | concurrent-with-end | after-start | after | after-end
      # RequestGroup.action.relatedAction.offset[x] : 0..1 Time offset for the relationship. One of these 2
        fhir:RequestGroup.action.relatedAction.offsetDuration [ Duration ]
        fhir:RequestGroup.action.relatedAction.offsetRange [ Range ]
    ], ...;
    # RequestGroup.action.timing[x] : 0..1 When the action should take place. One of these 5
      fhir:RequestGroup.action.timingDateTime [ dateTime ]
      fhir:RequestGroup.action.timingPeriod [ Period ]
      fhir:RequestGroup.action.timingDuration [ Duration ]
      fhir:RequestGroup.action.timingRange [ Range ]
      fhir:RequestGroup.action.timingTiming [ Timing ]
    fhir:RequestGroup.action.participant [ Reference(Patient|Person|Practitioner|RelatedPerson) ], ... ; # 0..* Who
    should perform the action
    fhir:RequestGroup.action.type [ Coding ]; # 0..1 create | update | remove | fire-event
    fhir:RequestGroup.action.groupingBehavior [ code ]; # 0..1 visual-group | logical-group | sentence-group
    fhir:RequestGroup.action.selectionBehavior [ code ]; # 0..1 any | all | all-or-none | exactly-one | at-most-one |
    one-or-more
    fhir:RequestGroup.action.requiredBehavior [ code ]; # 0..1 must | could | must-unless-documented
    fhir:RequestGroup.action.precheckBehavior [ code ]; # 0..1 yes | no
    fhir:RequestGroup.action.cardinalityBehavior [ code ]; # 0..1 single | multiple
    fhir:RequestGroup.action.resource [ Reference(Any) ]; # 0..1 The target of the action
    fhir:RequestGroup.action.action [ See RequestGroup.action ], ... ; # 0..* Sub action
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

14.4.2.1 Terminology Bindings

Path	Definition	Type	Reference
RequestGroup.status	Codes identifying the stage lifecycle stage of a request	Required	RequestStatus
RequestGroup.intent	Codes indicating the degree of authority/intentionality associated with a request	Required	RequestIntent
RequestGroup.priority	Identifies the level of importance to be assigned to actioning the request	Required	RequestPriority
RequestGroup.action.condition.kind	The kind of condition for the action	Required	ActionConditionKind
RequestGroup.action.relatedAction.relationship	Defines the types of relationships between actions	Required	ActionRelationshipType
RequestGroup.action.type	The type of action to be performed	Extensible	ActionType
RequestGroup.action.groupingBehavior	Defines organization behavior of a group	Required	ActionGroupingBehavior
RequestGroup.action.selectionBehavior	Defines selection behavior of a group	Required	ActionSelectionBehavior
RequestGroup.action.requiredBehavior	Defines requiredness behavior for selecting an action or an action group	Required	ActionRequiredBehavior
RequestGroup.action.precheckBehavior	Defines selection frequency behavior for an action or group	Required	ActionPrecheckBehavior
RequestGroup.action.cardinalityBehavior	Defines behavior for an action or a group for how many times that item may be repeated	Required	ActionCardinalityBehavior

14.4.2.2 Constraints

* rqg-1: On RequestGroup.action: Must have resource or action but not both (expression on RequestGroup.action: resource.exists() != action.exists())

14.4.3 Usage

The RequestGroup resource is used when there are temporal, co-occurrence or other dependencies between one or more steps of an overall workflow. For example, "do procedure A or procedure B, but not both" or "do procedure A after procedure B" or "Act on this ProcedureRequest, then use the value of that observation in the calculation of the dose of this subsequent MedicationRequest". RequestGroups that define actions (i.e. that are more than just narrative representations) will always reference other Request resources with an intent of "option".

Each "option" request can only be interpreted in the context of a RequestGroup that references it. This is because the RequestGroup defines the context in which the option request may/should/must occur, including any triggers, timing constraints, choices, sequencing requirements, etc. Typically such "option" requests will be contained resources due to this dependency. However, in some cases "option" requests may be stand-alone if they are immutable or tightly tied to a ActivityDefinition such that the option resources can safely be referenced without a risk of their content/intent changing

Elements in the "option" requests may include extensions for timing or other elements that allow calculation based on information found in the RequestGroup or other referenced "option" resources, as well as to expose elements within the "option" resource for referencing in other "option" resources. These extensions are:

* TODO

The RequestGroup and all of its referenced "option" Requests are treated as a single integrated Request whose status is the status of the RequestGroup. If there is a need to manage statuses of the different parts, separately, refer to the guidance

here.

14.4.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common
author | reference | The author of the request group | RequestGroup.author (Practitioner, Device)
authored | date | The date the request group was authored | RequestGroup.authoredOn
context | reference | The context the request group applies to | RequestGroup.context (EpisodeOfCare, Encounter)
definition | reference | The definition from which the request group is realized | RequestGroup.definition (Any)
encounter | reference | The encounter the request group applies to | RequestGroup.context (Encounter)
group-identifier | token | The group identifier for the request group | RequestGroup.groupIdentifier
identifier | token | External identifiers for the request group | RequestGroup.identifier
intent | token | The intent of the request group | RequestGroup.intent
participant | reference | The participant in the requests in the group | RequestGroup.action.participant (Practitioner, Patient, Person, RelatedPerson)
patient | reference | The identity of a patient to search for request groups | RequestGroup.subject (Patient)
priority | token | The priority of the request group | RequestGroup.priority
status | token | The status of the request group | RequestGroup.status
subject | reference | The subject that the request group is about | RequestGroup.subject (Group, Patient)

14.4.5 Resource RequestGroup - Examples

URL: https://hl7.org/fhir/STU3/requestgroup-examples.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.4.5 Resource RequestGroup - Examples

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Example List:

Simple example request group illustrating related actions with offsets | example | XML | JSON | Turtle
Example realization of the KDN5 PlanDefinition example | kdn5-example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.4.6 Resource RequestGroup - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/requestgroup-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.4.6 Resource RequestGroup - Detailed Descriptions

Clinical Decision Support Work Group | Maturity Level: 2 | Draft | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Detailed Descriptions for the elements in the RequestGroup resource.

RequestGroup
Definition | A group of related requests that can be used to capture intended activities that have inter-dependencies such as "give this medication after that one".
Control | 1..1
RequestGroup.identifier
Definition | Allows a service to provide a unique, business identifier for the request.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Allows identification of the request as it is known by various participating systems and in a way that remains consistent across servers.
Summary | true
RequestGroup.definition
Definition | A protocol, guideline, orderset or other definition that is adhered to in whole or in part by this request.
Control | 0..*
Type | Reference(Any)
RequestGroup.basedOn
Definition | A plan, proposal or order that is fulfilled in whole or in part by this request.
Control | 0..*
Type | Reference(Any)
Requirements | Allows tracing of authorization for the request and tracking whether proposals/recommendations were acted upon.
Alternate Names | fulfills
RequestGroup.replaces
Definition | Completed or terminated request(s) whose function is taken by this new request.

Control | 0..*
Type | Reference(Any)
Requirements | Allows tracing the continuation of a therapy or administrative process instantiated through multiple requests.
Alternate Names | supersedes; prior; renewed order
Comments | The replacement could be because the initial request was immediately rejected (due to an issue) or because the previous request was completed, but the need for the action described by the request remains ongoing.
RequestGroup.groupIdentifier
Definition | A shared identifier common to all requests that were authorized more or less simultaneously by a single author, representing the identifier of the requisition, prescription or similar form.
Control | 0..1
Type | Identifier
Requirements | Some business processes need to know if multiple items were ordered as part of the same "prescription" or "requisition" for billing or other purposes.
Alternate Names | grouperid; requisition
Summary | true
Comments | Requests are linked either by a "basedOn" relationship (i.e. one request is fulfilling another) or by having a common requisition. Requests that are part of the same requisition are generally treated independently from the perspective of changing their state or maintaining them after initial creation.
RequestGroup.status
Definition | The current state of the request. For request groups, the status reflects the status of all the requests in the group.
Control | 1..1
Terminology Binding | RequestStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
RequestGroup.intent
Definition | Indicates the level of authority/intentionality associated with the request and where the request fits into the workflow chain.
Control | 1..1
Terminology Binding | RequestIntent (Required)
Type | code
Is Modifier | true
Requirements | Proposals/recommendations, plans and orders all use the same structure and can exist in the same fulfillment chain.
Alternate Names | category
Summary | true
Comments | This element is labeled as a modifier because the intent alters when and how the resource is actually applicable.
RequestGroup.priority
Definition | Indicates how quickly the request should be addressed with respect to other requests.
Control | 0..1
Terminology Binding | RequestPriority (Required)
Type | code
Meaning if Missing | If missing, this request should be performed with normal priority
Summary | true
RequestGroup.subject
Definition | The subject for which the request group was created.
Control | 0..1
Type | Reference(Patient | Group)
RequestGroup.context
Definition | Describes the context of the request group, if any.
Control | 0..1
Type | Reference(Encounter | EpisodeOfCare)
RequestGroup.authoredOn
Definition | Indicates when the request group was created.
Control | 0..1
Type | dateTime
RequestGroup.author
Definition | Provides a reference to the author of the request group.
Control | 0..1
Type | Reference(Device | Practitioner)
RequestGroup.reason[x]
Definition | Indicates the reason the request group was created. This is typically provided as a parameter to the evaluation and echoed by the service, although for some use cases, such as subscription- or event-based scenarios, it may provide an indication of the cause for the response.
Control | 0..1
Type | CodeableConcept|Reference(Any)
[x] Note | See Choice of Data Types for further information about how to use [x]
RequestGroup.note
Definition | Provides a mechanism to communicate additional information about the response.
Control | 0..*
Type | Annotation
RequestGroup.action
Definition | The actions, if any, produced by the evaluation of the artifact.
Control | 0..*

Invariants | Defined on this element rqg-1: Must have resource or action but not both (expression : resource.exists() != action.exists(), xpath: exists(f:resource) != exists(f:action))
RequestGroup.action.label
Definition | A user-visible label for the action.
Control | 0..1
Type | string
RequestGroup.action.title
Definition | The title of the action displayed to a user.
Control | 0..1
Type | string
RequestGroup.action.description
Definition | A short description of the action used to provide a summary to display to the user.
Control | 0..1
Type | string
Summary | true
RequestGroup.action.textEquivalent
Definition | A text equivalent of the action to be performed. This provides a human-interpretable description of the action when the definition is consumed by a system that may not be capable of interpreting it dynamically.
Control | 0..1
Type | string
Summary | true
RequestGroup.action.code
Definition | A code that provides meaning for the action or action group. For example, a section may have a LOINC code for a the section of a documentation template.
Control | 0..*
Type | CodeableConcept
RequestGroup.action.documentation
Definition | Didactic or other informational resources associated with the action that can be provided to the CDS recipient. Information resources can include inline text commentary and links to web resources.
Control | 0..*
Type | RelatedArtifact
RequestGroup.action.condition
Definition | An expression that describes applicability criteria, or start/stop conditions for the action.
Control | 0..*
RequestGroup.action.condition.kind
Definition | The kind of condition.
Control | 1..1
Terminology Binding | ActionConditionKind (Required)
Type | code
Comments | Applicability criteria are used to determine immediate applicability when a plan definition is applied to a given context. Start and stop criteria are carried through application and used to describe when enter/exit criteria for an action.
RequestGroup.action.condition.description
Definition | A brief, natural language description of the condition that effectively communicates the intended semantics.
Control | 0..1
Type | string
RequestGroup.action.condition.language
Definition | The media type of the language for the expression.
Control | 0..1
Type | string
RequestGroup.action.condition.expression
Definition | An expression that returns true or false, indicating whether or not the condition is satisfied.
Control | 0..1
Type | string
Comments | The expression may be inlined, or may be a reference to a named expression within a logic library referenced by the library element.
RequestGroup.action.relatedAction
Definition | A relationship to another action such as "before" or "30-60 minutes after start of".
Control | 0..*
RequestGroup.action.relatedAction.actionId
Definition | The element id of the action this is related to.
Control | 1..1
Type | id
RequestGroup.action.relatedAction.relationship
Definition | The relationship of this action to the related action.
Control | 1..1
Terminology Binding | ActionRelationshipType (Required)
Type | code
RequestGroup.action.relatedAction.offset[x]
Definition | A duration or range of durations to apply to the relationship. For example, 30-60 minutes before.
Control | 0..1
Type | Duration|Range
[x] Note | See Choice of Data Types for further information about how to use [x]
RequestGroup.action.timing[x]
Definition | An optional value describing when the action should be performed.

Control | 0..1
Type | dateTime|Period|Duration|Range|Timing
[x] Note | See Choice of Data Types for further information about how to use [x]
RequestGroup.action.participant
Definition | The participant that should perform or be responsible for this action.
Control | 0..*
Type | Reference(Patient | Person | Practitioner | RelatedPerson)
RequestGroup.action.type
Definition | The type of action to perform (create, update, remove).
Control | 0..1
Terminology Binding | ActionType (Extensible)
Type | Coding
RequestGroup.action.groupingBehavior
Definition | Defines the grouping behavior for the action and its children.
Control | 0..1
Terminology Binding | ActionGroupingBehavior (Required)
Type | code
RequestGroup.action.selectionBehavior
Definition | Defines the selection behavior for the action and its children.
Control | 0..1
Terminology Binding | ActionSelectionBehavior (Required)
Type | code
RequestGroup.action.requiredBehavior
Definition | Defines the requiredness behavior for the action.
Control | 0..1
Terminology Binding | ActionRequiredBehavior (Required)
Type | code
RequestGroup.action.precheckBehavior
Definition | Defines whether the action should usually be preselected.
Control | 0..1
Terminology Binding | ActionPrecheckBehavior (Required)
Type | code
RequestGroup.action.cardinalityBehavior
Definition | Defines whether the action can be selected multiple times.
Control | 0..1
Terminology Binding | ActionCardinalityBehavior (Required)
Type | code
RequestGroup.action.resource
Definition | The resource that is the target of the action (e.g. CommunicationRequest).
Control | 0..1
Type | Reference(Any)
Comments | The target resource SHALL be a Request resource with a Request.intent set to "option".
Invariants | Affect this element rqg-1: Must have resource or action but not both (expression : resource.exists() != action.exists(), xpath: exists(f:resource) != exists(f:action))
RequestGroup.action.action
Definition | Sub actions.
Control | 0..*
Type | See RequestGroup.action
Invariants | Affect this element rqg-1: Must have resource or action but not both (expression : resource.exists() != action.exists(), xpath: exists(f:resource) != exists(f:action))

14.4.7 Resource RequestGroup - Mappings

URL: https://hl7.org/fhir/STU3/requestgroup-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.4.7 Resource RequestGroup - Mappings

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Device, Encounter, Patient, Practitioner, RelatedPerson

Mappings:

Workflow Mapping

Mappings for the requestgroup resource.

14.4.7.1 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

RequestGroup | Request
identifier | Request.identifier
definition | Request.identifier
basedOn | Request.basedOn
replaces | Request.replaces

groupIdentifier | Request.groupIdentifier
status | Request.status
intent | Request.intent
priority | Request.priority
subject | Request.subject
context | Request.context
authoredOn | Request.authoredOn
author | Request.author
reason[x] | Request.reason
note | Request.note
action
label
title
description
textEquivalent
code | Request.code
documentation | Request.supportingInfo
condition
kind
description
language
expression
relatedAction
actionId
relationship
offset[x]
timing[x] | Request.occurrence
participant | Request.performer
type
groupingBehavior
selectionBehavior
requiredBehavior
precheckBehavior
cardinalityBehavior
resource
action

14.4.8 Resource RequestGroup - Design Notes

URL: https://hl7.org/fhir/STU3/requestgroup-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.4.8 Resource RequestGroup - Design Notes

to do

14.4.9 Resource RequestGroup - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/requestgroup-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.4.9 Resource RequestGroup - Extensions & Profiles

Clinical Decision Support Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the RequestGroup resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

14.4.9.1 Profiles

No Profiles defined for this resource

14.4.9.2 Extensions

timing-daysOfCycle | daysOfCycle | for Common extensions for Timing data type

Extensions that reference this resource:

No Extensions refer to this resource

14.4.9.3 Search Extensions

No Search Extensions defined for this resource

14.5 Resource Measure - Content

URL: https://hl7.org/fhir/STU3/measure.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

The Measure resource provides the definition of a quality measure.

14.5.1 Scope and Usage

This resource is a definition resource from a FHIR workflow perspective - see Workflow, specifically Definition.

The Measure resource represents a structured computable definition of a clinical quality measure. A quality measure is a quantitative tool to assess the performance of an individual or organization with respect to a specified process or outcome via the measurement of actions, processes, or outcomes of clinical care. Quality measures are often derived from clinical guidelines and are designed to determine whether the appropriate care has been provided given a set of clinical criteria and an evidence base.

Note that the Measure itself does not contain any logic; rather a Library resource is referenced that contains the logic required by the measure, and the various expression elements, such as poulation criteria, reference named expressions within that library (or libraries). In addition, if the Measure references multiple libraries, then any expression references within the resource must be qualified with the name of the library that contains the referenced expression.

14.5.2 Boundaries and Relationships

The Measure resource describes a specific quality measure, providing the structure of the measure in terms of the calculation elements (the populations involved). The Group resource is also capable of describing a population, however, the complexity involved in specifying the criteria in the general case requires the use of a high-level query language such as Clinical Quality Language (CQL). As such, the Measure resource defines only the top-level populations and references CQL expressions for the actual criteria.

A Measure is also similar to an Observation resource, with the exception that the Measure is purely definitional, it contains no actual measurements, only a description of how to calculate a particular measurement.

A Measure is also similar to a clinical document, but as with the relationship to Observation, a Document is specific to a particular subject.

This resource is referenced by measurereport

14.5.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
Measure | | | DomainResource | A quality measure definition Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 0..1 | uri | Logical URI to reference this measure (globally unique)
identifier | ? | 0..* | Identifier | Additional identifier for the measure
version | ? | 0..1 | string | Business version of the measure
name | ? | 0..1 | string | Name for this measure (computer friendly)
title | ? | 0..1 | string | Name for this measure (human friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
description | ? | 0..1 | markdown | Natural language description of the measure
purpose | | 0..1 | markdown | Why this measure is defined
usage | | 0..1 | string | Describes the clinical usage of the measure
approvalDate | | 0..1 | date | When the measure was approved by publisher
lastReviewDate | | 0..1 | date | When the measure was last reviewed
effectivePeriod | ? | 0..1 | Period | When the measure is expected to be used
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for measure (if applicable) Jurisdiction ValueSet (Extensible)
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc DefinitionTopic (Example)
contributor | | 0..* | Contributor | A content contributor

contact | ? | 0..* | ContactDetail | Contact details for the publisher
copyright | | 0..1 | markdown | Use and/or publishing restrictions
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc
library | | 0..* | Reference(Library) | Logic used by the measure
disclaimer | ? | 0..1 | markdown | Disclaimer for use of the measure or its referenced content
scoring | ? | 0..1 | CodeableConcept | proportion | ratio | continuous-variable | cohort MeasureScoring (Extensible)
compositeScoring | ? | 0..1 | CodeableConcept | opportunity | all-or-nothing | linear | weighted CompositeMeasureScoring (Extensible)
type | ? | 0..* | CodeableConcept | process | outcome | structure | patient-reported-outcome | composite MeasureType (Extensible)
riskAdjustment | ? | 0..1 | string | How is risk adjustment applied for this measure
rateAggregation | ? | 0..1 | string | How is rate aggregation performed for this measure
rationale | ? | 0..1 | markdown | Why does this measure exist
clinicalRecommendationStatement | ? | 0..1 | markdown | Summary of clinical guidelines
improvementNotation | ? | 0..1 | string | Improvement notation for the measure, e.g. higher score indicates better quality
definition | ? | 0..* | markdown | Defined terms used in the measure documentation
guidance | ? | 0..1 | markdown | Additional guidance for implementers
set | ? | 0..1 | string | The measure set, e.g. Preventive Care and Screening
group | | 0..* | BackboneElement | Population criteria group
identifier | | 1..1 | Identifier | Unique identifier
name | | 0..1 | string | Short name
description | | 0..1 | string | Summary description
population | | 0..* | BackboneElement | Population criteria
identifier | | 0..1 | Identifier | Unique identifier
code | | 0..1 | CodeableConcept | initial-population | numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation MeasurePopulationType (Extensible)
name | | 0..1 | string | Short name
description | | 0..1 | string | The human readable description of this population criteria
criteria | | 1..1 | string | The name of a valid referenced CQL expression (may be namespaced) that defines this population criteria
stratifier | | 0..* | BackboneElement | Stratifier criteria for the measure
identifier | | 0..1 | Identifier | The identifier for the stratifier used to coordinate the reported data back to this stratifier
criteria | | 0..1 | string | How the measure should be stratified
path | | 0..1 | string | Path to the stratifier
supplementalData | | 0..* | BackboneElement | What other data should be reported with the measure
identifier | | 0..1 | Identifier | Identifier, unique within the measure
usage | | 0..* | CodeableConcept | supplemental-data | risk-adjustment-factor MeasureDataUsage (Extensible)
criteria | | 0..1 | string | Expression describing additional data to be reported
path | | 0..1 | string | Path to the supplemental data element
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Measure xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this measure (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the measure --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the measure -->
 <name value="[string]"/><!-- 0..1 Name for this measure (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this measure (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <description value="[markdown]"/><!-- 0..1 Natural language description of the measure -->
 <purpose value="[markdown]"/><!-- 0..1 Why this measure is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the measure -->
 <approvalDate value="[date]"/><!-- 0..1 When the measure was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the measure was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the measure is expected to be used --></effectivePeriod>
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for measure (if applicable) --></jurisdiction>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc --></topic>
 <contributor><!-- 0..* Contributor A content contributor --></contributor>
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc --></relatedArtifact>
 <library><!-- 0..* Reference(Library) Logic used by the measure --></library>
 <disclaimer value="[markdown]"/><!-- 0..1 Disclaimer for use of the measure or its referenced content -->
 <scoring><!-- 0..1 CodeableConcept proportion | ratio | continuous-variable | cohort --></scoring>
 <compositeScoring><!-- 0..1 CodeableConcept opportunity | all-or-nothing | linear | weighted --></compositeScoring>
 <type><!-- 0..* CodeableConcept process | outcome | structure | patient-reported-outcome | composite --></type>
 <riskAdjustment value="[string]"/><!-- 0..1 How is risk adjustment applied for this measure -->
 <rateAggregation value="[string]"/><!-- 0..1 How is rate aggregation performed for this measure -->
 <rationale value="[markdown]"/><!-- 0..1 Why does this measure exist -->
 <clinicalRecommendationStatement value="[markdown]"/><!-- 0..1 Summary of clinical guidelines -->
 <improvementNotation value="[string]"/><!-- 0..1 Improvement notation for the measure, e.g. higher score indicates
    better quality -->
 <definition value="[markdown]"/><!-- 0..* Defined terms used in the measure documentation -->
 <guidance value="[markdown]"/><!-- 0..1 Additional guidance for implementers -->
 <set value="[string]"/><!-- 0..1 The measure set, e.g. Preventive Care and Screening -->
 <group>  <!-- 0..* Population criteria group -->
```

```
  <identifier><!-- 1..1 Identifier Unique identifier --></identifier>
  <name value="[string]"/><!-- 0..1 Short name -->
  <description value="[string]"/><!-- 0..1 Summary description -->
  <population>  <!-- 0..* Population criteria -->
   <identifier><!-- 0..1 Identifier Unique identifier --></identifier>
   <code><!-- 0..1 CodeableConcept initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
    --></code>
   <name value="[string]"/><!-- 0..1 Short name -->
   <description value="[string]"/><!-- 0..1 The human readable description of this population criteria -->
   <criteria value="[string]"/><!-- 1..1 The name of a valid referenced CQL expression (may be namespaced) that defines
    this population criteria -->
  </population>
  <stratifier>  <!-- 0..* Stratifier criteria for the measure -->
   <identifier><!-- 0..1 Identifier The identifier for the stratifier used to coordinate the reported data back to this
    stratifier --></identifier>
   <criteria value="[string]"/><!-- 0..1 How the measure should be stratified -->
   <path value="[string]"/><!-- 0..1 Path to the stratifier -->
  </stratifier>
 </group>
 <supplementalData>  <!-- 0..* What other data should be reported with the measure -->
  <identifier><!-- 0..1 Identifier Identifier, unique within the measure --></identifier>
  <usage><!-- 0..* CodeableConcept supplemental-data | risk-adjustment-factor --></usage>
  <criteria value="[string]"/><!-- 0..1 Expression describing additional data to be reported -->
  <path value="[string]"/><!-- 0..1 Path to the supplemental data element -->
 </supplementalData>
</Measure>
```

JSON Template

```
{
  "resourceType" : "Measure",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this measure (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the measure
  "version" : "<string>", // Business version of the measure
  "name" : "<string>", // Name for this measure (computer friendly)
  "title" : "<string>", // Name for this measure (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "description" : "<markdown>", // Natural language description of the measure
  "purpose" : "<markdown>", // Why this measure is defined
  "usage" : "<string>", // Describes the clinical usage of the measure
  "approvalDate" : "<date>", // When the measure was approved by publisher
  "lastReviewDate" : "<date>", // When the measure was last reviewed
  "effectivePeriod" : { Period }, // When the measure is expected to be used
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for measure (if applicable)
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc
  "contributor" : [{ Contributor }], // A content contributor
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc
  "library" : [{ Reference(Library) }], // Logic used by the measure
  "disclaimer" : "<markdown>", // Disclaimer for use of the measure or its referenced content
  "scoring" : { CodeableConcept }, // proportion | ratio | continuous-variable | cohort
  "compositeScoring" : { CodeableConcept }, // opportunity | all-or-nothing | linear | weighted
  "type" : [{ CodeableConcept }], // process | outcome | structure | patient-reported-outcome | composite
  "riskAdjustment" : "<string>", // How is risk adjustment applied for this measure
  "rateAggregation" : "<string>", // How is rate aggregation performed for this measure
  "rationale" : "<markdown>", // Why does this measure exist
  "clinicalRecommendationStatement" : "<markdown>", // Summary of clinical guidelines
  "improvementNotation" : "<string>", // Improvement notation for the measure, e.g. higher score indicates better
    quality
  "definition" : ["<markdown>"], // Defined terms used in the measure documentation
  "guidance" : "<markdown>", // Additional guidance for implementers
  "set" : "<string>", // The measure set, e.g. Preventive Care and Screening
  "group" : [{ // Population criteria group
    "identifier" : { Identifier }, // R!  Unique identifier
    "name" : "<string>", // Short name
    "description" : "<string>", // Summary description
    "population" : [{ // Population criteria
      "identifier" : { Identifier }, // Unique identifier
      "code" : { CodeableConcept }, // initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
      "name" : "<string>", // Short name
      "description" : "<string>", // The human readable description of this population criteria
      "criteria" : "<string>" // R!  The name of a valid referenced CQL expression (may be namespaced) that defines this
     population criteria
    }],
    "stratifier" : [{ // Stratifier criteria for the measure
      "identifier" : { Identifier }, // The identifier for the stratifier used to coordinate the reported data back to
    this stratifier
      "criteria" : "<string>", // How the measure should be stratified
      "path" : "<string>" // Path to the stratifier
    }]
  }],
```

```
  "supplementalData" : [{ // What other data should be reported with the measure
    "identifier" : { Identifier }, // Identifier, unique within the measure
    "usage" : [{ CodeableConcept }], // supplemental-data | risk-adjustment-factor
    "criteria" : "<string>", // Expression describing additional data to be reported
    "path" : "<string>" // Path to the supplemental data element
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Measure;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Measure.url [ uri ]; # 0..1 Logical URI to reference this measure (globally unique)
  fhir:Measure.identifier [ Identifier ], ... ; # 0..* Additional identifier for the measure
  fhir:Measure.version [ string ]; # 0..1 Business version of the measure
  fhir:Measure.name [ string ]; # 0..1 Name for this measure (computer friendly)
  fhir:Measure.title [ string ]; # 0..1 Name for this measure (human friendly)
  fhir:Measure.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:Measure.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:Measure.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:Measure.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:Measure.description [ markdown ]; # 0..1 Natural language description of the measure
  fhir:Measure.purpose [ markdown ]; # 0..1 Why this measure is defined
  fhir:Measure.usage [ string ]; # 0..1 Describes the clinical usage of the measure
  fhir:Measure.approvalDate [ date ]; # 0..1 When the measure was approved by publisher
  fhir:Measure.lastReviewDate [ date ]; # 0..1 When the measure was last reviewed
  fhir:Measure.effectivePeriod [ Period ]; # 0..1 When the measure is expected to be used
  fhir:Measure.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:Measure.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for measure (if applicable)
  fhir:Measure.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc
  fhir:Measure.contributor [ Contributor ], ... ; # 0..* A content contributor
  fhir:Measure.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:Measure.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:Measure.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc
  fhir:Measure.library [ Reference(Library) ], ... ; # 0..* Logic used by the measure
  fhir:Measure.disclaimer [ markdown ]; # 0..1 Disclaimer for use of the measure or its referenced content
  fhir:Measure.scoring [ CodeableConcept ]; # 0..1 proportion | ratio | continuous-variable | cohort
  fhir:Measure.compositeScoring [ CodeableConcept ]; # 0..1 opportunity | all-or-nothing | linear | weighted
  fhir:Measure.type [ CodeableConcept ], ... ; # 0..* process | outcome | structure | patient-reported-outcome |
    composite
  fhir:Measure.riskAdjustment [ string ]; # 0..1 How is risk adjustment applied for this measure
  fhir:Measure.rateAggregation [ string ]; # 0..1 How is rate aggregation performed for this measure
  fhir:Measure.rationale [ markdown ]; # 0..1 Why does this measure exist
  fhir:Measure.clinicalRecommendationStatement [ markdown ]; # 0..1 Summary of clinical guidelines
  fhir:Measure.improvementNotation [ string ]; # 0..1 Improvement notation for the measure, e.g. higher score indicates
    better quality
  fhir:Measure.definition [ markdown ], ... ; # 0..* Defined terms used in the measure documentation
  fhir:Measure.guidance [ markdown ]; # 0..1 Additional guidance for implementers
  fhir:Measure.set [ string ]; # 0..1 The measure set, e.g. Preventive Care and Screening
  fhir:Measure.group [ # 0..* Population criteria group
    fhir:Measure.group.identifier [ Identifier ]; # 1..1 Unique identifier
    fhir:Measure.group.name [ string ]; # 0..1 Short name
    fhir:Measure.group.description [ string ]; # 0..1 Summary description
    fhir:Measure.group.population [ # 0..* Population criteria
      fhir:Measure.group.population.identifier [ Identifier ]; # 0..1 Unique identifier
      fhir:Measure.group.population.code [ CodeableConcept ]; # 0..1 initial-population | numerator | numerator-
    exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-
    exclusion | measure-observation
      fhir:Measure.group.population.name [ string ]; # 0..1 Short name
      fhir:Measure.group.population.description [ string ]; # 0..1 The human readable description of this population
    criteria
      fhir:Measure.group.population.criteria [ string ]; # 1..1 The name of a valid referenced CQL expression (may be
    namespaced) that defines this population criteria
    ], ...;
    fhir:Measure.group.stratifier [ # 0..* Stratifier criteria for the measure
      fhir:Measure.group.stratifier.identifier [ Identifier ]; # 0..1 The identifier for the stratifier used to
    coordinate the reported data back to this stratifier
      fhir:Measure.group.stratifier.criteria [ string ]; # 0..1 How the measure should be stratified
      fhir:Measure.group.stratifier.path [ string ]; # 0..1 Path to the stratifier
    ], ...;
  ], ...;
  fhir:Measure.supplementalData [ # 0..* What other data should be reported with the measure
    fhir:Measure.supplementalData.identifier [ Identifier ]; # 0..1 Identifier, unique within the measure
    fhir:Measure.supplementalData.usage [ CodeableConcept ], ... ; # 0..* supplemental-data | risk-adjustment-factor
    fhir:Measure.supplementalData.criteria [ string ]; # 0..1 Expression describing additional data to be reported
    fhir:Measure.supplementalData.path [ string ]; # 0..1 Path to the supplemental data element
  ], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
Measure | | | DomainResource | A quality measure definition Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
url | ? | 0..1 | uri | Logical URI to reference this measure (globally unique)
identifier | ? | 0..* | Identifier | Additional identifier for the measure
version | ? | 0..1 | string | Business version of the measure
name | ? | 0..1 | string | Name for this measure (computer friendly)
title | ? | 0..1 | string | Name for this measure (human friendly)
status | ?!? | 1..1 | code | draft | active | retired | unknown PublicationStatus (Required)
experimental | ?!? | 0..1 | boolean | For testing purposes, not real usage
date | ? | 0..1 | dateTime | Date this was last changed
publisher | ? | 0..1 | string | Name of the publisher (organization or individual)
description | ? | 0..1 | markdown | Natural language description of the measure
purpose | | 0..1 | markdown | Why this measure is defined
usage | | 0..1 | string | Describes the clinical usage of the measure
approvalDate | | 0..1 | date | When the measure was approved by publisher
lastReviewDate | | 0..1 | date | When the measure was last reviewed
effectivePeriod | ? | 0..1 | Period | When the measure is expected to be used
useContext | ? | 0..* | UsageContext | Context the content is intended to support
jurisdiction | ? | 0..* | CodeableConcept | Intended jurisdiction for measure (if applicable) Jurisdiction ValueSet (Extensible)
topic | | 0..* | CodeableConcept | E.g. Education, Treatment, Assessment, etc DefinitionTopic (Example)
contributor | | 0..* | Contributor | A content contributor
contact | ? | 0..* | ContactDetail | Contact details for the publisher
copyright | | 0..1 | markdown | Use and/or publishing restrictions
relatedArtifact | | 0..* | RelatedArtifact | Additional documentation, citations, etc
library | | 0..* | Reference(Library) | Logic used by the measure
disclaimer | ? | 0..1 | markdown | Disclaimer for use of the measure or its referenced content
scoring | ? | 0..1 | CodeableConcept | proportion | ratio | continuous-variable | cohort MeasureScoring (Extensible)
compositeScoring | ? | 0..1 | CodeableConcept | opportunity | all-or-nothing | linear | weighted CompositeMeasureScoring (Extensible)
type | ? | 0..* | CodeableConcept | process | outcome | structure | patient-reported-outcome | composite MeasureType (Extensible)
riskAdjustment | ? | 0..1 | string | How is risk adjustment applied for this measure
rateAggregation | ? | 0..1 | string | How is rate aggregation performed for this measure
rationale | ? | 0..1 | markdown | Why does this measure exist
clinicalRecommendationStatement | ? | 0..1 | markdown | Summary of clinical guidelines
improvementNotation | ? | 0..1 | string | Improvement notation for the measure, e.g. higher score indicates better quality
definition | ? | 0..* | markdown | Defined terms used in the measure documentation
guidance | ? | 0..1 | markdown | Additional guidance for implementers
set | ? | 0..1 | string | The measure set, e.g. Preventive Care and Screening
group | | 0..* | BackboneElement | Population criteria group
identifier | | 1..1 | Identifier | Unique identifier
name | | 0..1 | string | Short name
description | | 0..1 | string | Summary description
population | | 0..* | BackboneElement | Population criteria
identifier | | 0..1 | Identifier | Unique identifier
code | | 0..1 | CodeableConcept | initial-population | numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation MeasurePopulationType (Extensible)
name | | 0..1 | string | Short name
description | | 0..1 | string | The human readable description of this population criteria
criteria | | 1..1 | string | The name of a valid referenced CQL expression (may be namespaced) that defines this population criteria
stratifier | | 0..* | BackboneElement | Stratifier criteria for the measure
identifier | | 0..1 | Identifier | The identifier for the stratifier used to coordinate the reported data back to this stratifier
criteria | | 0..1 | string | How the measure should be stratified
path | | 0..1 | string | Path to the stratifier
supplementalData | | 0..* | BackboneElement | What other data should be reported with the measure
identifier | | 0..1 | Identifier | Identifier, unique within the measure
usage | | 0..* | CodeableConcept | supplemental-data | risk-adjustment-factor MeasureDataUsage (Extensible)
criteria | | 0..1 | string | Expression describing additional data to be reported
path | | 0..1 | string | Path to the supplemental data element
Documentation for this format

UML Diagram (Legend)

XML Template

```
<Measure xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <url value="[uri]"/><!-- 0..1 Logical URI to reference this measure (globally unique) -->
 <identifier><!-- 0..* Identifier Additional identifier for the measure --></identifier>
 <version value="[string]"/><!-- 0..1 Business version of the measure -->
 <name value="[string]"/><!-- 0..1 Name for this measure (computer friendly) -->
 <title value="[string]"/><!-- 0..1 Name for this measure (human friendly) -->
 <status value="[code]"/><!-- 1..1 draft | active | retired | unknown -->
 <experimental value="[boolean]"/><!-- 0..1 For testing purposes, not real usage -->
 <date value="[dateTime]"/><!-- 0..1 Date this was last changed -->
```

```
 <publisher value="[string]"/><!-- 0..1 Name of the publisher (organization or individual) -->
 <description value="[markdown]"/><!-- 0..1 Natural language description of the measure -->
 <purpose value="[markdown]"/><!-- 0..1 Why this measure is defined -->
 <usage value="[string]"/><!-- 0..1 Describes the clinical usage of the measure -->
 <approvalDate value="[date]"/><!-- 0..1 When the measure was approved by publisher -->
 <lastReviewDate value="[date]"/><!-- 0..1 When the measure was last reviewed -->
 <effectivePeriod><!-- 0..1 Period When the measure is expected to be used --></effectivePeriod>
 <useContext><!-- 0..* UsageContext Context the content is intended to support --></useContext>
 <jurisdiction><!-- 0..* CodeableConcept Intended jurisdiction for measure (if applicable) --></jurisdiction>
 <topic><!-- 0..* CodeableConcept E.g. Education, Treatment, Assessment, etc --></topic>
 <contributor><!-- 0..* Contributor A content contributor --></contributor>
 <contact><!-- 0..* ContactDetail Contact details for the publisher --></contact>
 <copyright value="[markdown]"/><!-- 0..1 Use and/or publishing restrictions -->
 <relatedArtifact><!-- 0..* RelatedArtifact Additional documentation, citations, etc --></relatedArtifact>
 <library><!-- 0..* Reference(Library) Logic used by the measure --></library>
 <disclaimer value="[markdown]"/><!-- 0..1 Disclaimer for use of the measure or its referenced content -->
 <scoring><!-- 0..1 CodeableConcept proportion | ratio | continuous-variable | cohort --></scoring>
 <compositeScoring><!-- 0..1 CodeableConcept opportunity | all-or-nothing | linear | weighted --></compositeScoring>
 <type><!-- 0..* CodeableConcept process | outcome | structure | patient-reported-outcome | composite --></type>
 <riskAdjustment value="[string]"/><!-- 0..1 How is risk adjustment applied for this measure -->
 <rateAggregation value="[string]"/><!-- 0..1 How is rate aggregation performed for this measure -->
 <rationale value="[markdown]"/><!-- 0..1 Why does this measure exist -->
 <clinicalRecommendationStatement value="[markdown]"/><!-- 0..1 Summary of clinical guidelines -->
 <improvementNotation value="[string]"/><!-- 0..1 Improvement notation for the measure, e.g. higher score indicates
    better quality -->
 <definition value="[markdown]"/><!-- 0..* Defined terms used in the measure documentation -->
 <guidance value="[markdown]"/><!-- 0..1 Additional guidance for implementers -->
 <set value="[string]"/><!-- 0..1 The measure set, e.g. Preventive Care and Screening -->
 <group>  <!-- 0..* Population criteria group -->
  <identifier><!-- 1..1 Identifier Unique identifier --></identifier>
  <name value="[string]"/><!-- 0..1 Short name -->
  <description value="[string]"/><!-- 0..1 Summary description -->
  <population>  <!-- 0..* Population criteria -->
   <identifier><!-- 0..1 Identifier Unique identifier --></identifier>
   <code><!-- 0..1 CodeableConcept initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
    --></code>
   <name value="[string]"/><!-- 0..1 Short name -->
   <description value="[string]"/><!-- 0..1 The human readable description of this population criteria -->
   <criteria value="[string]"/><!-- 1..1 The name of a valid referenced CQL expression (may be namespaced) that defines
    this population criteria -->
  </population>
  <stratifier>  <!-- 0..* Stratifier criteria for the measure -->
   <identifier><!-- 0..1 Identifier The identifier for the stratifier used to coordinate the reported data back to this
    stratifier --></identifier>
   <criteria value="[string]"/><!-- 0..1 How the measure should be stratified -->
   <path value="[string]"/><!-- 0..1 Path to the stratifier -->
  </stratifier>
 </group>
 <supplementalData>  <!-- 0..* What other data should be reported with the measure -->
  <identifier><!-- 0..1 Identifier Identifier, unique within the measure --></identifier>
  <usage><!-- 0..* CodeableConcept supplemental-data | risk-adjustment-factor --></usage>
  <criteria value="[string]"/><!-- 0..1 Expression describing additional data to be reported -->
  <path value="[string]"/><!-- 0..1 Path to the supplemental data element -->
 </supplementalData>
</Measure>
```

JSON Template

```
{
  "resourceType" : "Measure",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "url" : "<uri>", // Logical URI to reference this measure (globally unique)
  "identifier" : [{ Identifier }], // Additional identifier for the measure
  "version" : "<string>", // Business version of the measure
  "name" : "<string>", // Name for this measure (computer friendly)
  "title" : "<string>", // Name for this measure (human friendly)
  "status" : "<code>", // R!  draft | active | retired | unknown
  "experimental" : <boolean>, // For testing purposes, not real usage
  "date" : "<dateTime>", // Date this was last changed
  "publisher" : "<string>", // Name of the publisher (organization or individual)
  "description" : "<markdown>", // Natural language description of the measure
  "purpose" : "<markdown>", // Why this measure is defined
  "usage" : "<string>", // Describes the clinical usage of the measure
  "approvalDate" : "<date>", // When the measure was approved by publisher
  "lastReviewDate" : "<date>", // When the measure was last reviewed
  "effectivePeriod" : { Period }, // When the measure is expected to be used
  "useContext" : [{ UsageContext }], // Context the content is intended to support
  "jurisdiction" : [{ CodeableConcept }], // Intended jurisdiction for measure (if applicable)
  "topic" : [{ CodeableConcept }], // E.g. Education, Treatment, Assessment, etc
  "contributor" : [{ Contributor }], // A content contributor
  "contact" : [{ ContactDetail }], // Contact details for the publisher
  "copyright" : "<markdown>", // Use and/or publishing restrictions
  "relatedArtifact" : [{ RelatedArtifact }], // Additional documentation, citations, etc
  "library" : [{ Reference(Library) }], // Logic used by the measure
  "disclaimer" : "<markdown>", // Disclaimer for use of the measure or its referenced content
  "scoring" : { CodeableConcept }, // proportion | ratio | continuous-variable | cohort
  "compositeScoring" : { CodeableConcept }, // opportunity | all-or-nothing | linear | weighted
  "type" : [{ CodeableConcept }], // process | outcome | structure | patient-reported-outcome | composite
```

```
  "riskAdjustment" : "<string>", // How is risk adjustment applied for this measure
  "rateAggregation" : "<string>", // How is rate aggregation performed for this measure
  "rationale" : "<markdown>", // Why does this measure exist
  "clinicalRecommendationStatement" : "<markdown>", // Summary of clinical guidelines
  "improvementNotation" : "<string>", // Improvement notation for the measure, e.g. higher score indicates better
    quality
  "definition" : ["<markdown>"], // Defined terms used in the measure documentation
  "guidance" : "<markdown>", // Additional guidance for implementers
  "set" : "<string>", // The measure set, e.g. Preventive Care and Screening
  "group" : [{ // Population criteria group
    "identifier" : { Identifier }, // R!  Unique identifier
    "name" : "<string>", // Short name
    "description" : "<string>", // Summary description
    "population" : [{ // Population criteria
      "identifier" : { Identifier }, // Unique identifier
      "code" : { CodeableConcept }, // initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-observation
      "name" : "<string>", // Short name
      "description" : "<string>", // The human readable description of this population criteria
      "criteria" : "<string>" // R!  The name of a valid referenced CQL expression (may be namespaced) that defines this
     population criteria
    }],
    "stratifier" : [{ // Stratifier criteria for the measure
      "identifier" : { Identifier }, // The identifier for the stratifier used to coordinate the reported data back to
    this stratifier
      "criteria" : "<string>", // How the measure should be stratified
      "path" : "<string>" // Path to the stratifier
    }]
  }],
  "supplementalData" : [{ // What other data should be reported with the measure
    "identifier" : { Identifier }, // Identifier, unique within the measure
    "usage" : [{ CodeableConcept }], // supplemental-data | risk-adjustment-factor
    "criteria" : "<string>", // Expression describing additional data to be reported
    "path" : "<string>" // Path to the supplemental data element
  }]
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .


[ a fhir:Measure;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:Measure.url [ uri ]; # 0..1 Logical URI to reference this measure (globally unique)
  fhir:Measure.identifier [ Identifier ], ... ; # 0..* Additional identifier for the measure
  fhir:Measure.version [ string ]; # 0..1 Business version of the measure
  fhir:Measure.name [ string ]; # 0..1 Name for this measure (computer friendly)
  fhir:Measure.title [ string ]; # 0..1 Name for this measure (human friendly)
  fhir:Measure.status [ code ]; # 1..1 draft | active | retired | unknown
  fhir:Measure.experimental [ boolean ]; # 0..1 For testing purposes, not real usage
  fhir:Measure.date [ dateTime ]; # 0..1 Date this was last changed
  fhir:Measure.publisher [ string ]; # 0..1 Name of the publisher (organization or individual)
  fhir:Measure.description [ markdown ]; # 0..1 Natural language description of the measure
  fhir:Measure.purpose [ markdown ]; # 0..1 Why this measure is defined
  fhir:Measure.usage [ string ]; # 0..1 Describes the clinical usage of the measure
  fhir:Measure.approvalDate [ date ]; # 0..1 When the measure was approved by publisher
  fhir:Measure.lastReviewDate [ date ]; # 0..1 When the measure was last reviewed
  fhir:Measure.effectivePeriod [ Period ]; # 0..1 When the measure is expected to be used
  fhir:Measure.useContext [ UsageContext ], ... ; # 0..* Context the content is intended to support
  fhir:Measure.jurisdiction [ CodeableConcept ], ... ; # 0..* Intended jurisdiction for measure (if applicable)
  fhir:Measure.topic [ CodeableConcept ], ... ; # 0..* E.g. Education, Treatment, Assessment, etc
  fhir:Measure.contributor [ Contributor ], ... ; # 0..* A content contributor
  fhir:Measure.contact [ ContactDetail ], ... ; # 0..* Contact details for the publisher
  fhir:Measure.copyright [ markdown ]; # 0..1 Use and/or publishing restrictions
  fhir:Measure.relatedArtifact [ RelatedArtifact ], ... ; # 0..* Additional documentation, citations, etc
  fhir:Measure.library [ Reference(Library) ], ... ; # 0..* Logic used by the measure
  fhir:Measure.disclaimer [ markdown ]; # 0..1 Disclaimer for use of the measure or its referenced content
  fhir:Measure.scoring [ CodeableConcept ]; # 0..1 proportion | ratio | continuous-variable | cohort
  fhir:Measure.compositeScoring [ CodeableConcept ]; # 0..1 opportunity | all-or-nothing | linear | weighted
  fhir:Measure.type [ CodeableConcept ], ... ; # 0..* process | outcome | structure | patient-reported-outcome |
    composite
  fhir:Measure.riskAdjustment [ string ]; # 0..1 How is risk adjustment applied for this measure
  fhir:Measure.rateAggregation [ string ]; # 0..1 How is rate aggregation performed for this measure
  fhir:Measure.rationale [ markdown ]; # 0..1 Why does this measure exist
  fhir:Measure.clinicalRecommendationStatement [ markdown ]; # 0..1 Summary of clinical guidelines
  fhir:Measure.improvementNotation [ string ]; # 0..1 Improvement notation for the measure, e.g. higher score indicates
    better quality
  fhir:Measure.definition [ markdown ], ... ; # 0..* Defined terms used in the measure documentation
  fhir:Measure.guidance [ markdown ]; # 0..1 Additional guidance for implementers
  fhir:Measure.set [ string ]; # 0..1 The measure set, e.g. Preventive Care and Screening
  fhir:Measure.group [ # 0..* Population criteria group
    fhir:Measure.group.identifier [ Identifier ]; # 1..1 Unique identifier
    fhir:Measure.group.name [ string ]; # 0..1 Short name
    fhir:Measure.group.description [ string ]; # 0..1 Summary description
    fhir:Measure.group.population [ # 0..* Population criteria
      fhir:Measure.group.population.identifier [ Identifier ]; # 0..1 Unique identifier
```

```
    fhir:Measure.group.population.code [ CodeableConcept ]; # 0..1 initial-population | numerator | numerator-
  exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-
  exclusion | measure-observation
    fhir:Measure.group.population.name [ string ]; # 0..1 Short name
    fhir:Measure.group.population.description [ string ]; # 0..1 The human readable description of this population
  criteria
    fhir:Measure.group.population.criteria [ string ]; # 1..1 The name of a valid referenced CQL expression (may be
  namespaced) that defines this population criteria
  ], ...;
  fhir:Measure.group.stratifier [ # 0..* Stratifier criteria for the measure
    fhir:Measure.group.stratifier.identifier [ Identifier ]; # 0..1 The identifier for the stratifier used to
  coordinate the reported data back to this stratifier
    fhir:Measure.group.stratifier.criteria [ string ]; # 0..1 How the measure should be stratified
    fhir:Measure.group.stratifier.path [ string ]; # 0..1 Path to the stratifier
  ], ...;
], ...;
fhir:Measure.supplementalData [ # 0..* What other data should be reported with the measure
  fhir:Measure.supplementalData.identifier [ Identifier ]; # 0..1 Identifier, unique within the measure
  fhir:Measure.supplementalData.usage [ CodeableConcept ], ... ; # 0..* supplemental-data | risk-adjustment-factor
  fhir:Measure.supplementalData.criteria [ string ]; # 0..1 Expression describing additional data to be reported
  fhir:Measure.supplementalData.path [ string ]; # 0..1 Path to the supplemental data element
], ...;
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

14.5.3.1 Terminology Bindings

Path	Definition	Type	Reference
Measure.status	The lifecycle status of a Value Set or Concept Map.	Required	PublicationStatus
Measure.jurisdiction	Countries and regions within which this artifact is targeted for use	Extensible	Jurisdiction ValueSet
Measure.topic	High-level categorization of the definition, used for searching, sorting, and filtering	Example	DefinitionTopic
Measure.scoring	The scoring type of the measure	Extensible	MeasureScoring
Measure.compositeScoring	The composite scoring method of the measure	Extensible	CompositeMeasureScoring
Measure.type	The type of measure (includes codes from 2.16.840.1.113883.1.11.20368)	Extensible	MeasureType
Measure.group.population.code	The type of population	Extensible	MeasurePopulationType
Measure.supplementalData.usage	The intended usage for supplemental data elements in the measure	Extensible	MeasureDataUsage

14.5.4 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name	Type	Description	Expression	In Common
composed-of	reference	What resource is being referenced	Measure.relatedArtifact.where(type='composed-of').resource (Any)	
date	date	The measure publication date	Measure.date	
depends-on	reference	What resource is being referenced	Measure.relatedArtifact.where(type='depends-on').resource	Measure.library (Library, Any)
derived-from	reference	What resource is being referenced	Measure.relatedArtifact.where(type='derived-from').resource (Any)	
description	string	The description of the measure	Measure.description	
effective	date	The time during which the measure is intended to be in use	Measure.effectivePeriod	
identifier	token	External identifier for the measure	Measure.identifier	
jurisdiction	token	Intended jurisdiction for the measure	Measure.jurisdiction	
name	string	Computationally friendly name of the measure	Measure.name	
predecessor	reference	What resource is being referenced	Measure.relatedArtifact.where(type='predecessor').resource (Any)	
publisher	string	Name of the publisher of the measure	Measure.publisher	
status	token	The current status of the measure	Measure.status	
successor	reference	What resource is being referenced	Measure.relatedArtifact.where(type='successor').resource (Any)	
title	string	The human-friendly name of the measure	Measure.title	
topic	token	Topics associated with the module	Measure.topic	
url	uri	The uri that identifies the measure	Measure.url	
version	token	The business version of the measure	Measure.version	

14.5.5 Resource Measure - Examples

URL: https://hl7.org/fhir/STU3/measure-examples.html

This example illustrates the use of the Measure resource to describe the CMS9v4 Exclusive Breastfeeding measure. The logic for the measure is specified in the reference library, Exclusive Breatfeeding Measurement Logic.

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.5.5 Resource Measure - Examples

Clinical Quality Information Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Example List:

Exclusive Breastfeeding Measure | measure-exclusive-breastfeeding | XML | JSON | Turtle
Appropriate Testing for Children with Pharyngitis | measure-cms146-example | XML | JSON | Turtle
Screening for Alcohol Misuse - Component A of a Composite Measure | component-a-example | XML | JSON | Turtle
Screening for Depression - Component B of a Composite Measure Example | component-b-example | XML | JSON | Turtle
Behavioral Health Assessment - Composite Measure Example | composite-example | XML | JSON | Turtle
Predecessor example - Exclusive Breastfeeding | measure-predecessor-example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.5.6 Resource Measure - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/measure-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.5.6 Resource Measure - Detailed Descriptions

Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

Detailed Descriptions for the elements in the Measure resource.

Measure
Definition | The Measure resource provides the definition of a quality measure.
Control | 1..1
Measure.url
Definition | An absolute URI that is used to identify this measure when it is referenced in a specification, model, design or an instance. This SHALL be a URL, SHOULD be globally unique, and SHOULD be an address at which this measure is (or will be) published. The URL SHOULD include the major version of the measure. For more information see Technical and Business Versions.
Control | 0..1
Type | uri
Requirements | Allows the measure to be referenced by a single globally unique identifier.
Summary | true
Comments | Can be a urn:uuid: or a urn:oid:, but real http: addresses are preferred. Multiple instances may share the same url if they have a distinct version.
Measure.identifier
Definition | A formal identifier that is used to identify this measure when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..*
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 V3 II (instance identifier) data type, e.g., to identify this measure outside of FHIR, where it is not possible to use the logical URI.
Measure.version
Definition | The identifier that is used to identify this version of the measure when it is referenced in a specification, model, design or instance. This is an arbitrary value managed by the measure author and is not expected to be globally unique. For example, it might be a timestamp (e.g. yyyymmdd) if a managed version is not available. There is also no expectation that versions can be placed in a lexicographical sequence. To provide a version consistent with the Decision Support Service specification, use the format Major.Minor.Revision (e.g. 1.0.0). For more information on versioning knowledge assets, refer to the Decision Support Service specification. Note that a version is required for non-experimental active artifacts.
Note | This is a business versionId, not a resource version id (see discussion)
Control | 0..1
Type | string
Summary | true
Comments | There may be different measure instances that have the same identifier but different versions. The version can be appended to the url in a reference to allow a refrence to a particular business version of the measure with the format [url]|[version].
Measure.name
Definition | A natural language name identifying the measure. This name should be usable as an identifier for the module by machine processing applications such as code generation.
Control | 0..1
Type | string
Requirements | Support human navigation and code generation.
Summary | true
Comments | The name is not expected to be globally unique. The name should be a simple alpha-numeric type name to ensure that it is computable friendly.
Measure.title
Definition | A short, descriptive, user-friendly title for the measure.
Control | 0..1
Type | string
Summary | true
Comments | This name does not need to be machine-processing friendly and may contain punctuation, white-space, etc.
Measure.status
Definition | The status of this measure. Enables tracking the life-cycle of the content.

Control | 1..1
Terminology Binding | PublicationStatus (Required)
Type | code
Is Modifier | true
Summary | true
Comments | Allows filtering of measures that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use a retired measure without due consideration.
Measure.experimental
Definition | A boolean value to indicate that this measure is authored for testing purposes (or education/evaluation/marketing), and is not intended to be used for genuine usage.
Control | 0..1
Type | boolean
Is Modifier | true
Requirements | Enables experimental content to be developed following the same lifecycle that would be used for a production-level measure.
Summary | true
Comments | Allows filtering of measure that are appropriate for use vs. not. This is labeled as "Is Modifier" because applications should not use an experimental measure in production.
Measure.date
Definition | The date (and optionally time) when the measure was published. The date must change if and when the business version changes and it must change if the status code changes. In addition, it should change when the substantive content of the measure changes.
Control | 0..1
Type | dateTime
Alternate Names | Revision Date
Summary | true
Comments | Note that this is not the same as the resource last-modified-date, since the resource may be a secondary representation of the measure. Additional specific dates may be added as extensions or be found by consulting Provenances associated with past versions of the resource.
Measure.publisher
Definition | The name of the individual or organization that published the measure.
Control | 0..1
Type | string
Requirements | Helps establish the "authority/credibility" of the measure. May also allow for contact.
Summary | true
Comments | Usually an organization, but may be an individual. The publisher (or steward) of the measure is the organization or individual primarily responsible for the maintenance and upkeep of the measure. This is not necessarily the same individual or organization that developed and initially authored the content. The publisher is the primary point of contact for questions or issues with the measure. This item SHOULD be populated unless the information is available from context.
Measure.description
Definition | A free text natural language description of the measure from a consumer's perspective.
Control | 0..1
Type | markdown
Summary | true
Comments | This description can be used to capture details such as why the measure was built, comments about misuse, instructions for clinical use and interpretation, literature references, examples from the paper world, etc. It is not a rendering of the measure as conveyed in the 'text' field of the resource itself. This item SHOULD be populated unless the information is available from context (e.g. the language of the profile is presumed to be the predominant language in the place the profile was created).
Measure.purpose
Definition | Explaination of why this measure is needed and why it has been designed as it has.
Control | 0..1
Type | markdown
Comments | This element does not describe the usage of the measure Instead it provides traceability of "why" the resource is either needed or "why" it is defined as it is. This may be used to point to source materials or specifications that drove the structure of this measure.
Measure.usage
Definition | A detailed description of how the measure is used from a clinical perspective.
Control | 0..1
Type | string
Measure.approvalDate
Definition | The date on which the resource content was approved by the publisher. Approval happens once when the content is officially approved for usage.
Control | 0..1
Type | date
Comments | The 'date' element may be more recent than the approval date because of minor changes / editorial corrections.
Measure.lastReviewDate
Definition | The date on which the resource content was last reviewed. Review happens periodically after approval, but doesn't change the original approval date.
Control | 0..1
Type | date
Requirements | Gives a sense of how "current" the content is. Resources that have not been reviewed in a long time may have a risk of being less appropriate/relevant.
Comments | If specified, this is usually after the approval date.
Measure.effectivePeriod
Definition | The period during which the measure content was or is planned to be in active use.
Control | 0..1
Type | Period

Requirements | Allows establishing a transition before a resource comes into effect and also allows for a sunsetting process when new versions of a the measure are or are expected to be used instead.
Summary | true
Comments | The effective period for a measure determines when the content is applicable for usage and is independent of publication and review dates. For example, a measure intended to be used for the year 2016 might be published in 2015.
Measure.useContext
Definition | The content was developed with a focus and intent of supporting the contexts that are listed. These terms may be used to assist with indexing and searching for appropriate measure instances.
Control | 0..*
Type | UsageContext
Requirements | Assist in searching for appropriate content.
Summary | true
Comments | When multiple useContexts are specified, there is no expectation whether all or any of the contexts apply.
Measure.jurisdiction
Definition | A legal or geographic region in which the measure is intended to be used.
Control | 0..*
Terminology Binding | Jurisdiction ValueSet (Extensible)
Type | CodeableConcept
Summary | true
Comments | It may be possible for the measure to be used in jurisdictions other than those for which it was originally designed or intended.
Measure.topic
Definition | Descriptive topics related to the content of the measure. Topics provide a high-level categorization of the type of the measure that can be useful for filtering and searching.
Control | 0..*
Terminology Binding | DefinitionTopic (Example)
Type | CodeableConcept
Requirements | Repositories must be able to determine how to categorize the measure so that it can be found by topical searches.
Measure.contributor
Definition | A contributor to the content of the measure, including authors, editors, reviewers, and endorsers.
Control | 0..*
Type | Contributor
Requirements | Consumers of the content must be able to quickly determine who contributed to the content of the measure.
Measure.contact
Definition | Contact details to assist a user in finding and communicating with the publisher.
Control | 0..*
Type | ContactDetail
Summary | true
Comments | May be a web site, an email address, a telephone number, etc.
Measure.copyright
Definition | A copyright statement relating to the measure and/or its contents. Copyright statements are generally legal restrictions on the use and publishing of the measure.
Control | 0..1
Type | markdown
Requirements | Consumers must be able to determine any legal restrictions on the use of the measure and/or its content.
Alternate Names | License; Restrictions
Measure.relatedArtifact
Definition | Related artifacts such as additional documentation, justification, or bibliographic references.
Control | 0..*
Type | RelatedArtifact
Requirements | Measures must be able to provide enough information for consumers of the content (and/or interventions or results produced by the content) to be able to determine and understand the justification for and evidence in support of the content.
Comments | Each related artifact is either an attachment, or a reference to another resource, but not both.
Measure.library
Definition | A reference to a Library resource containing the formal logic used by the measure.
Control | 0..*
Type | Reference(Library)
Measure.disclaimer
Definition | Notices and disclaimers regarding the use of the measure, or related to intellectual property (such as code systems) referenced by the measure.
Control | 0..1
Type | markdown
Requirements | Some organizations require that notices and disclosures be included when publishing measures that reference their intellecutal property.
Summary | true
Measure.scoring
Definition | Indicates how the calculation is performed for the measure, including proportion, ratio, continuous variable, and cohort. The value set is extensible, allowing additional measure scoring types to be represented.
Control | 0..1
Terminology Binding | MeasureScoring (Extensible)
Type | CodeableConcept
Summary | true
Measure.compositeScoring
Definition | If this is a composite measure, the scoring method used to combine the component measures to determine the composite score.

Control | 0..1
Terminology Binding | CompositeMeasureScoring (Extensible)
Type | CodeableConcept
Summary | true
Measure.type
Definition | Indicates whether the measure is used to examine a process, an outcome over time, a patient-reported outcome, or a structure measure such as utilization.
Control | 0..*
Terminology Binding | MeasureType (Extensible)
Type | CodeableConcept
Summary | true
Measure.riskAdjustment
Definition | A description of the risk adjustment factors that may impact the resulting score for the measure and how they may be accounted for when computing and reporting measure results.
Control | 0..1
Type | string
Summary | true
Comments | Describes the method of adjusting for clinical severity and conditions present at the start of care that can influence patient outcomes for making valid comparisons of outcome measures across providers. Indicates whether an eMeasure is subject to the statistical process for reducing, removing, or clarifying the influences of confounding factors to allow more useful comparisons.
Measure.rateAggregation
Definition | Describes how to combine the information calculated, based on logic in each of several populations, into one summarized result.
Control | 0..1
Type | string
Summary | true
Comments | The measure rate for an organization or clinician is based upon the entities' aggregate data and summarizes the performance of the entity over a given time period (e.g., monthly, quarterly, yearly). The aggregated data are derived from the results of a specific measure algorithm and, if appropriate, the application of specific risk adjustment models. Can also be used to describe how to risk adjust the data based on supplemental data elements described in the eMeasure (e.g., pneumonia hospital measures antibiotic selection in the ICU versus non-ICU and then the roll-up of the two). This could be applied to aggregated cohort measure definitions (e.g., CDC's aggregate reporting for TB at the state level).
Measure.rationale
Definition | Provides a succint statement of the need for the measure. Usually includes statements pertaining to importance criterion: impact, gap in care, and evidence.
Control | 0..1
Type | markdown
Summary | true
Measure.clinicalRecommendationStatement
Definition | Provides a summary of relevant clinical guidelines or other clinical recommendations supporting the measure.
Control | 0..1
Type | markdown
Summary | true
Measure.improvementNotation
Definition | Information on whether an increase or decrease in score is the preferred result (e.g., a higher score indicates better quality OR a lower score indicates better quality OR quality is whthin a range).
Control | 0..1
Type | string
Requirements | Measure consumers and implementers must be able to determine how to interpret a measure score.
Summary | true
Measure.definition
Definition | Provides a description of an individual term used within the measure.
Control | 0..*
Type | markdown
Requirements | Measure developers must be able to provide precise definitions of terms used within measure descriptions and guidance to help communicate the intent of the measure.
Summary | true
Measure.guidance
Definition | Additional guidance for the measure including how it can be used in a clinical context, and the intent of the measure.
Control | 0..1
Type | markdown
Requirements | Measure developers must be able to provide additional guidance for implementers to understand the intent with greater specificity than that provided in the logic of the measure.
Summary | true
Measure.set
Definition | The measure set, e.g. Preventive Care and Screening.
Control | 0..1
Type | string
Summary | true
Measure.group
Definition | A group of population criteria for the measure.
Control | 0..*
Measure.group.identifier
Definition | A unique identifier for the group. This identifier will used to report data for the group in the measure report.
Note | This is a business identifer, not a resource identifier (see discussion)

Control | 1..1
Type | Identifier
Measure.group.name
Definition | Optional name or short description of this group.
Control | 0..1
Type | string
Measure.group.description
Definition | The human readable description of this population group.
Control | 0..1
Type | string
Measure.group.population
Definition | A population criteria for the measure.
Control | 0..*
Measure.group.population.identifier
Definition | A unique identifier for the population criteria. This identifier is used to report data against this criteria within the measure report.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Measure.group.population.code
Definition | The type of population criteria.
Control | 0..1
Terminology Binding | MeasurePopulationType (Extensible)
Type | CodeableConcept
Measure.group.population.name
Definition | Optional name or short description of this population.
Control | 0..1
Type | string
Measure.group.population.description
Definition | The human readable description of this population criteria.
Control | 0..1
Type | string
Measure.group.population.criteria
Definition | The name of a valid referenced CQL expression (may be namespaced) that defines this population criteria.
Control | 1..1
Type | string
Comments | In the case of a continuous variable or ratio measure, this may be the name of a function that calculates the value of the individual observation for each patient or event in the population. For these types of measures, individual observations are reported as observation resources included in the evaluatedResources bundle for each patient. See the MeasureReport resource or the Quality Reporting topic for more information.
Measure.group.stratifier
Definition | The stratifier criteria for the measure report, specified as either the name of a valid CQL expression defined within a referenced library, or a valid FHIR Resource Path.
Control | 0..*
Measure.group.stratifier.identifier
Definition | The identifier for the stratifier used to coordinate the reported data back to this stratifier.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Measure.group.stratifier.criteria
Definition | The criteria for the stratifier. This must be the name of an expression defined within a referenced library.
Control | 0..1
Type | string
Measure.group.stratifier.path
Definition | The path to an element that defines the stratifier, specified as a valid FHIR resource path.
Control | 0..1
Type | string
Measure.supplementalData
Definition | The supplemental data criteria for the measure report, specified as either the name of a valid CQL expression within a referenced library, or a valid FHIR Resource Path.
Control | 0..*
Comments | Note that supplemental data are reported as observations for each patient and included in the evaluatedResources bundle. See the MeasureReport resource or the Quality Reporting topic for more information.
Measure.supplementalData.identifier
Definition | An identifier for the supplemental data.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Measure.supplementalData.usage
Definition | An indicator of the intended usage for the supplemental data element. Supplemental data indicates the data is additional information requested to augment the measure information. Risk adjustment factor indicates the data is additional information used to calculate risk adjustment factors when applying a risk model to the measure calculation.
Control | 0..*
Terminology Binding | MeasureDataUsage (Extensible)
Type | CodeableConcept

Measure.supplementalData.criteria
Definition | The criteria for the supplemental data. This must be the name of a valid expression defined within a referenced library, and defines the data to be returned for this element.
Control | 0..1
Type | string
Measure.supplementalData.path
Definition | The supplemental data to be supplied as part of the measure response, specified as a valid FHIR Resource Path.
Control | 0..1
Type | string

14.5.7 Resource Measure - Mappings

URL: https://hl7.org/fhir/STU3/measure-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.5.7 Resource Measure - Mappings

Clinical Quality Information Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Not linked to any defined compartments

Mappings:

Workflow Mapping

Mappings for the measure resource.

14.5.7.1 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

Measure | Definition
url | Definition.url
identifier | Definition.identifier
version | Definition.version
name
title | Definition.title
status | Definition.status
experimental | Definition.experimental
date | Definition.date
publisher | Definition.publisher
description | Definition.description
purpose | Definition.purpose
usage
approvalDate | Definition.approvalDate
lastReviewDate | Definition.lastReviewDate
effectivePeriod | Definition.effectivePeriod
useContext | Definition.useContext
jurisdiction | Definition.jurisdiction
topic
contributor
contact | Definition.contact
copyright | Definition.copyright
relatedArtifact
library
disclaimer
scoring
compositeScoring
type | Definition.subjectCode
riskAdjustment
rateAggregation
rationale
clinicalRecommendationStatement
improvementNotation
definition
guidance
set
group
identifier
name
description
population
identifier
code
name
description
criteria
stratifier

identifier
criteria
path
supplementalData
identifier
usage
criteria
path

14.5.8 Resource Measure - Design Notes

URL: https://hl7.org/fhir/STU3/measure-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.5.8 Resource Measure - Design Notes

to do

14.5.9 Resource Measure - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/measure-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.5.9 Resource Measure - Extensions & Profiles

Clinical Quality Information Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the Measure resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

14.5.9.1 Profiles

No Profiles defined for this resource

14.5.9.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

cqif-measureInfo | measureInfo | for CQIF Extensions

14.5.9.3 Search Extensions

No Search Extensions defined for this resource

14.5.10 Resource Measure - Operations

URL: https://hl7.org/fhir/STU3/measure-operations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.5.10 Resource Measure - Operations

Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Compartments: Not linked to any defined compartments

This resource has 2 operations associated with it:

$evaluate-measure | Evaluate Measure
$data-requirements | Data Requirements

For more information about operations, including how they are invoked, see Operations.

14.5.10.1 Evaluate Measure

The evaluate-measure operation is used to invoke an eMeasure and obtain the results

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Measure-evaluate-measure
```

Formal Definition (as a OperationDefinition).

URL: [base]/Measure/$evaluate-measure

URL: [base]/Measure/[id]/$evaluate-measure

This is not an idempotent operation

In Parameters:
Name | Cardinality | Type | Binding | Profile | Documentation

periodStart | 1..1 | date | | | The start of the measurement period. In keeping with the semantics of the date parameter used in the FHIR search operation, the period will start at the beginning of the period implied by the supplied timestamp. E.g. a value of 2014 would set the period start to be 2014-01-01T00:00:00 inclusive

periodEnd | 1..1 | date | | | The end of the measurement period. The period will end at the end of the period implied by the supplied timestamp. E.g. a value of 2014 would set the period end to be 2014-12-31T23:59:59 inclusive

measure | 0..1 | Reference(Measure) | | | The measure to evaluate. This parameter is only required when the operation is invoked on the resource type, it is not used when invoking the operation on a Measure instance

reportType | 0..1 | code | | | The type of measure report, patient, patient-list, or population. If not specified, a default value of patient will be used if the patient parameter is supplied, otherwise, population will be used

patient | 0..1 | Reference(Patient) | | | Patient to evaluate against. If not specified, the measure will be evaluated for all patients that meet the requirements of the measure. If specified, only the referenced patient will be evaluated

practitioner | 0..1 | Reference(Practitioner) | | | Practitioner to evaluate. If specified, the measure will be evaluated only for patients whose primary practitioner is the identified practitioner

lastReceivedOn | 0..1 | dateTime | | | The date the results of this measure were last received. This parameter is only valid for patient-level reports and is used to indicate when the last time a result for this patient was received. This information can be used to limit the set of resources returned for a patient-level report

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | MeasureReport | | | The results of the measure calculation. See the MeasureReport resource for a complete description of the output of this operation Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The effect of invoking this operation is to calculate the measure for the given patient, or all patients if no patient is supplied, and return the results as a MeasureReport resource of the appropriate type

14.5.10.1.1 Examples

Request evaluation of the CMS146 measure for patient 124 over a measurement period of the first 3 months of 2014 (Request):

```
GET [base]/Measure/CMS146/$evaluate-measure?periodStart=2014&periodEnd=2014

GET [base]/Measure/CMS146/$evaluate-measure?patient-124&periodStart=2014-01&periodend=2014-03
```

Returned on successful evaluation (Response):

```
HTTP/1.1 200 OK

<?xml version="1.0" encoding="UTF-8"?>
<MeasureReport xmlns="http://hl7.org/fhir">
  <id value="evaluate-measure-operation-response"/>
  <text>
    <status value="generated"/>
    <div xmlns="http://www.w3.org/1999/xhtml">
  <!-- Generated... -->
    </div>
  </text>
  <contained>
    <Organization>
      <id value="reporter"/>
      <name value="Good Health Hospital"/>
    </Organization>
  </contained>
  <measure>
    <reference value="Measure/CMS146"/>
  </measure>
  <type value="summary"/>
  <period>
      <start value="2014-01-01" />
      <end value="2014-03-31" />
  </period>
  <status value="complete"/>
  <reportingOrganization>
    <reference value="#reporter"/>
  </reportingOrganization>
  <group>
    <identifier><value value="CMS146-group-1"/></identifier>
    <population>
      <code><coding><code value="initial-population"/></coding></code>
      <count value="500"/>
    </population>
    <population>
      <code><coding><code value="numerator"/></coding></code>
      <count value="200"/>
    </population>
    <population>
      <code><coding><code value="denominator"/></coding></code>
      <count value="500"/>
    </population>
    <population>
      <code><coding><code value="denominator-exclusion"/></coding></code>
      <count value="100"/>
    </population>
    <stratifier>
      <identifier><value value="stratifier-ages-up-to-9"/></identifier>
      <stratum>
        <value value="true"/>
```

```
    <population>
      <code><coding><code value="initial-population"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="numerator"/></coding></code>
      <count value="100"/>
    </population>
    <population>
      <code><coding><code value="denominator"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="denominator-exclusion"/></coding></code>
      <count value="50"/>
    </population>
  </stratum>
  <stratum>
    <value value="false"/>
    <population>
      <code><coding><code value="initial-population"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="numerator"/></coding></code>
      <count value="100"/>
    </population>
    <population>
      <code><coding><code value="denominator"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="denominator-exclusion"/></coding></code>
      <count value="50"/>
    </population>
  </stratum>
</stratifier>
<stratifier>
  <identifier><value value="stratifier-ages-10-plus"/></identifier>
  <stratum>
    <value value="true"/>
    <population>
      <code><coding><code value="initial-population"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="numerator"/></coding></code>
      <count value="100"/>
    </population>
    <population>
      <code><coding><code value="denominator"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="denominator-exclusion"/></coding></code>
      <count value="50"/>
    </population>
  </stratum>
  <stratum>
    <value value="false"/>
    <population>
      <code><coding><code value="initial-population"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="numerator"/></coding></code>
      <count value="100"/>
    </population>
    <population>
      <code><coding><code value="denominator"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="denominator-exclusion"/></coding></code>
      <count value="50"/>
    </population>
  </stratum>
</stratifier>
<stratifier>
  <identifier><value value="stratifier-gender"/></identifier>
  <stratum>
    <value value="male"/>
    <population>
      <code><coding><code value="initial-population"/></coding></code>
      <count value="250"/>
    </population>
    <population>
      <code><coding><code value="numerator"/></coding></code>
      <count value="100"/>
```

```
        </population>
        <population>
          <code><coding><code value="denominator"/></coding></code>
          <count value="250"/>
        </population>
        <population>
          <code><coding><code value="denominator-exclusion"/></coding></code>
          <count value="50"/>
        </population>
      </stratum>
      <stratum>
        <value value="female"/>
        <population>
          <code><coding><code value="initial-population"/></coding></code>
          <count value="250"/>
        </population>
        <population>
          <code><coding><code value="numerator"/></coding></code>
          <count value="100"/>
        </population>
        <population>
          <code><coding><code value="denominator"/></coding></code>
          <count value="250"/>
        </population>
        <population>
          <code><coding><code value="denominator-exclusion"/></coding></code>
          <count value="50"/>
        </population>
      </stratum>
      <stratum>
        <value value="other"/>
        <population>
          <code><coding><code value="initial-population"/></coding></code>
          <count value="0"/>
        </population>
        <population>
          <code><coding><code value="numerator"/></coding></code>
          <count value="0"/>
        </population>
        <population>
          <code><coding><code value="denominator"/></coding></code>
          <count value="0"/>
        </population>
        <population>
          <code><coding><code value="denominator-exclusion"/></coding></code>
          <count value="0"/>
        </population>
      </stratum>
      <stratum>
        <value value="unknown"/>
        <population>
          <code><coding><code value="initial-population"/></coding></code>
          <count value="0"/>
        </population>
        <population>
          <code><coding><code value="numerator"/></coding></code>
          <count value="0"/>
        </population>
        <population>
          <code><coding><code value="denominator"/></coding></code>
          <count value="0"/>
        </population>
        <population>
          <code><coding><code value="denominator-exclusion"/></coding></code>
          <count value="0"/>
        </population>
      </stratum>
    </stratifier>
  </group>
</MeasureReport>
```

14.5.10.2 Data Requirements

The data-requirements operation aggregates and returns the parameters and data requirements for the measure and all its dependencies as a single module definition

The official URL for this operation definition is

```
http://hl7.org/fhir/OperationDefinition/Measure-data-requirements
```

Formal Definition (as a OperationDefinition).

URL: [base]/Measure/[id]/$data-requirements

This is not an idempotent operation

In Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

periodStart | 1..1 | date | | | The start of the measurement period. In keeping with the semantics of the date parameter used in the FHIR search operation, the period will start at the beginning of the period implied by the supplied timestamp. E.g. a value of 2014 would set the period start to be

2014-01-01T00:00:00 inclusive

periodEnd | 1..1 | date | | | The end of the measurement period. The period will end at the end of the period implied by the supplied timestamp. E.g. a value of 2014 would set the period end to be 2014-12-31T23:59:59 inclusive

Out Parameters:

Name | Cardinality | Type | Binding | Profile | Documentation

return | 1..1 | Library | | | The result of the requirements gathering is a module-definition Library that describes the aggregate parameters, data requirements, and dependencies of the measure Note: as this the only out parameter, it is a resource, and it has the name 'return', the result of this operation is returned directly as a resource

The effect of invoking this operation is to determine the aggregate set of data requirements and dependencies for the measure. The result is a Library resource with a type of module-definition that contains all the parameter definitions and data requirements of the libraries referenced by the measures. Implementations SHOULD aggregate data requirements intelligently (i.e. by collapsing overlapping data requirements)

14.5.10.2.1 Examples

Request data requirements for the CMS146 measure (Request):

```
GET [base]/Measure/CMS146/$data-requirements?periodStart=2014&periodEnd=2014
```

Returned on successful evaluation (Response):

```
HTTP/1.1 200 OK

{
  "resourceType": "Library",
  "type": {
    "coding": [
      {
        "code": "module-definition"
      }
    ]
  },
  "relatedArtifact": [
    {
      "type": "depends-on",
      "resource": {
        "reference": "Library/library-quick-model-definition"
      }
    }
  ],
  "dataRequirement": [
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "code",
          "valueSetString": "Other Female Reproductive Conditions"
        }
      ]
    },
    {
      "type": "Patient"
    },
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "category",
          "valueCode": [
            "diagnosis"
          ]
        },
        {
          "path": "clinicalStatus",
          "valueCode": [
            "confirmed"
          ]
        },
        {
          "path": "code",
          "valueSetString": "2.16.840.1.113883.3.464.1003.102.12.1011"
        }
      ]
    },
    {
      "type": "Condition",
      "codeFilter": [
        {
          "path": "category",
          "valueCode": [
            "diagnosis"
          ]
        },
        {
          "path": "clinicalStatus",
          "valueCode": [
            "confirmed"
```

```
          ]
        },
        {
          "path": "code",
          "valueSetString": "2.16.840.1.113883.3.464.1003.102.12.1012"
        }
      ]
    },
    {
      "type": "Encounter",
      "codeFilter": [
        {
          "path": "status",
          "valueCode": [
            "finished"
          ]
        },
        {
          "path": "class",
          "valueCode": [
            "ambulatory"
          ]
        },
        {
          "path": "type",
          "valueSetString": "2.16.840.1.113883.3.464.1003.101.12.1061"
        }
      ]
    },
    {
      "type": "DiagnosticReport",
      "codeFilter": [
        {
          "path": "diagnosis",
          "valueSetString": "2.16.840.1.113883.3.464.1003.198.12.1012"
        }
      ]
    },
    {
      "type": "Medication",
      "codeFilter": [
        {
          "path": "code",
          "valueSetString": "2.16.840.1.113883.3.464.1003.196.12.1001"
        }
      ]
    },
    {
      "type": "MedicationRequest",
      "codeFilter": [
        {
          "path": "status",
          "valueCode": [
            "active"
          ]
        },
        {
          "path": "medication.code",
          "valueSetString": "2.16.840.1.113883.3.464.1003.196.12.1001"
        }
      ]
    },
    {
      "type": "MedicationStatement",
      "codeFilter": [
        {
          "path": "status",
          "valueCode": [
            "completed"
          ]
        },
        {
          "path": "medication.code",
          "valueSetString": "2.16.840.1.113883.3.464.1003.196.12.1001"
        }
      ]
    }
  ]
}
```


14.6 Resource MeasureReport - Content

URL: https://hl7.org/fhir/STU3/measurereport.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient

The MeasureReport resource contains the results of evaluating a measure.

14.6.1 Scope and Usage

This resource is an event resource from a FHIR workflow perspective - see Workflow, specifically Event.

The MeasureReport resource represents the results of evaluating a measure for a specific patient or group of patients. The $evaluate-measure operation of the Measure resource is defined to return a MeasureReport. The resource is capable of representing three different levels of report: patient-level, patient-list, and summary.

The resource draws requirements from the HL7 CDA R2 Implementation Guide: Quality Reporting Document Architecture - Category 1 (QRDA I) DSTU Release 3 (US Realm) and the HL7 Implementation Guide for CDA Release 2: Quality Reporting Document Architecture - Category III (QRDA III), DSTU Release 1 implementation guides.

Note that this resource is a special case of the more general notion of a query evaluation result. However, because the general case requires the ability to represent arbitrary content, this resource uses a simple indicator structure to describe population sizes for each population type defined in the measure. The intent is to be able to represent the more general case as well, either by generalizing this resource, or by making this structure a profile of a more general resource, and we are actively seeking comments about what approaches might be taken to achieve that aim.

14.6.2 Boundaries and Relationships

Although the MeasureReport is conceptually an Observation, there is enough specific information required to support the quality reporting use case to warrant a separate resource.

The resource is differentiated from a general purpose query result because it communicates specific information related to quality measurement evaluation that would be difficult to convey generally without imposing some other structure on top of the general results.

14.6.3 Resource Content

* Structure

* UML

* XML

* JSON

* Turtle

* R2 Diff

* All

Structure

Name | Flags | Card. | Type | Description & Constraints
MeasureReport | | | DomainResource | Results of a measure evaluation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..1 | Identifier | Additional identifier for the Report
status | ?!? | 1..1 | code | complete | pending | error MeasureReportStatus (Required)
type | ? | 1..1 | code | individual | patient-list | summary MeasureReportType (Required)
measure | ? | 1..1 | Reference(Measure) | What measure was evaluated
patient | ? | 0..1 | Reference(Patient) | What patient the report is for
date | ? | 0..1 | dateTime | When the report was generated
reportingOrganization | ? | 0..1 | Reference(Organization) | Who is reporting the data
period | ? | 1..1 | Period | What period the report covers
group | | 0..* | BackboneElement | Measure results for each group
identifier | | 1..1 | Identifier | What group of the measure
population | | 0..* | BackboneElement | The populations in the group
identifier | ? | 0..1 | Identifier | Population identifier as defined in the measure
code | ? | 0..1 | CodeableConcept | initial-population | numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-score MeasurePopulationType (Extensible)
count | | 0..1 | integer | Size of the population
patients | | 0..1 | Reference(List) | For patient-list reports, the patients in this population
measureScore | ? | 0..1 | decimal | What score this group achieved
stratifier | | 0..* | BackboneElement | Stratification results
identifier | | 0..1 | Identifier | What stratifier of the group
stratum | | 0..* | BackboneElement | Stratum results, one for each unique value in the stratifier
value | | 1..1 | string | The stratum value, e.g. male
population | | 0..* | BackboneElement | Population results in this stratum
identifier | ? | 0..1 | Identifier | Population identifier as defined in the measure
code | ? | 0..1 | CodeableConcept | initial-population | numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-score MeasurePopulationType (Extensible)
count | | 0..1 | integer | Size of the population
patients | | 0..1 | Reference(List) | For patient-list reports, the patients in this population
measureScore | ? | 0..1 | decimal | What score this stratum achieved
evaluatedResources | | 0..1 | Reference(Bundle) | What data was evaluated to produce the measure score
Documentation for this format

UML Diagram (Legend)

XML Template

```
<MeasureReport xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Additional identifier for the Report --></identifier>
 <status value="[code]"/><!-- 1..1 complete | pending | error -->
 <type value="[code]"/><!-- 1..1 individual | patient-list | summary -->
 <measure><!-- 1..1 Reference(Measure) What measure was evaluated --></measure>
 <patient><!-- 0..1 Reference(Patient) What patient the report is for --></patient>
 <date value="[dateTime]"/><!-- 0..1 When the report was generated -->
 <reportingOrganization><!-- 0..1 Reference(Organization) Who is reporting the data --></reportingOrganization>
 <period><!-- 1..1 Period What period the report covers --></period>
 <group>  <!-- 0..* Measure results for each group -->
  <identifier><!-- 1..1 Identifier What group of the measure --></identifier>
  <population>  <!-- 0..* The populations in the group -->
   <identifier><!-- 0..1 Identifier Population identifier as defined in the measure --></identifier>
   <code><!-- 0..1 CodeableConcept initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-score --></code>
   <count value="[integer]"/><!-- 0..1 Size of the population -->
   <patients><!-- 0..1 Reference(List) For patient-list reports, the patients in this population --></patients>
  </population>
  <measureScore value="[decimal]"/><!-- 0..1 What score this group achieved -->
  <stratifier>  <!-- 0..* Stratification results -->
   <identifier><!-- 0..1 Identifier What stratifier of the group --></identifier>
   <stratum>  <!-- 0..* Stratum results, one for each unique value in the stratifier -->
    <value value="[string]"/><!-- 1..1 The stratum value, e.g. male -->
    <population>  <!-- 0..* Population results in this stratum -->
     <identifier><!-- 0..1 Identifier Population identifier as defined in the measure --></identifier>
     <code><!-- 0..1 CodeableConcept initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-score --></code>
     <count value="[integer]"/><!-- 0..1 Size of the population -->
     <patients><!-- 0..1 Reference(List) For patient-list reports, the patients in this population --></patients>
    </population>
    <measureScore value="[decimal]"/><!-- 0..1 What score this stratum achieved -->
   </stratum>
  </stratifier>
 </group>
 <evaluatedResources><!-- 0..1 Reference(Bundle) What data was evaluated to produce the measure score
    --></evaluatedResources>
</MeasureReport>
```

JSON Template

```
{
  "resourceType" : "MeasureReport",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Additional identifier for the Report
  "status" : "<code>", // R!  complete | pending | error
  "type" : "<code>", // R!  individual | patient-list | summary
  "measure" : { Reference(Measure) }, // R!  What measure was evaluated
  "patient" : { Reference(Patient) }, // What patient the report is for
  "date" : "<dateTime>", // When the report was generated
  "reportingOrganization" : { Reference(Organization) }, // Who is reporting the data
  "period" : { Period }, // R!  What period the report covers
  "group" : [{ // Measure results for each group
    "identifier" : { Identifier }, // R!  What group of the measure
    "population" : [{ // The populations in the group
      "identifier" : { Identifier }, // Population identifier as defined in the measure
      "code" : { CodeableConcept }, // initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-score
      "count" : <integer>, // Size of the population
      "patients" : { Reference(List) } // For patient-list reports, the patients in this population
    }],
    "measureScore" : <decimal>, // What score this group achieved
    "stratifier" : [{ // Stratification results
      "identifier" : { Identifier }, // What stratifier of the group
      "stratum" : [{ // Stratum results, one for each unique value in the stratifier
        "value" : "<string>", // R!  The stratum value, e.g. male
        "population" : [{ // Population results in this stratum
          "identifier" : { Identifier }, // Population identifier as defined in the measure
          "code" : { CodeableConcept }, // initial-population | numerator | numerator-exclusion | denominator |
    denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-score
          "count" : <integer>, // Size of the population
          "patients" : { Reference(List) } // For patient-list reports, the patients in this population
        }],
        "measureScore" : <decimal> // What score this stratum achieved
      }]
    }]
  }],
  "evaluatedResources" : { Reference(Bundle) } // What data was evaluated to produce the measure score
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .

[ a fhir:MeasureReport;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root
```

```
  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MeasureReport.identifier [ Identifier ]; # 0..1 Additional identifier for the Report
  fhir:MeasureReport.status [ code ]; # 1..1 complete | pending | error
  fhir:MeasureReport.type [ code ]; # 1..1 individual | patient-list | summary
  fhir:MeasureReport.measure [ Reference(Measure) ]; # 1..1 What measure was evaluated
  fhir:MeasureReport.patient [ Reference(Patient) ]; # 0..1 What patient the report is for
  fhir:MeasureReport.date [ dateTime ]; # 0..1 When the report was generated
  fhir:MeasureReport.reportingOrganization [ Reference(Organization) ]; # 0..1 Who is reporting the data
  fhir:MeasureReport.period [ Period ]; # 1..1 What period the report covers
  fhir:MeasureReport.group [ # 0..* Measure results for each group
    fhir:MeasureReport.group.identifier [ Identifier ]; # 1..1 What group of the measure
    fhir:MeasureReport.group.population [ # 0..* The populations in the group
      fhir:MeasureReport.group.population.identifier [ Identifier ]; # 0..1 Population identifier as defined in the
    measure
      fhir:MeasureReport.group.population.code [ CodeableConcept ]; # 0..1 initial-population | numerator | numerator-
    exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-
    exclusion | measure-score
      fhir:MeasureReport.group.population.count [ integer ]; # 0..1 Size of the population
      fhir:MeasureReport.group.population.patients [ Reference(List) ]; # 0..1 For patient-list reports, the patients in
     this population
    ], ...;
    fhir:MeasureReport.group.measureScore [ decimal ]; # 0..1 What score this group achieved
    fhir:MeasureReport.group.stratifier [ # 0..* Stratification results
      fhir:MeasureReport.group.stratifier.identifier [ Identifier ]; # 0..1 What stratifier of the group
      fhir:MeasureReport.group.stratifier.stratum [ # 0..* Stratum results, one for each unique value in the stratifier
        fhir:MeasureReport.group.stratifier.stratum.value [ string ]; # 1..1 The stratum value, e.g. male
        fhir:MeasureReport.group.stratifier.stratum.population [ # 0..* Population results in this stratum
          fhir:MeasureReport.group.stratifier.stratum.population.identifier [ Identifier ]; # 0..1 Population identifier
     as defined in the measure
          fhir:MeasureReport.group.stratifier.stratum.population.code [ CodeableConcept ]; # 0..1 initial-population |
    numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population |
     measure-population-exclusion | measure-score
          fhir:MeasureReport.group.stratifier.stratum.population.count [ integer ]; # 0..1 Size of the population
          fhir:MeasureReport.group.stratifier.stratum.population.patients [ Reference(List) ]; # 0..1 For patient-list
    reports, the patients in this population
        ], ...;
        fhir:MeasureReport.group.stratifier.stratum.measureScore [ decimal ]; # 0..1 What score this stratum achieved
      ], ...;
    ], ...;
  ], ...;
  fhir:MeasureReport.evaluatedResources [ Reference(Bundle) ]; # 0..1 What data was evaluated to produce the measure
    score
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Structure

Name | Flags | Card. | Type | Description & Constraints
MeasureReport | | | DomainResource | Results of a measure evaluation Elements defined in Ancestors: id, meta, implicitRules, language, text, contained, extension, modifierExtension
identifier | ? | 0..1 | Identifier | Additional identifier for the Report
status | ?!? | 1..1 | code | complete | pending | error MeasureReportStatus (Required)
type | ? | 1..1 | code | individual | patient-list | summary MeasureReportType (Required)
measure | ? | 1..1 | Reference(Measure) | What measure was evaluated
patient | ? | 0..1 | Reference(Patient) | What patient the report is for
date | ? | 0..1 | dateTime | When the report was generated
reportingOrganization | ? | 0..1 | Reference(Organization) | Who is reporting the data
period | ? | 1..1 | Period | What period the report covers
group | | 0..* | BackboneElement | Measure results for each group
identifier | | 1..1 | Identifier | What group of the measure
population | | 0..* | BackboneElement | The populations in the group
identifier | ? | 0..1 | Identifier | Population identifier as defined in the measure
code | ? | 0..1 | CodeableConcept | initial-population | numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-score MeasurePopulationType (Extensible)
count | | 0..1 | integer | Size of the population
patients | | 0..1 | Reference(List) | For patient-list reports, the patients in this population
measureScore | ? | 0..1 | decimal | What score this group achieved
stratifier | | 0..* | BackboneElement | Stratification results
identifier | | 0..1 | Identifier | What stratifier of the group
stratum | | 0..* | BackboneElement | Stratum results, one for each unique value in the stratifier
value | | 1..1 | string | The stratum value, e.g. male
population | | 0..* | BackboneElement | Population results in this stratum
identifier | ? | 0..1 | Identifier | Population identifier as defined in the measure
code | ? | 0..1 | CodeableConcept | initial-population | numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-score MeasurePopulationType (Extensible)
count | | 0..1 | integer | Size of the population
patients | | 0..1 | Reference(List) | For patient-list reports, the patients in this population
measureScore | ? | 0..1 | decimal | What score this stratum achieved

evaluatedResources | | 0..1 | Reference(Bundle) | What data was evaluated to produce the measure score

Documentation for this format

UML Diagram (Legend)

XML Template

```
<MeasureReport xmlns="http://hl7.org/fhir">
 <!-- from Resource: id, meta, implicitRules, and language -->
 <!-- from DomainResource: text, contained, extension, and modifierExtension -->
 <identifier><!-- 0..1 Identifier Additional identifier for the Report --></identifier>
 <status value="[code]"/><!-- 1..1 complete | pending | error -->
 <type value="[code]"/><!-- 1..1 individual | patient-list | summary -->
 <measure><!-- 1..1 Reference(Measure) What measure was evaluated --></measure>
 <patient><!-- 0..1 Reference(Patient) What patient the report is for --></patient>
 <date value="[dateTime]"/><!-- 0..1 When the report was generated -->
 <reportingOrganization><!-- 0..1 Reference(Organization) Who is reporting the data --></reportingOrganization>
 <period><!-- 1..1 Period What period the report covers --></period>
 <group>  <!-- 0..* Measure results for each group -->
  <identifier><!-- 1..1 Identifier What group of the measure --></identifier>
  <population>  <!-- 0..* The populations in the group -->
   <identifier><!-- 0..1 Identifier Population identifier as defined in the measure --></identifier>
   <code><!-- 0..1 CodeableConcept initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-score --></code>
   <count value="[integer]"/><!-- 0..1 Size of the population -->
   <patients><!-- 0..1 Reference(List) For patient-list reports, the patients in this population --></patients>
  </population>
  <measureScore value="[decimal]"/><!-- 0..1 What score this group achieved -->
  <stratifier>  <!-- 0..* Stratification results -->
   <identifier><!-- 0..1 Identifier What stratifier of the group --></identifier>
   <stratum>  <!-- 0..* Stratum results, one for each unique value in the stratifier -->
    <value value="[string]"/><!-- 1..1 The stratum value, e.g. male -->
    <population>  <!-- 0..* Population results in this stratum -->
     <identifier><!-- 0..1 Identifier Population identifier as defined in the measure --></identifier>
     <code><!-- 0..1 CodeableConcept initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-score --></code>
     <count value="[integer]"/><!-- 0..1 Size of the population -->
     <patients><!-- 0..1 Reference(List) For patient-list reports, the patients in this population --></patients>
    </population>
    <measureScore value="[decimal]"/><!-- 0..1 What score this stratum achieved -->
   </stratum>
  </stratifier>
 </group>
 <evaluatedResources><!-- 0..1 Reference(Bundle) What data was evaluated to produce the measure score
    --></evaluatedResources>
</MeasureReport>
```

JSON Template

```
{
  "resourceType" : "MeasureReport",
  // from Resource: id, meta, implicitRules, and language
  // from DomainResource: text, contained, extension, and modifierExtension
  "identifier" : { Identifier }, // Additional identifier for the Report
  "status" : "<code>", // R!  complete | pending | error
  "type" : "<code>", // R!  individual | patient-list | summary
  "measure" : { Reference(Measure) }, // R!  What measure was evaluated
  "patient" : { Reference(Patient) }, // What patient the report is for
  "date" : "<dateTime>", // When the report was generated
  "reportingOrganization" : { Reference(Organization) }, // Who is reporting the data
  "period" : { Period }, // R!  What period the report covers
  "group" : [{ // Measure results for each group
    "identifier" : { Identifier }, // R!  What group of the measure
    "population" : [{ // The populations in the group
      "identifier" : { Identifier }, // Population identifier as defined in the measure
      "code" : { CodeableConcept }, // initial-population | numerator | numerator-exclusion | denominator | denominator-
    exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-score
      "count" : <integer>, // Size of the population
      "patients" : { Reference(List) } // For patient-list reports, the patients in this population
    }],
    "measureScore" : <decimal>, // What score this group achieved
    "stratifier" : [{ // Stratification results
      "identifier" : { Identifier }, // What stratifier of the group
      "stratum" : [{ // Stratum results, one for each unique value in the stratifier
        "value" : "<string>", // R!  The stratum value, e.g. male
        "population" : [{ // Population results in this stratum
          "identifier" : { Identifier }, // Population identifier as defined in the measure
          "code" : { CodeableConcept }, // initial-population | numerator | numerator-exclusion | denominator |
    denominator-exclusion | denominator-exception | measure-population | measure-population-exclusion | measure-score
          "count" : <integer>, // Size of the population
          "patients" : { Reference(List) } // For patient-list reports, the patients in this population
        }],
        "measureScore" : <decimal> // What score this stratum achieved
      }]
    }]
  }],
  "evaluatedResources" : { Reference(Bundle) } // What data was evaluated to produce the measure score
}
```

Turtle Template

```
@prefix fhir: <http://hl7.org/fhir/> .
```

```
[ a fhir:MeasureReport;
  fhir:nodeRole fhir:treeRoot; # if this is the parser root

  # from Resource: .id, .meta, .implicitRules, and .language
  # from DomainResource: .text, .contained, .extension, and .modifierExtension
  fhir:MeasureReport.identifier [ Identifier ]; # 0..1 Additional identifier for the Report
  fhir:MeasureReport.status [ code ]; # 1..1 complete | pending | error
  fhir:MeasureReport.type [ code ]; # 1..1 individual | patient-list | summary
  fhir:MeasureReport.measure [ Reference(Measure) ]; # 1..1 What measure was evaluated
  fhir:MeasureReport.patient [ Reference(Patient) ]; # 0..1 What patient the report is for
  fhir:MeasureReport.date [ dateTime ]; # 0..1 When the report was generated
  fhir:MeasureReport.reportingOrganization [ Reference(Organization) ]; # 0..1 Who is reporting the data
  fhir:MeasureReport.period [ Period ]; # 1..1 What period the report covers
  fhir:MeasureReport.group [ # 0..* Measure results for each group
    fhir:MeasureReport.group.identifier [ Identifier ]; # 1..1 What group of the measure
    fhir:MeasureReport.group.population [ # 0..* The populations in the group
      fhir:MeasureReport.group.population.identifier [ Identifier ]; # 0..1 Population identifier as defined in the
    measure
      fhir:MeasureReport.group.population.code [ CodeableConcept ]; # 0..1 initial-population | numerator | numerator-
    exclusion | denominator | denominator-exclusion | denominator-exception | measure-population | measure-population-
    exclusion | measure-score
      fhir:MeasureReport.group.population.count [ integer ]; # 0..1 Size of the population
      fhir:MeasureReport.group.population.patients [ Reference(List) ]; # 0..1 For patient-list reports, the patients in
     this population
    ], ...;
    fhir:MeasureReport.group.measureScore [ decimal ]; # 0..1 What score this group achieved
    fhir:MeasureReport.group.stratifier [ # 0..* Stratification results
      fhir:MeasureReport.group.stratifier.identifier [ Identifier ]; # 0..1 What stratifier of the group
      fhir:MeasureReport.group.stratifier.stratum [ # 0..* Stratum results, one for each unique value in the stratifier
        fhir:MeasureReport.group.stratifier.stratum.value [ string ]; # 1..1 The stratum value, e.g. male
        fhir:MeasureReport.group.stratifier.stratum.population [ # 0..* Population results in this stratum
          fhir:MeasureReport.group.stratifier.stratum.population.identifier [ Identifier ]; # 0..1 Population identifier
     as defined in the measure
          fhir:MeasureReport.group.stratifier.stratum.population.code [ CodeableConcept ]; # 0..1 initial-population |
    numerator | numerator-exclusion | denominator | denominator-exclusion | denominator-exception | measure-population |
     measure-population-exclusion | measure-score
          fhir:MeasureReport.group.stratifier.stratum.population.count [ integer ]; # 0..1 Size of the population
          fhir:MeasureReport.group.stratifier.stratum.population.patients [ Reference(List) ]; # 0..1 For patient-list
    reports, the patients in this population
        ], ...;
        fhir:MeasureReport.group.stratifier.stratum.measureScore [ decimal ]; # 0..1 What score this stratum achieved
      ], ...;
    ], ...;
  ], ...;
  fhir:MeasureReport.evaluatedResources [ Reference(Bundle) ]; # 0..1 What data was evaluated to produce the measure
    score
]
```

Changes since DSTU2

This resource did not exist in Release 2

This analysis is available as XML or JSON.

Alternate definitions: Master Definition (XML, JSON), XML Schema/Schematron (for) + JSON Schema, ShEx (for Turtle)

14.6.3.1 Terminology Bindings

Path | Definition | Type | Reference

MeasureReport.status | The status of the measure report | Required | MeasureReportStatus

MeasureReport.type | The type of the measure report | Required | MeasureReportType

MeasureReport.group.population.code MeasureReport.group.stratifier.stratum.population.code | The type of population (e.g. initial, numerator, denominator, etc.) | Extensible | MeasurePopulationType

14.6.4 MeasureReport Search

Although the MeasureReport resource does define search parameters, it is up to the individual measure evaluation service whether or not historical records of measure evaluation requests are preserved. A service may not support searching on MeasureReports at all, or it may support searching only for a pre-defined expiration period. The search parameters are defined to provide consumers with a consistent interface to searching if it is available for a specific service implementation. In any case, services should detail the support they do provide using a CapabilityStatement.

14.6.5 Search Parameters

Search parameters for this resource. The common parameters also apply. See Searching for more information about searching in REST, messaging, and services.

Name | Type | Description | Expression | In Common

identifier | token | External identifier of the measure report to be returned | MeasureReport.identifier

patient | reference | The identity of a patient to search for individual measure report results for | MeasureReport.patient (Patient)

status | token | The status of the measure report | MeasureReport.status

14.6.6 Resource MeasureReport - Examples

URL: https://hl7.org/fhir/STU3/measurereport-examples.html

These examples illustrate the use of the MeasureReport resource to report individual, patient-listing, and summary-level results for measure calculation.

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.6.6 Resource MeasureReport - Examples

Clinical Quality Information Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Example List:

CMS 146 Summary Report | measurereport-cms146-cat3-example | XML | JSON | Turtle
CMS 146 Patient Listing Report | measurereport-cms146-cat2-example | XML | JSON | Turtle
CMS 146 Individual Report | measurereport-cms146-cat1-example | XML | JSON | Turtle

Usage note: every effort has been made to ensure that the examples are correct and useful, but they are not a normative part of the specification.

14.6.7 Resource MeasureReport - Detailed Descriptions

URL: https://hl7.org/fhir/STU3/measurereport-definitions.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.6.7 Resource MeasureReport - Detailed Descriptions

Clinical Quality Information Work Group | Maturity Level: 2 | Trial Use | Compartments: Patient

Detailed Descriptions for the elements in the MeasureReport resource.

MeasureReport
Definition | The MeasureReport resource contains the results of evaluating a measure.
Control | 1..1
MeasureReport.identifier
Definition | A formal identifier that is used to identify this report when it is represented in other formats, or referenced in a specification, model, design or an instance.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Requirements | Allows externally provided and/or usable business identifiers to be easily associated with the module.
Summary | true
Comments | Typically, this is used for identifiers that can go in an HL7 v3 II data type - e.g. to identify this {{title}} outside of FHIR, where the logical URL is not possible to use.
MeasureReport.status
Definition | The report status. No data will be available until the report status is complete.
Control | 1..1
Terminology Binding | MeasureReportStatus (Required)
Type | code
Is Modifier | true
Summary | true
MeasureReport.type
Definition | The type of measure report. This may be an individual report, which provides a single patient's score for the measure; a patient listing, which returns the list of patients that meet the various criteria in the measure; or a summary report, which returns a population count for each of the criteria in the measure.
Control | 1..1
Terminology Binding | MeasureReportType (Required)
Type | code
Summary | true
Comments | This element is labeled as a modifier because the status contains codes that mark the resource as not currently valid.
MeasureReport.measure
Definition | A reference to the Measure that was evaluated to produce this report.
Control | 1..1
Type | Reference(Measure)
Summary | true
MeasureReport.patient
Definition | Optional Patient if the report was requested for a single patient.
Control | 0..1
Type | Reference(Patient)
Summary | true
MeasureReport.date
Definition | The date this measure report was generated.
Control | 0..1
Type | dateTime
Summary | true
MeasureReport.reportingOrganization
Definition | Reporting Organization.

Control | 0..1
Type | Reference(Organization)
Summary | true
MeasureReport.period
Definition | The reporting period for which the report was calculated.
Control | 1..1
Type | Period
Summary | true
MeasureReport.group
Definition | The results of the calculation, one for each population group in the measure.
Control | 0..*
Summary | false
MeasureReport.group.identifier
Definition | The identifier of the population group as defined in the measure definition.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 1..1
Type | Identifier
Summary | false
MeasureReport.group.population
Definition | The populations that make up the population group, one for each type of population appropriate for the measure.
Control | 0..*
Summary | false
MeasureReport.group.population.identifier
Definition | The identifier of the population being reported, as defined by the population element of the measure.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Summary | true
MeasureReport.group.population.code
Definition | The type of the population.
Control | 0..1
Terminology Binding | MeasurePopulationType (Extensible)
Type | CodeableConcept
Summary | true
MeasureReport.group.population.count
Definition | The number of members of the population.
Control | 0..1
Type | integer
Summary | false
MeasureReport.group.population.patients
Definition | This element refers to a List of patient level MeasureReport resources, one for each patient in this population.
Control | 0..1
Type | Reference(List)
Summary | false
MeasureReport.group.measureScore
Definition | The measure score for this population group, calculated as appropriate for the measure type and scoring method, and based on the contents of the populations defined in the group.
Control | 0..1
Type | decimal
Summary | true
MeasureReport.group.stratifier
Definition | When a measure includes multiple stratifiers, there will be a stratifier group for each stratifier defined by the measure.
Control | 0..*
Summary | false
MeasureReport.group.stratifier.identifier
Definition | The identifier of this stratifier, as defined in the measure definition.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Summary | false
MeasureReport.group.stratifier.stratum
Definition | This element contains the results for a single stratum within the stratifier. For example, when stratifying on administrative gender, there will be four strata, one for each possible gender value.
Control | 0..*
Summary | false
MeasureReport.group.stratifier.stratum.value
Definition | The value for this stratum, expressed as a string. When defining stratifiers on complex values, the value must be rendered such that the value for each stratum within the stratifier is unique.
Control | 1..1
Type | string
Summary | false
MeasureReport.group.stratifier.stratum.population

Definition | The populations that make up the stratum, one for each type of population appropriate to the measure.
Control | 0..*
Summary | false
MeasureReport.group.stratifier.stratum.population.identifier
Definition | The identifier of the population being reported, as defined by the population element of the measure.
Note | This is a business identifer, not a resource identifier (see discussion)
Control | 0..1
Type | Identifier
Summary | true
MeasureReport.group.stratifier.stratum.population.code
Definition | The type of the population.
Control | 0..1
Terminology Binding | MeasurePopulationType (Extensible)
Type | CodeableConcept
Summary | true
MeasureReport.group.stratifier.stratum.population.count
Definition | The number of members of the population in this stratum.
Control | 0..1
Type | integer
Summary | false
MeasureReport.group.stratifier.stratum.population.patients
Definition | This element refers to a List of patient level MeasureReport resources, one for each patient in this population in this stratum.
Control | 0..1
Type | Reference(List)
Summary | false
MeasureReport.group.stratifier.stratum.measureScore
Definition | The measure score for this stratum, calculated as appropriate for the measure type and scoring method, and based on only the members of this stratum.
Control | 0..1
Type | decimal
Summary | true
MeasureReport.evaluatedResources
Definition | A reference to a Bundle containing the Resources that were used in the evaluation of this report.
Control | 0..1
Type | Reference(Bundle)
Summary | false

14.6.8 Resource MeasureReport - Mappings

URL: https://hl7.org/fhir/STU3/measurereport-mappings.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.6.8 Resource MeasureReport - Mappings

Clinical Quality Information Work Group | Maturity Level: N/A | Ballot Status: Informative | Compartments: Patient

Mappings:

Workflow Mapping

Mappings for the measurereport resource.

14.6.8.1 Mappings for Workflow Mapping (http://hl7.org/fhir/workflow)

MeasureReport | Event
identifier
status | Event.status
type | Event.code
measure | Event.definition
patient | Event.subject
date | Event.occurrenceDateTime
reportingOrganization
period
group
identifier
population
identifier
code
count
patients
measureScore
stratifier
identifier
stratum

value
population
identifier
code
count
patients
measureScore
evaluatedResources

14.6.9 Resource MeasureReport - Design Notes

URL: https://hl7.org/fhir/STU3/measurereport-explanations.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.6.9 Resource MeasureReport - Design Notes

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14.6.10 Resource MeasureReport - Extensions & Profiles

URL: https://hl7.org/fhir/STU3/measurereport-profiles.html

This page is part of the FHIR Specification (v3.0.2: STU 3). The current version which supercedes this version is 4.0.1. For a full list of available versions, see the Directory of published versions

14.6.10 Resource MeasureReport - Extensions & Profiles

Clinical Quality Information Work Group | Maturity Level: N/A | Ballot Status: Informative

This table lists profiles and extensions for the MeasureReport resource. For background information, see Profiling Resources and Extensibility. Additional profiles and resources may be found in published Implementation Guides , or in the Conformance resource registry .

14.6.10.1 Profiles

No Profiles defined for this resource

14.6.10.2 Extensions

No Extensions defined for this resource

Extensions that reference this resource:

No Extensions refer to this resource

14.6.10.3 Search Extensions

No Search Extensions defined for this resource